

Small Foundation Models of Human Cognition and Behaviour

Nick Oh

 socius labs

London, UK

nick.sh.oh@socius.org

Fernand Gobet



London School of Economics and Political Science

London, UK

f.gobet@lse.ac.uk

Abstract

Large language models fine-tuned on human behavioural data have emerged as general-purpose cognitive proxies, but the scale this requires, and whether these models process task structure or exploit statistical shortcuts, remain open questions. We train fourteen models from 135M to 14B parameters across four architecture families on Psych-101, a dataset of 10.7 million trial-level choices from 160 experiments. For in-distribution simulations, scale barely matters. The models fall within a narrow band, as though against a ceiling, and 0.6B to 1B parameters suffice to match a 70B baseline on held-out participants. Out-of-distribution, that band opens into a markedly steeper scaling gradient, with larger models clearly advantaged in generalisation to novel task structure. To determine what information these models use, we run two diagnostics. We progressively strip four prompt channels – task instructions, experimental stimuli, outcome feedback, and choice history – across 27 experiments, and permute trial order. Masking the content of stimuli and feedback destroys 75.7% of learned information and pushes models below chance, demonstrating that choice history alone does not account for performance. Permutation reveals invariance on tasks with independent trials but sensitivity where trial order is determined by prior responses. Small cognitively fine-tuned models therefore show promise as noise ceiling estimators for psychological experiments, though their scope remains bounded by the paradigms seen in training.



<https://github.com/socius-org/Centauri>



<https://hf.co/collections/socius/centauri>

1 Introduction

Unified theories of cognition (Newell, 1990) aspire to account for the breadth of human mental function within a single framework, and decades of work on symbolic and hybrid architectures such as ACT-R (Anderson & Lebiere, 1998), CHREST (Gobet et al., 2001), and Soar (Laird, 2012) have laid a robust foundation. Yet a comprehensive account remains elusive, in part because each new experimental paradigm typically demands new model components, and cross-domain comparison requires reconciling incommensurable formalisms (Busemeyer & Diederich, 2010).

Deep neural networks (DNNs) trained on human behavioural data offer a complementary path, learning regularities directly from choice data and serving as benchmarks against which interpretable models can be evaluated (Agrawal et al., 2020; Peterson et al., 2021; Kuperwaits et al., 2023). Centaur (Binz et al., 2025) scaled this idea to a general-purpose cognitive proxy by fine-tuning Llama-3.1-70B on Psych-101, a dataset of 10.7 million trial-level choices from over 60,000 participants across 160 experiments. Subsequent work has extended this paradigm to new domains including social science surveys (Kolluri et al., 2025), behavioural economics (Xie et al., 2025), and social media-derived behavioural prediction

(Lei et al., 2026), as well as new post-training methods (Zhu et al., 2025b) and in-context model generation (Rmus et al., 2025).

Two questions about this paradigm remain open, and we address them in turn. The first concerns *scale*. Asking whether 70B parameters are necessary is not by itself a well-posed question, because predictive accuracy could be limited by the capacity of the base model, the capacity of the adapter that fine-tuning is permitted to move, the quantity of behavioural training data, or the irreducible stochasticity of human responses. We therefore vary these factors separately, training fourteen models from 135M to 14B parameters across four families – Llama-3.1/3.2 (Grattafiori et al., 2024), Qwen3 (Yang et al., 2025), SmoLLM2/3 (Allal et al., 2025; Bakouch et al., 2025), and OLMo2/3 (Walsh et al., 2025; Ettinger et al., 2026) – sweeping LoRA rank from 4 to 64, and fine-tuning on nested subsets of Psych-101. In-distribution, model scale is not what limits prediction. The size needed to match a reproduced Centaur-70B on held-out participants falls from 8B to 0.6B as adapter rank rises from 4 to 32. Out-of-distribution the advantage of scale reappears, so what additional parameters buy is not a better fit to Psych-101 but better transfer beyond it. Model size is therefore a choice about transfer rather than about in-distribution fit (Section 4.1).

The second question concerns *what these models learn*. Recent work has raised the concern that they exploit statistical shortcuts in choice sequences rather than process task structure (Xie & Zhu, 2025; Liu & Ding, 2025; Namazova et al., 2025; Schlegelmilch & Dome, 2026), yet no previous study has systematically removed subcomponents of the prompts. We decompose the prompts into four information channels (instruction *I*, stimuli *S*, feedback *F*, and choice history *C*) and conduct structural ablations across 32 experiments, alongside order permutation tests on 2 experiments with contrasting exchangeability structure. The results show a pattern of task-adaptive information use that is difficult to reconcile with pure shortcut learning (Section 4.2).

Taken together, these results support a modest reading of what Centaur-style supervised fine-tuning produces. These models are not theories of cognition in the sense that ACT-R, CHREST, and Soar are, since they posit no architecture, commit to no mechanism, and explain nothing about how a choice is generated. What they offer instead is an estimate of the noise ceiling¹ (Agrawal et al., 2020; Kuperwajs et al., 2023), an empirical bound on how much of the behaviour in a given paradigm is predictable at all, and thus on how much room a process-level theory of that paradigm still has to improve. Such a bound is informative only if the model reaches it by using the information the experiment actually provides – which our study establishes – and it extends only as far as the paradigms the training data covers. We return to this position in Section 5.

2 Related work

2.1 Neural network as a cognitive and behavioural proxy

Cognitive models have traditionally been handcrafted and refined through cycles of intuition and model comparison (Kuperwajs et al., 2023). Neural networks trained on human behavioural data are reshaping this process by providing a data-driven reference point against which interpretable theories can be benchmarked. Frank & Goodman (2026) argue that the key advance of modern AI models is their stimulus computability, the ability to operate over the same stimuli that humans experience, which enables their use as scientific models of mind. Agrawal et al. (2020) formalised one instance of this idea as *scientific regret minimisation*, in which the per-datapoint gap between what a DNN can predict and what an interpretable theory predicts identifies precisely where theoretical improvement is possible.

This approach has proven informative across domains. In *planning*, Kuperwajs et al. (2023) trained feedforward DNNs on 82.8 million moves made by 1.2 million players of 4-in-a-row, establishing a noise ceiling that exposed systematic failures in a heuristic-search cognitive model undetectable from raw data alone. Jensen et al. (2024) took a different but

¹A noise ceiling estimates the best achievable prediction performance on a dataset after accounting for irreducible stochasticity in human responses.

complementary approach, training a meta-reinforcement learning agent that learns when to plan, allocating more deliberation to novel situations and producing rollout patterns resembling rodent hippocampal replays. In *decision-making*, Peterson et al. (2021) found that a DNN trained on approximately 13,000 risky choice problems captured regularities that all 21 existing theories missed, independently recovering Prospect Theory’s S-shaped utility function while revealing unformalised context-sensitivity in human value assignment. Zhu et al. (2025a) extended this to strategic interaction across 90,000 decisions in 2,400 matrix games, showing context-dependent variation in strategic sophistication beyond what fixed-parameter theories can accommodate.

2.2 Cognitive and behavioural foundation models

Where the preceding work uses neural networks as tools for improving domain-specific cognitive models, a recent line of work goes further, fine-tuning large language models directly on human behavioural data to create general-purpose foundation models of human cognition and behaviour. The resulting systems vary widely in their training data, adaptation strategy, and target granularity of prediction. A structured comparison of these models, along with full training, infrastructure, and evaluation details, is provided in Appendix A.

Centaur (Binz et al., 2025) established the paradigm by fine-tuning Llama-3.1-70B on Psych-101, outperforming domain-specific cognitive models on held-out participants while generalising to unseen tasks and domains. Subsequent work has extended this approach in two directions. The first broadens the data foundation: Be.FM (Xie et al., 2025), Socrates (Kolluri et al., 2025), and HumanLLM (Lei et al., 2026) move beyond controlled experiments to behavioural science literature, large-scale social science surveys with demographic metadata, and social media corpora respectively, targeting capabilities from population-level distributional alignment to individualised behaviour prediction. The second explores alternatives to supervised fine-tuning: Zhu et al. (2025b) applied reinforcement learning (GRPO) to risky choice data, showing that chain-of-thought traces are causally linked to prediction accuracy, while GeCCo (Rmus et al., 2025) avoided fine-tuning entirely, using in-context learning to generate interpretable parametric cognitive models as executable code that matched or exceeded both Centaur and the best handcrafted models.

3 Methods

Data. We train on Psych-101 (Binz et al., 2025), a corpus of 10.7 million trial-level choices from over 60,000 participants across 160 psychological experiments, transcribed into natural language. Each prompt encodes a single participant’s complete trial-by-trial session, and we use the same train-test splits and prompt templates as Centaur without modification. Experiments fall into six task types, namely decision-making, bandit, Markov decision processes (MDP), memory, supervised learning, and miscellaneous.

Models. We fine-tune fourteen pre-trained base² (non-instruct) models drawn from four families: five Qwen3-Base models (0.6B, 1.7B, 4B, 8B, 14B) (Yang et al., 2025), three Llama-3 models (1B, 3B, 8B) (Grattafiori et al., 2024), four SmolLM models (SmolLM2 at 135M, 360M, and 1.7B (Allal et al., 2025), SmolLM3 at 3B (Bakouch et al., 2025)), and two OLMo models (OLMo-2 at 1B (Walsh et al., 2025), OLMo-3 at 7B (Ettinger et al., 2026)). We refer to the resulting fine-tuned models as **Qwentaur**, **Llama-Centaur**, **Smoltaur**, and **Olmotaur** respectively.

Training. Following Binz et al. (2025), we apply supervised fine-tuning with cross-entropy loss masked to human response tokens only, so that the model is not optimised on task instructions or context. We use rank-stabilised LoRA applied to all linear layers, trained for one epoch with AdamW (learning rate 5×10^{-5} , weight decay 0.01) and a linear warmup of 100 steps. All models are trained in bf16 precision on a single NVIDIA A100 80GB GPU, with training times ranging from 3 to 49 hours.

²We fine-tune base rather than post-trained models. Binz et al. (2026) find that post-training reduces alignment with human behaviour across model families, sizes, and objectives; Gao et al. (2025) find that alignment with human brain responses improves monotonically with scale in base models, while instruction-tuned variants show no advantage over base models of the same size.

Varying capacity and data. Model scale, adapter capacity, and training-set size are varied independently, each while the other two are held. Adapter capacity is swept over $r \in \{4, 8, 16, 32, 64\}$ at full data. Training-set size is varied over nested, experiment-stratified subsets of Psych-101 (1/16, 1/8, 1/4, 1/2, and the full set) at fixed rank ($r=16$). Subsets are nested so that each larger fraction strictly contains the smaller ones, and stratified so that all 160 experiments remain represented at every size, ensuring that reducing the quantity of training data does not also reduce paradigm coverage.

Comparability. Psych-101 sessions are long, with a median of 4,183 tokens and a maximum of 44,599, so a model’s context window determines how much of a session it can condition on. Qwen3, Llama-3, SmolLM3 and OLMo-3 are fine-tuned/evaluated at 32,768 tokens and lose 2.5% of trials to truncation, whereas SmolLM2 is capped at 8,192 (25.0%) and OLMo-2 at 4,096 (39.6%). In both families that span generations, SmolLM2/3 and OLMo-2/3, the larger model is also the newer generation with the larger window, so scaling within those families confounds parameter count with generation and context. We therefore mainly focus on the eight Qwentaaur and Llama-Centaur models in Section 4, comparing Smoltaur and Olmotaur against them at matched size where it is informative.

4 Experiments

With fourteen models trained on Psych-101, we ask two questions. The first is how much of Centaur’s predictive accuracy survives at small scale (Section 4.1), and the second is what information in the prompt the models actually use (Section 4.2). Adapter rank and training-set size vary only in Section 4.1.1, where they are themselves the object of study. Sections 4.1.2 and 4.2 use one configuration throughout, LoRA rank 16 on the full dataset.

4.1 Cognitive benchmark performance

We evaluate on two benchmarks. Psych-101 tests prediction for held out participants in-distribution; Psych-201-RT, a subset of Psych-201 (Binz et al., 2026), tests transfer to out-of-distribution experiments. Full per-experiment results for both are reported in Appendices C.1 and C.2. Figures report mean negative log-likelihood, in nats, over the 38 of 46 Psych-101 tasks for which Binz et al. (2025) publish a domain-specific cognitive model. Lower is better throughout.

4.1.1 Psych-101 (in-distribution performance)

Comparing each fine-tuned model with its own base checkpoint of the same size separates what fine-tuning contributes from what the base model already provides. Qwen3-0.6B scores 0.731 before fine-tuning and 0.528 after, a gain of 0.203 nats; across the eight models the average gain is 0.188 (range 0.138 to 0.295). Scale buys far less. Going from Qwentaaur-0.6B to Qwentaaur-14B gains 0.019 nats, and from Llama-Centaur-1B to Llama-Centaur-8B 0.024 nats. On Psych-101, whether a model has been fine-tuned matters roughly ten times more than how large it is. The fine-tuning gain shrinks steadily as the base model grows, from 0.203 at Qwentaaur-0.6B to 0.138 at Qwentaaur-14B, and from 0.295 at Llama-Centaur-1B to 0.158 at Llama-Centaur-8B. A larger model already predicts Psych-101 better before any fine-tuning, so less is left for fine-tuning to supply.

At rank 8, the adapter capacity Centaur itself used (QLoRA, 4-bit base weights), four models match or beat our reproduction of Centaur-70B (0.524; see Appendix C.3): Qwentaaur-8B and Llama-Centaur-8B (both 0.517), Qwentaaur-4B (0.520) and Smoltaur-3B (0.523), with Qwentaaur-1.7B close behind (0.528). Three to eight billion parameters therefore suffice to match a seventy-billion-parameter model trained on the same data with the same adapter budget, though at higher base-weight precision.

That threshold is not a fixed property of the task. Adapter capacity substitutes for parameters, so a small model at high rank reaches a much larger model at low rank in both matched families. Qwentaaur-0.6B at $r=64$ (0.513) matches Qwentaaur-14B at $r=4$ (0.516) with twenty-three times fewer parameters, and Llama-Centaur-1B at $r=64$ (0.520)

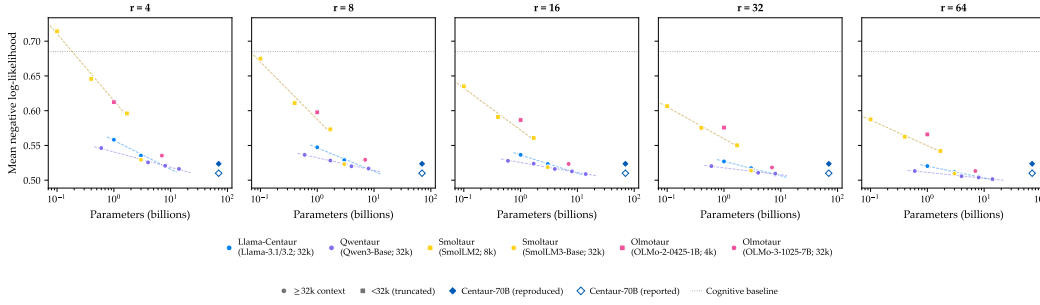


Figure 1: **Adapter rank against model size on Psych-101.** Mean negative log-likelihood over the 38 of 46 Psych-101 tasks for which Binz et al. (2025) publish a domain-specific cognitive model, plotted against parameter count, one panel per adapter rank ($r = 4-64$). Trend lines are fitted only within groups matched on generation and context window, so Olmotaur has none and the Smoltau fit covers the SmolLM2 models only.

matches Llama-Centaur-8B at $r=4$ (0.521) with eight times fewer. The size needed to match Centaur-70B moves accordingly, from 8B at $r=4$ to 3B at $r=8$, 1.7B at $r=16$, and 0.6B at $r=32$ and above (Figure 1).

Training data is a third lever on the same quantity, and its returns are visibly diminishing. Across nested subsets of Psych-101 the gain from each doubling declines steadily, with Qwentaaur-14B gaining 0.016, 0.013, 0.009 and 0.008 nats over successive doublings, so further participants of the same kind would add little (Appendix B).

4.1.2 Psych-201 (out-of-distribution performance)

The in-distribution results leave the models looking almost interchangeable. On Psych-101 the eight matched models (at rank 16 on the full dataset) span just 0.028 nats, from 0.509 to 0.537, a band that also contains our Centaur-70B reproduction. On the eighteen held-out experiments of Psych-201-RT the same eight models run from 0.788 to 1.033, a spread of 0.244 (Figure 2).

Before fine-tuning, a larger model is better on both benchmarks. A tenfold larger Qwen3 predicts Psych-101 0.062 nats better and Psych-201 0.118 better; for Llama-3 the figures are 0.177 and 0.204. After fine-tuning, the in-distribution advantage nearly disappears (0.014 and 0.026) while the out-of-distribution advantage largely persists (0.104 and 0.172). In other words, once a model has been fine-tuned on Psych-101, making it larger buys very little further accuracy on Psych-101 itself, but as much accuracy on unseen experiments as it did before fine-tuning. Model size should therefore be chosen for transfer rather than for in-distribution fit.

Set against each other, the two benchmarks show that what bounds these models as instruments is the range of paradigms they were trained on rather than their capacity. In-distribution they sit in a narrow band, as though against a ceiling; out-of-distribution they remain far apart, and the largest model in each family is still clearly the best, with no sign of a ceiling. All fourteen fine-tuned models beat the best base model in the study in-distribution, an un-fine-tuned Qwen3-14B at 0.647, whereas only three do out of distribution, against the same model at 0.869. A noise-ceiling estimate taken from one of these models is therefore trustworthy for paradigms in Psych-101, and unvalidated beyond them. We cannot, however, separate the benchmark from the training objective. Psych-101 may contain structure that supervised fine-tuning on next-choice prediction is unable to extract, in which case the limit belongs to the method rather than the data (Section 6).

4.2 Prompt decomposition and diagnostic testing

A model is useful as an instrument only if its predictions rest on the information the participant used, and aggregate fit cannot separate that from exploiting whichever regularity

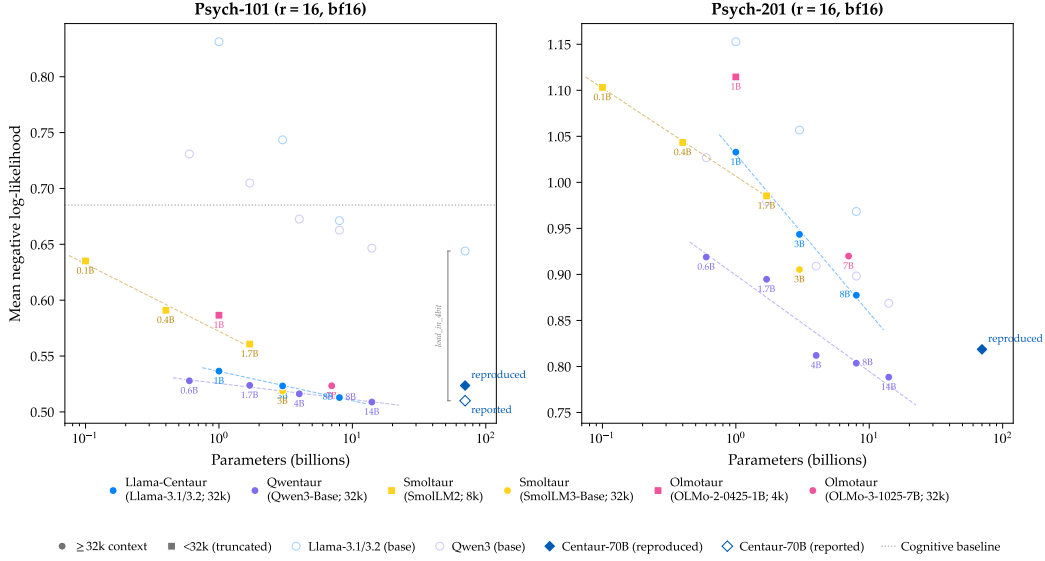


Figure 2: **In-distribution versus out-of-distribution scaling at rank 16.** (a) Psych-101, mean NLL over the 38 tasks with a reported cognitive model. (b) Psych-201-RT, all 18 held-out experiments; no cognitive baseline is published for Psych-201. Trend lines are fitted only within groups matched on generation and context window.

in the prompt is most predictive (Bowers et al., 2025; Orr et al., 2025). For Centaur-style supervised fine-tuning the concern is concrete, since recent work reports that most of its accuracy survives reducing the prompt to choice history or removing task instructions (Xie & Zhu, 2025; Liu & Ding, 2025). We therefore decompose the Psych-101 prompt into information channels and ask which of them the model relies on (Section 4.2.2) and whether it uses them as each task’s structure requires (Section 4.2.3).

4.2.1 Decomposing the training objective

Xie & Zhu (2025) formalise Centaur’s training objective as maximising $P(C_{t+1} \mid C_{1:t}, T)$ at the token level, where each C represents a tokenised participant choice and T denotes task-related information in the prompt. This formulation bundles all non-choice information into a single variable T , which conflates conceptually distinct sources of information that can be independently manipulated. We propose a finer-grained decomposition of the prompt into four information channels, $P(C_{t+1} \mid I, S_{1:t+1}, F_{1:t}, C_{1:t})$, where I is the **task instruction**, $S_{1:t+1}$ the **experimental stimuli** including those of the upcoming trial, $F_{1:t}$ the **outcome feedback** after each past choice, and $C_{1:t}$ the participant’s **choice history**.

The decomposition lets us state the shortcut hypothesis precisely. In its strongest form it asserts *history sufficiency*, that choice history already carries all the information available about the next choice, $C_{t+1} \perp\!\!\!\perp \{I, S, F\} \mid C_{1:t}$, so a model could reach its full-prompt accuracy while ignoring instruction, stimuli and feedback outright. Weaker forms allow the model to read the remaining channels but deny that it uses their content, and they differ in which channel they exempt. The one our design is built to test is *format sufficiency*, that the model locates the choice from the recurring arrangement of stimulus, choice and feedback statements without using what any of those statements say.

Existing studies test the strong form, and only in part. Xie & Zhu (2025) evaluate a history-only condition and Liu & Ding (2025) test instruction-ablated, choice-only and adversarial-instruction conditions, but each covers only four experiments, and neither varies the granularity of what is removed, so neither separates content from template (Appendix F.1). We extend both scope and granularity (Figure 3). First, we evaluate four structural ablation conditions, original, instruction-ablated, content-masked and history-only, across 32 sequential

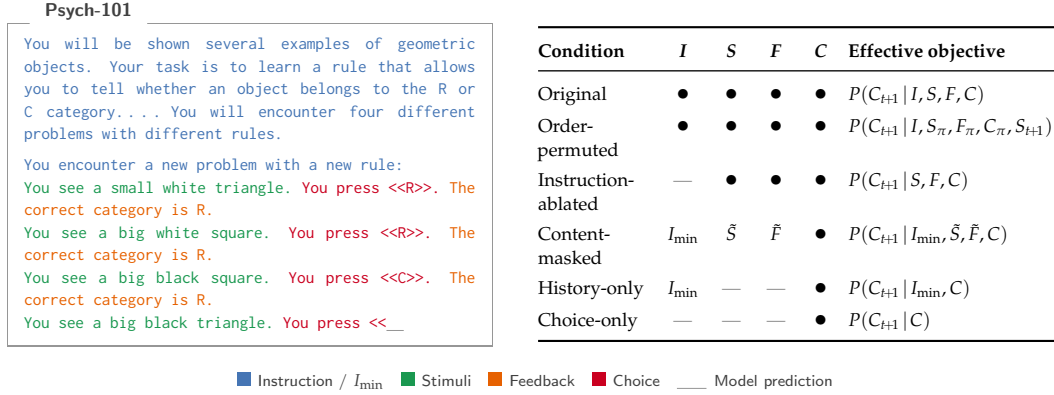


Figure 3: **Prompt decomposition into four information channels. (Left)** Example prompt from Psych-101 (Binz et al., 2025) with channels colour-coded. **(Right)** Ablation conditions and their effective prediction objectives. Per-experiment prompts under all four ablation conditions are shown in Figures 19–50 (Appendix F.2).

experiments. The content-masked condition is new to this work and is the one that discriminates the two hypotheses: it replaces specific stimulus values and feedback outcomes with generic placeholders (e.g. “some points”, “a shape”) and reduces the instruction to a minimal action-space definition ($I_{\min}$), while preserving each trial’s formatting, so whatever degradation it causes is attributable to informational content rather than to positional template. Second, we test order permutation on two experiments, one exchangeable and one adaptive as a negative control. Only one exchangeable task is available, as most Psych-101 experiments carry sequential dependencies.

4.2.2 Structural ablation (sequential tasks)

We measure each condition by information retention $R = (\ln k - \text{NLL}_{\text{ablation}}) / (\ln k - \text{NLL}_{\text{original}})$, where $\ln k$ is the chance-level cross-entropy for k response options, so $R = 1$ when nothing is lost and $R = 0$ at chance. Retention falls at every step (Figure 4a), from original (1.00) to instruction-ablated (0.84), content-masked (−0.12) and history-only (−0.26), and each drop is significant (Friedman $\chi^2(3) = 63.76$, $p = 9.3 \times 10^{-14}$; pairwise Wilcoxon $p < 2 \times 10^{-3}$ throughout) and holds for all eight models. Accordingly, removing the instruction costs little, whereas masking the content of stimuli and feedback costs almost everything and pushes models below chance. Neither form of the shortcut hypothesis survives; *choice history alone does not suffice, and neither does the trial template.*

Partitioning the total loss across the three steps, the instruction accounts for 12.5%, the content of stimuli and feedback for 75.7%, and the template for the remaining 11.7%. What the models use is what the participant saw and was told. This follows two earlier findings. The small cost of removing instructions reported by Liu & Ding (2025) replicates here, but does not extend to the other channels; and the accuracy Xie & Zhu (2025) recover from choice history alone holds on the two sequential tasks they examined, both of which are in our set, but not on average, since mean retention under history-only is below chance across the 27 experiments.

At the per-task level, 18 of 27 tasks (67%) degrade strictly monotonically across the four conditions (Figure 4b). Of the nine exceptions, five invert only between two conditions that are already at or below chance and four are genuine reversals (further analysis in Appendix F.2.1).

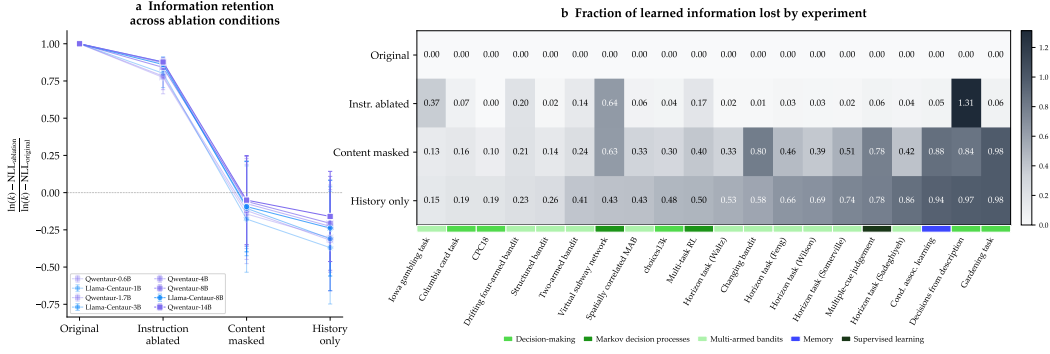


Figure 4: Structural ablation results across eight models and 27 experiments with well-defined response options. Five experiments with mixed or continuous formats are excluded. **(a)** Mean information retention $R = (\ln k - \mathcal{L}_c) / (\ln k - \mathcal{L}_{orig})$ by condition, averaged over experiments; $R = 1$: no loss, $R = 0$: chance. Circles: Llama-Centaur; squares: Qwentaaur; colour intensity scales with model size; error bars: SEM. **(b)** Per-experiment fraction of learned information lost, $\delta = 1 - R$, averaged across models. Columns sorted by history-only δ ; colour strip indicates task type. Experiments with $\delta > 1$ under the history-only condition are omitted; see Figure 18 (Appendix F.2) for the complete set.

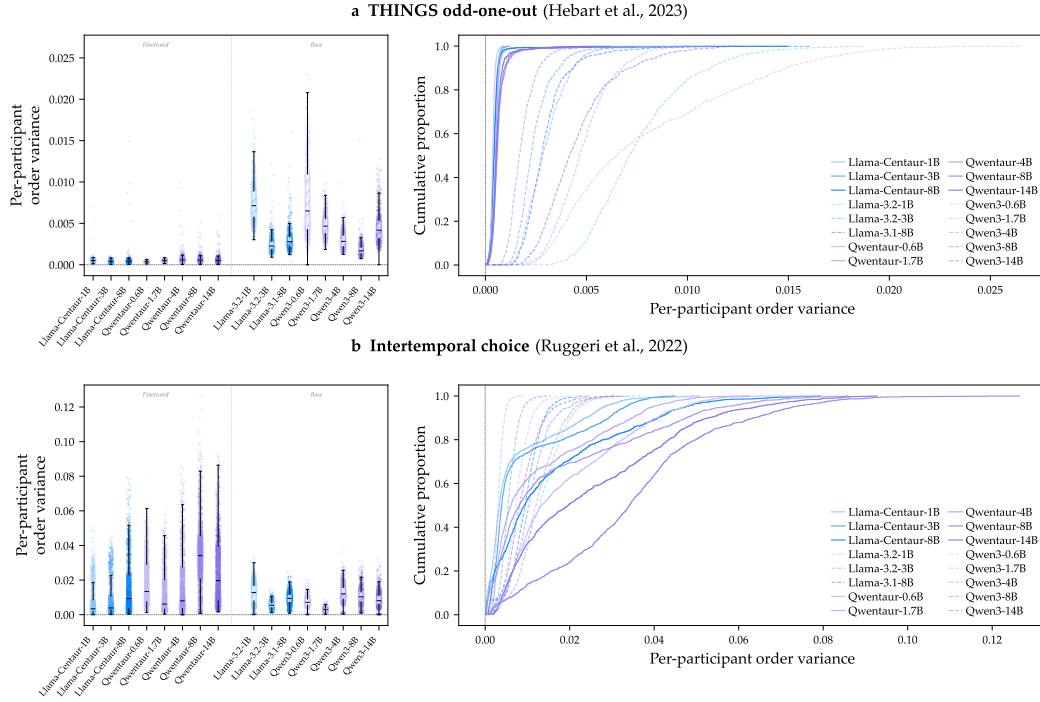


Figure 5: Per-participant order variance across models and experiments. **(a)** THINGS odd-one-out (exchangeable; primary test). **(b)** Intertemporal choice (adaptive staircase; negative control). **Left:** violin and box plots; fine-tuned (solid) vs. base (hatched); y-axis scales differ. **Right:** ECDFs; solid = fine-tuned, dashed = base; steeper curves near zero indicate greater order invariance.

4.2.3 Order permutation (exchangeability)

The ablations show which channels the model reads, not whether it reads them in a way the task requires. A model can depend heavily on stimuli and still treat the trials as an ordered

sequence where their order carries no information (Namazova et al., 2025; Schlegelmilch & Dome, 2026). Exchangeability makes this testable. Where trials are independent, permuting the context must leave the prediction unchanged; where each trial follows from the previous response, it must not. For each (participant, target trial) pair we draw 50 random permutations of the context trials and measure the variance of each response-token probability across them, low variance indicating order invariance.

Two experiments are chosen to anchor opposite ends of the exchangeability spectrum. **THINGS odd-one-out** (Hebart et al., 2023) (primary test; 768 participants) presents triplets of objects for similarity judgement. Triplets are sampled independently, with no feedback, no adaptation, and no sequential dependency. Because the order in which previous triplets appeared carries no information about the current judgement, context trials are exchangeable by construction. A cognitive proxy that has genuinely learned this task structure, rather than defaulting to sequential heuristics, should produce order-invariant predictions. **Intertemporal choice** (Ruggeri et al., 2022) (negative control; 1,295 participants) uses an adaptive, response-contingent procedure to locate each participant’s indifference point, where the value offered on each trial depends on the previous response. Permuting trial order breaks the staircase logic entirely, so a faithful behavioural proxy should be *order-sensitive*.

Figure 5 matches both expectations. On THINGS odd-one-out, fine-tuned models cluster near zero variance, so they have learned the exchangeability of independent similarity judgements. On intertemporal choice, fine-tuned models are order-sensitive, as the design requires, since each offer is generated from the answer before it and permutation then yields a sequence the procedure itself could never have produced.

Together with the structural ablation results, the picture that emerges is one of *task-adaptive information use*. Cognitive fine-tuning produces proxies that exploit sequential regularities where they are genuinely predictive, process stimuli where stimuli are informative, and respect exchangeability where the experimental design requires it. The critique that these models are merely exploiting statistical regularities is, in one sense, correct, but the regularities they exploit are precisely the ones that reflect the computational structure of the underlying cognitive tasks.

5 Discussion

Our results bound what these models are for. Scale is not what limits prediction within the paradigms a model was trained on, but it is what limits transfer beyond them, and the ablations show that the predictions rest on the content of what participants were shown and told, which licenses using the models as instruments. That licence also rests on the gain being specific to cognitive data rather than to fine-tuning as such, which the non-cognitive and behavioural controls confirm (Appendix D).

Yet predicting held-out choices, however accurately, is not explaining the processes that generate them. In the tradition of Newell (1990) and Simon (1969), a cognitive model posits explicit mechanisms within a decomposable architecture, and reproducing behavioural regularities alone is what Simon called *explanation from the outside* (Simon, 1969; Simon & Newell, 1971; Newell, 1990; Simon & Gobet, 2000). Unlike ACT-R (Anderson & Lebiere, 1998) or Soar (Laird, 2012), these models commit to no architecture and no mechanism, and that absence is what makes them general, since they cover many paradigms by relying on scale and learned statistics rather than designed-in assumptions.

What we claim instead is the role of a *noise ceiling estimator* for in-distribution paradigms (Agrawal et al., 2020; Kuperwajs et al., 2023), which is more substantive than the phrase suggests. Estimated per datapoint, a noise ceiling maps where systematic structure in behaviour exists and where it does not, so the gap between what the model captures and what an interpretable theory captures marks where behaviour is more structured than current theory can account for, and therefore where theory-building has most to gain. A model used only for prediction returns a number; used as a ceiling it returns a map of where an existing theory falls short, which is the logic Agrawal et al. (2020) call scientific regret minimisation. It is in this sense, and not as a claim to mechanism, that we read our

title’s reference to cognition *and* behaviour, since to call the work behaviour alone would understate the stakes and to call it a theory of cognition would overstate them.

Whether fine-tuning moves these models any distance inward is a question for mechanistic and representational analysis, and one worth asking, since networks trained on behaviour have recovered known architectures of memory (Salvatore & Zhang, 2025; Ji-An et al., 2024), prefrontal-like specialisation (Yang et al., 2019), and planning mechanisms with neural counterparts (Jensen et al., 2024).

6 Limitations and future work

Data structure. Two features of Psych-101 bound what we can conclude from it. The first concerns which axis of the data we varied. Our subsets are drawn at the level of whole participant sessions and stratified so that all 160 experiments contribute to every fraction, which isolates data quantity but holds paradigm coverage fixed by construction. The diminishing returns we report are therefore returns to depth, more participants per experiment, and the return to breadth is unmeasured. Given that the models saturate within trained paradigms and remain far from saturated outside them, breadth is likely the more informative axis, and subsetting by experiment rather than by participant would estimate how many distinct paradigms a cognitive foundation model needs. Psych-201 (Binz et al., 2026) is a promising step in that direction, 3.5 times the size of Psych-101, but it does not escape the second limitation, which is structural. Participants are nested within experiments in Psych-201 as in Psych-101, so no individual is observed across tasks, which precludes the cross-experiment person-task decomposition that Gobet & Ritter (2000) and Orr et al. (2025) identify as necessary for bounding predictability. Datasets in which the same participants complete multiple paradigms would address both limitations at once.

Training and architecture. Adaptation here is low-rank throughout. The rank sweep shows that adapter capacity substitutes for parameters over the range we test, but it does not tell us where either axis ends. Full fine-tuning at the smallest scales would separate a genuine ceiling on the learnable signal from one imposed by low-rank adaptation, and the data ablation shows returns already diminishing, so the three may exhaust at different points. Architecture coverage is narrow in a further sense, since all four families are dense decoder-only transformers, leaving open whether the plateau shifts under mixture-of-experts (Jiang et al., 2024), state-space (Gu & Dao, 2024), or hybrid designs (Blakeman et al., 2025) that allocate capacity differently. Finally, SFT optimises for imitation of the training distribution, at a measurable cost to capabilities distant from it (Appendix E). Kolluri et al. (2025) show that contrastive DPO on demographic preference pairs improves individual response accuracy over SFT, while Zhu et al. (2025b) find that GRPO elicits chain-of-thought reasoning that in turn improves choice prediction. RL is unlikely to surface genuinely novel cognitive mechanisms, since it redistributes probability mass over behaviours already represented in the pre-trained weights, but it generalises where SFT memorises (Chu et al., 2025), which is the regime our Psych-201 results identify as the harder one. Pairing RL with cross-task datasets would allow fine-tuning toward person-level behavioural profiles rather than task-level distributions.

7 Conclusion

Seventy billion parameters are not what Centaur’s accuracy on Psych-101 rests on. Across fourteen models from 135M to 14B, whether a model has been fine-tuned matters far more than how large it is. Only beyond the training paradigms does scale begin to earn its cost, and there larger models transfer better with no sign of saturation. Accuracy alone, however, would settle nothing; an instrument must also be right for the right reasons. Prediction collapses when the content of stimuli and feedback is masked, and responds to trial order only where the design makes order informative. Small cognitively fine-tuned models are therefore adequate for estimating noise ceilings within the paradigms they have seen, and what binds them is the reach of that coverage rather than their capacity.

Acknowledgements

We thank Marcel Binz for answering our questions about Psych-201 and the Centaur evaluation setup, and Seungbin Joo for insightful discussions on the different forms of the shortcut hypothesis.

Ethics Statement

All models are trained on Psych-101 (Binz et al., 2025), a publicly released dataset of anonymised trial-level responses aggregated from published psychological experiments. We collected no new human data and made no attempts to re-identify participants. Our models predict laboratory task responses and are intended as scientific instruments. We acknowledge that models capable of predicting human choices could in principle be misused; however, the training domain (abstract laboratory paradigms) limit this risk. We release all code and model weights to support reproducibility and scientific progress, and encourage users to employ these models within the scientific context for which they were developed.

However, their utility as scientific instruments is currently bounded by the paradigm coverage of the training data, and whether they extrapolate reliably to conditions producing more extreme outcomes than any in Psych-101 remains untested (Schlegelmilch & Dome, 2026). More speculatively, D’Alessandro & Thompson (2025) caution that widespread reliance on cheap AI-generated behavioural data could induce a “street light effect” in the cognitive sciences, with researchers gravitating toward questions that fall within these models’ capabilities and neglecting phenomena that resist efficient simulation.

Reproducibility Statement

All code for training, evaluation, and visualisation is publicly available at <https://github.com/socius-org/Centauri>. This includes the training scripts for all four model families, the evaluation pipelines, and all figure-generation code. Fine-tuned LoRA adapters are released on Hugging Face at <https://hf.co/collections/socius/Centauri>. The training data, Psych-101, is publicly available through Binz et al. (2025), and we use the same train-test splits and prompt templates without modification. All models were trained on a single NVIDIA A100 80GB GPU using identical hyperparameters (Table 2): rank-stabilised LoRA applied to all linear layers at $r = \alpha \in \{4, 8, 16, 32, 64\}$, with $r=16$ as the reference configuration, AdamW with learning rate 5×10^{-5} , weight decay 0.01, linear warmup of 100 steps, and one epoch of training in bf16 precision. The structural ablation and order permutation experiments use deterministic prompt manipulation scripts included in the GitHub repository; permutation tests use 50 random permutations per (participant, target trial) pair. Appendix A provides full infrastructure details including training times and batch configurations.

References

- Mayank Agrawal, Joshua C Peterson, and Thomas L Griffiths. Scaling up psychology via scientific regret minimization. *Proceedings of the National Academy of Sciences*, 117(16): 8825–8835, 2020.
- Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, Gabriel Martín Blázquez, Guilherme Penedo, Lewis Tunstall, Andrés Marafioti, Hynek Kydlíček, Agustín Piqueres Lajarín, Vaibhav Srivastav, Joshua Lochner, Caleb Fahlgren, Xuan-Son Nguyen, Clémentine Fourrier, Ben Burtenshaw, Hugo Larcher, Haojun Zhao, Cyril Zakka, Mathieu Morlon, Colin Raffel, Leandro von Werra, and Thomas Wolf. Smollm2: When smol goes big – data-centric training of a small language model, 2025. URL <https://arxiv.org/abs/2502.02737>.
- John R. Anderson and Christian Lebiere. *The Atomic Components of Thought*. Lawrence Erlbaum Associates, Mahwah, NJ, 1998.
- Hernán Anlló, Sophie Bavard, Fatima Ezzahra Benmarrakchi, Darla Bonagura, Fabien Cerrotti, Mirona Cicue, Maelle Gueguen, Eugenio José Guzmán, Dzerassa Kadieva, Maiko Kobayashi, et al. Comparing experience-and description-based economic preferences across 11 countries. *Nature Human Behaviour*, 8(8):1554–1567, 2024.
- Stephen P Badham, Adam N Sanborn, and Elizabeth A Maylor. Deficits in category learning in older adults: Rule-based versus clustering accounts. *Psychology and Aging*, 32(5):473, 2017.
- Bahador Bahrami, Joaquin Navajas, Yuyan Wang, and Shangyi Wang. 4 arm bandit task dataset. <https://doi.org/10.17605/OSF.IO/F3T2A>, 2020. OSF. Updated February 8, 2022. Licensed under CC BY 4.0.
- Elie Bakouch, Loubna Ben Allal, Anton Lozhkov, Nouamane Tazi, Lewis Tunstall, Carlos Miguel Patiño, Edward Beeching, Aymeric Roucher, Aksel Joonas Reedi, Quentin Gallouédec, Kashif Rasul, Nathan Habib, Clémentine Fourrier, Hynek Kydlíček, Guilherme Penedo, Hugo Larcher, Mathieu Morlon, Vaibhav Srivastav, Joshua Lochner, Xuan-Son Nguyen, Colin Raffel, Leandro von Werra, and Thomas Wolf. SmolLM3: smol, multilingual, long-context reasoner. <https://huggingface.co/blog/smollm3>, 2025.
- Sophie Bavard, Aldo Rustichini, and Stefano Palminteri. Two sides of the same coin: Beneficial and detrimental consequences of range adaptation in human reinforcement learning. *Science Advances*, 7(14):eabe0340, 2021.
- Akhiad Bercovich, Itay Levy, Izik Golan, Mohammad Dabbah, Ran El-Yaniv, Omri Puny, Ido Galil, Zach Moshe, Tomer Ronen, Najeib Nabwani, Ido Shahaf, Oren Tropp, Ehud Karpas, Ran Zilberstein, Jiaqi Zeng, Soumye Singhal, Alexander Bukharin, Yian Zhang, Tugrul Konuk, Gerald Shen, Ameya Sunil Mahabaleshwarkar, Bilal Kartal, Yoshi Suhara, Olivier Delalleau, Zijia Chen, Zhilin Wang, David Mosallanezhad, Adi Renduchintala, Haifeng Qian, Dima Rekish, Fei Jia, Somshubra Majumdar, Vahid Noroozi, Wasi Uddin Ahmad, Sean Narenthiran, Aleksander Ficek, Mehrzad Samadi, Jocelyn Huang, Siddhartha Jain, Igor Gitman, Ivan Moshkov, Wei Du, Shubham Toshniwal, George Armstrong, Branislav Kisanin, Matvei Novikov, Daria Gitman, Evelina Bakhturina, Jane Polak Scowcroft, John Kamalu, Dan Su, Kezhi Kong, Markus Kliegl, Rabeeh Karimi, Ying Lin, Sanjeev Satheesh, Jupinder Parmar, Pritam Gundecha, Brandon Norick, Joseph Jennings, Shrimai Prabhume, Syeda Nahida Akter, Mostofa Patwary, Abhinav Khattar, Deepak Narayanan, Roger Waleffe, Jimmy Zhang, Bor-Ying Su, Guyue Huang, Terry Kong, Parth Chadha, Sahil Jain, Christine Harvey, Elad Segal, Jining Huang, Sergey Kashirsky, Robert McQueen, Izzy Putterman, George Lam, Arun Venkatesan, Sherry Wu, Vinh Nguyen, Manoj Kilaru, Andrew Wang, Anna Warno, Abhilash Somasamudramath, Sandip Bhaskar, Maka Dong, Nave Assaf, Shahar Mor, Omer Ullman Argov, Scot Junkin, Oleksandr Romanenko, Pedro Larroy, Monika Katariya, Marco Rovinelli, Viji Balas, Nicholas Edelman, Anahita Bhiwandiwalla, Muthu Subramanian, Smita Ithape, Karthik Ramamoorthy, Yuting Wu, Suguna Varshini Velury, Omri Almog, Joyjit Daw, Denys Fridman, Erick Galinkin, Michael Evans, Katherine Luna, Leon Derczynski, Nikki Pope, Eileen Long, Seth Schneider, Guillermo Siman, Tomasz Grzegorzec, Pablo Ribalta, Monika Katariya, Joey Conway,

Trisha Saar, Ann Guan, Krzysztof Pawelec, Shyamala Prayaga, Oleksii Kuchaiev, Boris Ginsburg, Oluwatobi Olabiyi, Kari Briski, Jonathan Cohen, Bryan Catanzaro, Jonah Alben, Yonatan Geifman, Eric Chung, and Chris Alexiuk. Llama-nemotron: Efficient reasoning models, 2025. URL <https://arxiv.org/abs/2505.00949>.

Marcel Binz, Elif Akata, Matthias Bethge, Franziska Brändle, Fred Callaway, Julian Coda-Forno, Peter Dayan, Can Demircan, Maria K. Eckstein, Noémi Éltető, Thomas L. Griffiths, Susanne Haridi, Akshay K. Jagadish, Li Ji-An, Alexander Kipnis, Sreejan Kumar, Tobias Ludwig, Marvin Mathony, Marcelo Mattar, Alireza Modirshanechi, Surabhi S. Nath, Joshua C. Peterson, Milena Rmus, Evan M. Russek, Tankred Saanum, Johannes A. Schubert, Luca M. Schulze Buschoff, Nishad Singhi, Xin Sui, Mirko Thalmann, Fabian Theis, Vuong Truong, Vishaal Udandara, Konstantinos Voudouris, Robert Wilson, Kristin Witte, Shuchen Wu, Dirk Wulff, Huadong Xiong, and Eric Schulz. A foundation model to predict and capture human cognition. *Nature*, 644(8078):1002–1009, 2025.

Marcel Binz, Elif Akata, Abdullah Almaatouq, Mohammed Alsobay, Oleksii Ariasov, Franziska Brändle, David Broska, Jason W. Burton, Nuno Busch, Frederick Callaway, Vanessa Cheung, Brian Christian, Julian Coda-Forno, Can Demircan, Vittoria Dentella, Maria K. Eckstein, Noémi Éltető, Michael Franke, Thomas L. Griffiths, Fritz Günther, Susanne Haridi, Sebastian Hellmann, Stefan Herytash, Linus Hof, Eleanor Holton, Isabelle Hoxha, Zak Hussain, Akshay Jagadish, Elif Kara, Valentin Kriegmair, Evelina Leivada, Li Ji-An, Tobias Ludwig, Maximilian Maier, Marcelo G. Mattar, Marvin Mathony, Alireza Modirshanechi, Robin Na, Mariia Nadverniuk, Antonios Nasioulas, Surabhi S. Nath, Helen Niemeyer, Kate Nussenbaum, Sebastian Olschewski, Thorsten Pachur, Stefano Palminteri, Aliona Petrenco, Camille V. Phaneuf-Hadd, Angelo Pirrone, Manuel Rausch, Laura Raveling, Shashank Reddy, Milena Rmus, Evan M. Russek, Tankred Saanum, Kai Sandbrink, Louis Schiekiera, Johannes A. Schubert, Luca M. Schulze Buschoff, Nishad Singhi, Leah H. Somerville, Mikhail S. Spektor, Xin Sui, Christopher Summerfield, Mirko Thalmann, Anna I. Thoma, Taisiia Tikhomirova, Vuong Truong, Polina Tsvilodub, Konstantinos Voudouris, Kristin Witte, Shuchen Wu, Dirk U. Wulff, Hua-Dong Xiong, Songlin Xu, Lance Ying, Xinyu Zhang, Jian-Qiao Zhu, and Eric Schulz. Post-training makes large language models less human-like, 2026. URL <https://arxiv.org/abs/2605.07632>.

Aaron Blakeman, Aarti Basant, Abhinav Khattar, Adithya Renduchintala, Akhiad Bercovich, Aleksander Ficek, Alexis Bjorlin, Ali Taghibakhshi, Amala Sanjay Deshmukh, Ameya Sunil Mahabaleshwarkar, Andrew Tao, Anna Shors, Ashwath Aithal, Ashwin Poojary, Ayush Dattagupta, Balaram Buddharaju, Bobby Chen, Boris Ginsburg, Boxin Wang, Brandon Norick, Brian Butterfield, Bryan Catanzaro, Carlo del Mundo, Chengyu Dong, Christine Harvey, Christopher Parisien, Dan Su, Daniel Korzekwa, Danny Yin, Daria Gitman, David Mosallanezhad, Deepak Narayanan, Denys Fridman, Dima Reakesh, Ding Ma, Dmytro Pykhtar, Dong Ahn, Duncan Riach, Dusan Stosic, Eileen Long, Elad Segal, Ellie Evans, Eric Chung, Erick Galinkin, Evelina Bakhturina, Ewa Dobrowolska, Fei Jia, Fuxiao Liu, Gargi Prasad, Gerald Shen, Guilin Liu, Guo Chen, Haifeng Qian, Helen Ngo, Hongbin Liu, Hui Li, Igor Gitman, Ilia Karmanov, Ivan Moshkov, Izik Golan, Jan Kautz, Jane Polak Scowcroft, Jared Casper, Jarno Seppanen, Jason Lu, Jason Sewall, Jiaqi Zeng, Jiakuan You, Jimmy Zhang, Jing Zhang, Jining Huang, Jinze Xue, Jocelyn Huang, Joey Conway, John Kamalu, Jon Barker, Jonathan Cohen, Joseph Jennings, Jupiter Parmar, Karan Sapra, Kari Briski, Kateryna Chumachenko, Katherine Luna, Keshav Santhanam, Kezhi Kong, Kirthi Sivamani, Krzysztof Pawelec, Kumar Anik, Kunlun Li, Lawrence McAfee, Leon Derczynski, Lindsey Pavao, Luis Vega, Lukas Voegtli, Maciej Bala, Maer Rodrigues de Melo, Makesh Narsimhan Sreedhar, Marcin Chochowski, Markus Kliegl, Marta Stepniewska-Dziubinska, Matthieu Le, Matvei Novikov, Mehrzad Samadi, Michael Andersch, Michael Evans, Miguel Martinez, Mike Chrzanowski, Mike Ranzinger, Mikolaj Blaz, Misha Smelyanskiy, Mohamed Fawzy, Mohammad Shoeby, Mostofa Patwary, Nayeon Lee, Nima Tajbakhsh, Ning Xu, Oleg Rybakov, Oleksii Kuchaiev, Olivier Delalleau, Osvald Nitski, Parth Chadha, Pasha Shamis, Paulius Micikevicius, Pavlo Molchanov, Peter Dykas, Philipp Fischer, Pierre-Yves Aquilanti, Piotr Bialecki, Prasoon Varshney, Pritam Gundecha, Przemek Tredak, Rabeeh Karimi, Rahul Kandu, Ran El-Yaniv, Raviraj Joshi, Roger Waleffe, Ruoxi Zhang, Sabrina Kavanaugh, Sahil Jain,

- Samuel Krizan, Sangkug Lym, Sanjeev Satheesh, Saurav Muralidharan, Sean Narenthiran, Selvaraj Anandaraj, Seonmyeong Bak, Sergey Kashirsky, Seungju Han, Shantanu Acharya, Shaona Ghosh, Sharath Turuvekere Sreenivas, Sharon Clay, Shelby Thomas, Shrimai Prabhumoye, Shubham Pachori, Shubham Toshniwal, Shyamala Prayaga, Siddhartha Jain, Sirshak Das, Slawek Kierat, Somshubra Majumdar, Song Han, Soumye Singhal, Sriharsha Niverty, Stefania Alborghetti, Suseella Panguluri, Swetha Bhendigeri, Syeda Nahida Akter, Szymon Migacz, Tal Shiri, Terry Kong, Timo Roman, Tomer Ronen, Trisha Saar, Tugrul Konuk, Tuomas Rintamaki, Tyler Poon, Ushnish De, Vahid Noroozi, Varun Singh, Vijay Korthikanti, Vitaly Kurin, Wasi Uddin Ahmad, Wei Du, Wei Ping, Wenliang Dai, Wonmin Byeon, Xiaowei Ren, Yao Xu, Yejin Choi, Yian Zhang, Ying Lin, Yoshi Suhara, Zhiding Yu, Zhiqi Li, Zhiyu Li, Zhongbo Zhu, Zhuolin Yang, and Zijia Chen. Nemotron-h: A family of accurate and efficient hybrid mamba-transformer models, 2025. URL <https://arxiv.org/abs/2504.03624>.
- JS Bowers, G Puebla, S Thorat, K Tsetsos, and CJH Ludwig. On the misuse of llms as models of mind: A case study of centaur. 2025.
- Nuno Busch, Thomas Geyer, and Artyom Zinchenko. Individual peak alpha frequency does not index individual differences in inhibitory cognitive control. *Psychophysiology*, 61(8): e14586, 2024a.
- Nuno Busch, Thomas Geyer, and Artyom Zinchenko. Individual peak alpha frequency does not index individual differences in inhibitory cognitive control. *Psychophysiology*, 61(8): e14586, 2024b.
- Jerome R. Busemeyer and Adele Diederich. *Cognitive Modeling*. Sage, Los Angeles, 2010.
- Pedro Castro-Rodrigues, Thomas Akam, Ivar Snorasson, Marta Camacho, Vitor Paixão, Ana Maia, J Bernardo Barahona-Corrêa, Peter Dayan, H Blair Simpson, Rui M Costa, et al. Explicit knowledge of task structure is a primary determinant of human model-based action. *Nature human behaviour*, 6(8):1126–1141, 2022.
- Tianzhe Chu, Yuexiang Zhai, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V Le, Sergey Levine, and Yi Ma. Sft memorizes, rl generalizes: A comparative study of foundation model post-training. *arXiv preprint arXiv:2501.17161*, 2025.
- Anne GE Collins, Jaime K Brown, James M Gold, James A Waltz, and Michael J Frank. Working memory contributions to reinforcement learning impairments in schizophrenia. *Journal of Neuroscience*, 34(41):13747–13756, 2014.
- August Collsiö, Peter Juslin, and Anders Winman. Is numerical information always beneficial? verbal and numerical cue-integration in additive and non-additive tasks. *Cognition*, 240:105584, 2023.
- Gregory E Cox, Pernille Hemmer, William R Aue, and Amy H Criss. Information and processes underlying semantic and episodic memory across tasks, items, and individuals. *Journal of Experimental Psychology: General*, 147(4):545, 2018.
- William D’Alessandro and Jessica Thompson. Ai surrogacy in psychological research. 2025.
- Vittoria Dentella, Fritz Günther, and Evelina Leivada. Systematic testing of three language models reveals low language accuracy, absence of response stability, and a yes-response bias. *Proceedings of the National Academy of Sciences*, 120(51):e2309583120, 2023.
- A Zeynep Enkavi, Ian W Eisenberg, Patrick G Bissett, Gina L Mazza, David P MacKinnon, Lisa A Marsch, and Russell A Poldrack. Large-scale analysis of test-retest reliabilities of self-regulation measures. *Proceedings of the National Academy of Sciences*, 116(12):5472–5477, 2019.
- Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, David Graham, David Heineman, Dirk Groeneveld, Faeze Brahman, Finbarr Timbers, Hamish Ivison, Jacob Morrison, Jake Poznanski, Kyle Lo, Luca Soldaini, Matt Jordan, Mayee Chen, Michael Noukhovitch, Nathan Lambert, Pete Walsh, Pradeep Dasigi, Robert Berry, Saumya Malik, Saurabh Shah,

- Scott Geng, Shane Arora, Shashank Gupta, Taira Anderson, Teng Xiao, Tyler Murray, Tyler Romero, Victoria Graf, Akari Asai, Akshita Bhagia, Alexander Wettig, Alisa Liu, Aman Rangapur, Chloe Anastasiades, Costa Huang, Dustin Schwenk, Harsh Trivedi, Ian Magnusson, Jaron Lochner, Jiacheng Liu, Lester James V. Miranda, Maarten Sap, Malia Morgan, Michael Schmitz, Michal Guerquin, Michael Wilson, Regan Huff, Ronan Le Bras, Rui Xin, Rulin Shao, Sam Skjonsberg, Shannon Zejiang Shen, Shuyue Stella Li, Tucker Wilde, Valentina Pyatkin, Will Merrill, Yapei Chang, Yuling Gu, Zhiyuan Zeng, Ashish Sabharwal, Luke Zettlemoyer, Pang Wei Koh, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. Olmo 3, 2026. URL <https://arxiv.org/abs/2512.13961>.
- Haoxue Fan, Samuel J Gershman, and Elizabeth A Phelps. Trait somatic anxiety is associated with reduced directed exploration and underestimation of uncertainty. *Nature Human Behaviour*, 7(1):102–113, 2023.
- Samuel F Feng, Siyu Wang, Sylvia Zarnescu, and Robert C Wilson. The dynamics of explore–exploit decisions reveal a signal-to-noise mechanism for random exploration. *Scientific reports*, 11(1):3077, 2021.
- Timo Flesch, Jan Balaguer, Ronald Dekker, Hamed Nili, and Christopher Summerfield. Comparing continual task learning in minds and machines. *Proceedings of the National Academy of Sciences*, 115(44):E10313–E10322, 2018.
- Michael C Frank and Noah D Goodman. Cognitive modeling using artificial intelligence. *Annual Review of Psychology*, 77, 2026.
- Michael Franke and Judith Degen. Reasoning in reference games: Individual-vs. population-level probabilistic modeling. *PloS one*, 11(5):e0154854, 2016.
- Michael Franke, Polina Tsvilodub, and Fausto Carcassi. Bayesian statistical modeling with predictors from llms. *arXiv preprint arXiv:2406.09012*, 2024.
- Renato Frey, Andreas Pedroni, Rui Mata, Jörg Rieskamp, and Ralph Hertwig. Risk preference shares the psychometric structure of major psychological traits. *Science advances*, 3(10): e1701381, 2017.
- Changjiang Gao, Zhengwu Ma, Jiajun Chen, Ping Li, Shujian Huang, and Jixing Li. Increasing alignment of large language models with language processing in the human brain. *Nature computational science*, 5(11):1080–1090, 2025.
- Basile Garcia, Maël Lebreton, Sacha Bourgeois-Gironde, and Stefano Palminteri. Experiential values are underweighted in decisions involving symbolic options. *Nature human behaviour*, 7(4):611–626, 2023.
- Samuel J Gershman. Deconstructing the human algorithms for exploration. *Cognition*, 173: 34–42, 2018.
- Fernand Gobet and Frank E. Ritter. Individual data analysis and unified theories of cognition: A methodological proposal. In Niels A. Taatgen and Jakub Aasman (eds.), *Proceedings of the Third International Conference on Cognitive Modeling (ICCM 2000)*, pp. 150–157, Veenendaal, The Netherlands, 2000. Universal Press.
- Fernand Gobet, Peter C. R. Lane, Stephen Croker, Peter C.-H. Cheng, Gareth Jones, Ian Oliver, and Julian M. Pine. Chunking mechanisms in human learning. *Trends in Cognitive Sciences*, 5(6):236–243, 2001. doi: 10.1016/S1364-6613(00)01662-4.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny

Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuweke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla, Kushal Lakhotia, Lauren Rantala-Yearly, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang, Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shaoliang Nie, Sharan Narang, Sharath R-parthy, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gonguet, Virginie Do, Vish Vogeti, Vitor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyan Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao Jia, Xuwei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changhan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkan Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippas Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damlaj, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan

McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabisa, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshv, Maxim Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojian Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The llama 3 herd of models, 2024. URL <https://arxiv.org/abs/2407.21783>.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *First conference on language modeling*, 2024.

Fritz Günther, Marco Alessandro Petilli, and Marco Marelli. Semantic transparency is not invisibility: A computational model of perceptually-grounded conceptual combination in word processing. *Journal of Memory and Language*, 112:104104, 2020.

Martin N Hebart, Oliver Contier, Lina Teichmann, Adam H Rockter, Charles Y Zheng, Alexis Kidder, Anna Corriveau, Maryam Vaziri-Pashkam, and Chris I Baker. Things-data, a multimodal collection of large-scale datasets for investigating object representations in human brain and behavior. *Elife*, 12:e82580, 2023.

Benjamin E Hilbig and Morten Moshagen. Generalized outcome-based strategy classification: Comparing deterministic and probabilistic choice models. *Psychonomic bulletin & review*, 21(6):1431–1443, 2014.

Rachel A Jansen, Anna N Rafferty, and Thomas L Griffiths. A rational model of the dunning-kruger effect supports insensitivity to evidence in low performers. *Nature Human Behaviour*, 5(6):756–763, 2021.

Kristopher T Jensen, Guillaume Hennequin, and Marcelo G Mattar. A recurrent network model of planning explains hippocampal replay and human behavior. *Nature neuroscience*, 27(7):1340–1348, 2024.

Li Ji-An, Corey Y Zhou, Marcus K Benna, and Marcelo G Mattar. Linking in-context learning in transformers to human episodic memory. *Advances in neural information processing systems*, 37:6180–6212, 2024.

- Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, Gianna Lengyel, Guillaume Bour, Guillaume Lample, L lio Renard Lavaud, Lucile Saulnier, Marie-Anne Lachaux, Pierre Stock, Sandeep Subramanian, Sophia Yang, Szymon Antoniak, Teven Le Scao, Th ophile Gervet, Thibaut Lavril, Thomas Wang, Timoth e Lacroix, and William El Sayed. Mixtral of experts, 2024. URL <https://arxiv.org/abs/2401.04088>.
- Akaash Kolluri, Shengguang Wu, Joon Sung Park, and Michael S Bernstein. Finetuning llms for human behavior prediction in social science experiments. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 30084–30099, 2025.
- Wouter Kool, Fiery A Cushman, and Samuel J Gershman. When does model-based control pay off? *PLoS computational biology*, 12(8):e1005090, 2016.
- Wouter Kool, Samuel J Gershman, and Fiery A Cushman. Cost-benefit arbitration between multiple reinforcement-learning systems. *Psychological science*, 28(9):1321–1333, 2017.
- Paul M Krueger, Frederick Callaway, Sayan Gul, Thomas L Griffiths, and Falk Lieder. Identifying resource-rational heuristics for risky choice. *Psychological Review*, 131(4):905, 2024.
- Sreejan Kumar, Ishita Dasgupta, Nathaniel D Daw, Jonathan D Cohen, and Thomas L Griffiths. Disentangling abstraction from statistical pattern matching in human and machine learning. *PLoS computational biology*, 19(8):e1011316, 2023.
- Ionatan Kuperwajs, Heiko H Sch tt, and Wei Ji Ma. Using deep neural networks as a guide for modeling human planning. *Scientific reports*, 13(1):20269, 2023.
- John E. Laird. *The Soar Cognitive Architecture*. MIT Press, Cambridge, MA, 2012. ISBN 9780262122963.
- Germain Lefebvre, Ma l Lebreton, Florent Meyniel, Sacha Bourgeois-Gironde, and Stefano Palminteri. Behavioural and neural characterization of optimistic reinforcement learning. *Nature human behaviour*, 1(4):0067, 2017.
- Yuxuan Lei, Tianfu Wang, Jianxun Lian, Zhengyu Hu, Defu Lian, and Xing Xie. Humanllm: Towards personalized understanding and simulation of human nature. *arXiv preprint arXiv:2601.15793*, 2026.
- Kimery R Levering, Nolan Conaway, and Kenneth J Kurtz. Revisiting the linear separability constraint: New implications for theories of human category learning. *Memory & cognition*, 48(3):335–347, 2020.
- Wei Liu and Nai Ding. Can centaur truly simulate human cognition? the fundamental limitation of instruction understanding. *PsyArXiv preprint*, 2025. Version 1, submitted August 12, 2025. Retrieved from <https://osf.io/preprints/psyarxiv/zfhv9.v1>.
- T Ludwig, M Siegel, and E Schulz. Human multi-task learning: the why and what. In *Conference on Cognitive Computational Neuroscience (CCN 2023)*, pp. 418–421, 2023.
- Sabrina Namazova, Alessandra Brondetta, Younes Strittmatter, Matthew Nassar, and Sebastian Musslick. Not yet alphafold for the mind: Evaluating centaur as a synthetic participant. *arXiv preprint arXiv:2508.07887*, 2025.
- Allen Newell. *Unified Theories of Cognition*. Harvard University Press, Cambridge, MA, 1990.
- Mark Orr, Drew Cranford, Ken Ford, Kevin Gluck, Will Hancock, Christian Lebiere, Pete Pirolli, Frank Ritter, and Andrea Stocco. Not even wrong: On the limits of prediction as explanation in cognitive science. *arXiv preprint arXiv:2510.03311*, 2025.
- Stefano Palminteri, Germain Lefebvre, Emma J Kilford, and Sarah-Jayne Blakemore. Confirmation bias in human reinforcement learning: Evidence from counterfactual feedback processing. *PLoS computational biology*, 13(8):e1005684, 2017.

- Joshua C Peterson, David D Bourgin, Mayank Agrawal, Daniel Reichman, and Thomas L Griffiths. Using large-scale experiments and machine learning to discover theories of human decision-making. *Science*, 372(6547):1209–1214, 2021.
- Ori Plonsky, Reut Apel, Ido Erev, Eyal Ert, and Moshe Tennenholtz. When and how can social scientists add value to data scientists? a choice prediction competition for human decision making. *Unpublished Manuscript*, 2018.
- Vencislav Popov and Hannah Dames. Intent matters: Resolving the intentional versus incidental learning paradox in episodic long-term memory. *Journal of Experimental Psychology: General*, 152(1):268, 2023.
- Milena Rmus, Akshay K Jagadish, Marvin Mathony, Tobias Ludwig, and Eric Schulz. Generating computational cognitive models using large language models. *arXiv preprint arXiv:2502.00879*, 2025.
- Kai Ruggeri, Amma Panin, Milica Vdovic, Bojana Većkalov, Nazeer Abdul-Salaam, Jascha Achterberg, Carla Akil, Jolly Amatya, Kanchan Amatya, Thomas Lind Andersen, et al. The globalizability of temporal discounting. *Nature Human Behaviour*, 6(10):1386–1397, 2022.
- RB Rutledge. Risky decision and happiness task: The great brain experiment smartphone app, 2023.
- Hashem Sadeghiyeh, Siyu Wang, Maxwell R Alberhasky, Hannah M Kylo, Amitai Shenhav, and Robert C Wilson. Temporal discounting correlates with directed exploration but not with random exploration. *Scientific reports*, 10(1):4020, 2020.
- Nikolaus Salvatore and Qiong Zhang. Sequence-to-sequence models with attention mechanistically map to the architecture of human memory search. *Communications Psychology*, 3(1):146, 2025.
- René Schlegelmilch and Lenard Dome. Snake oil or panacea? how to misuse ai in scientific inquiries of the human mind. *Behavioral Sciences*, 16(2):219, 2026.
- Eric Schulz, Nicholas T Franklin, and Samuel J Gershman. Finding structure in multi-armed bandits. *Cognitive psychology*, 119:101261, 2020.
- Nitzan Shahar, Tobias U Hauser, Michael Moutoussis, Rani Moran, Mehdi Keramati, Nspn Consortium, and Raymond J Dolan. Improving the reliability of model-based decision-making estimates in the two-stage decision task with reaction-times and drift-diffusion modeling. *PLoS computational biology*, 15(2):e1006803, 2019.
- Herbert A. Simon. *The Sciences of the Artificial*. The MIT Press, Cambridge, MA, 1969.
- Herbert A. Simon and Fernand Gobet. Expertise effects in memory recall: Comment on vicente and wang (1998). *Psychological Review*, 107(3):593–600, 2000. doi: 10.1037/0033-295X.107.3.593.
- Herbert A. Simon and Allen Newell. Human problem solving: The state of the theory in 1970. *American Psychologist*, 26(2):145–159, February 1971.
- Leah H Somerville, Stephanie F Sasse, Megan C Garrad, Andrew T Drysdale, Nadine Abi Akar, Catherine Insel, and Robert C Wilson. Charting the expansion of strategic exploratory behavior during adolescence. *Journal of experimental psychology: general*, 146(2):155, 2017.
- Maarten Speekenbrink, Shelley Channon, and David R Shanks. Learning strategies in amnesia. *Neuroscience & Biobehavioral Reviews*, 32(2):292–310, 2008.
- Mikhail S Spektor, David Kellen, Jörg Rieskamp, and Karl Christoph Klauer. Absolute and relative stability of loss aversion across contexts. *Journal of Experimental Psychology: General*, 153(2):454, 2024.

- Helen Steingroever, Daniel J Fridberg, Annette Horstmann, Kimberly L Kjome, Veena Kumari, Scott D Lane, Tiago V Maia, James L McClelland, Thorsten Pachur, Preethi Premkumar, et al. Data from 617 healthy participants performing the iowa gambling task: A “many labs” collaboration. *Journal of Open Psychology Data*, 3(1):e5–e5, 2015.
- Ryan Teknium, Jeffrey Quesnelle, and Chen Guang. Hermes 3 technical report, 2024. URL <https://arxiv.org/abs/2408.11857>.
- Ryan Teknium, Roger Jin, Jai Suphavadeeprasit, Dakota Mahan, Jeffrey Quesnelle, Joe Li, Chen Guang, Shannon Sands, and Karan Malhotra. Hermes 4 technical report, 2025. URL <https://arxiv.org/abs/2508.18255>.
- Momchil S Tomov, Samyukta Yagati, Agni Kumar, Wanqian Yang, and Samuel J Gershman. Discovery of hierarchical representations for efficient planning. *PLoS computational biology*, 16(4):e1007594, 2020.
- Momchil S Tomov, Eric Schulz, and Samuel J Gershman. Multi-task reinforcement learning in humans. *Nature Human Behaviour*, 5(6):764–773, 2021.
- Polina Tsvilodub, Bob van Tiel, and Michael Franke. The role of relevance, competence, and priors for scalar inferences. *Experiments in Linguistic Meaning*, 2:288–298, 2023.
- Henri Vandendriessche, Amel Demmou, Sophie Bavard, Julien Yadak, Cédric Lemogne, Thomas Mauras, and Stefano Palminteri. Contextual influence of reinforcement learning performance of depression: evidence for a negativity bias? *Psychological Medicine*, 53(10): 4696–4706, 2023.
- Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, Allyson Ettinger, Michal Guerquin, David Heineman, Hamish Ivison, Pang Wei Koh, Jiacheng Liu, Saumya Malik, William Merrill, Lester James V. Miranda, Jacob Morrison, Tyler Murray, Crystal Nam, Jake Poznanski, Valentina Pyatkin, Aman Rangapur, Michael Schmitz, Sam Skjonsberg, David Wadden, Christopher Wilhelm, Michael Wilson, Luke Zettlemoyer, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. 2 olmo 2 furious, 2025. URL <https://arxiv.org/abs/2501.00656>.
- James A Waltz, Robert C Wilson, Matthew A Albrecht, Michael J Frank, and James M Gold. Differential effects of psychotic illness on directed and random exploration. *Computational Psychiatry (Cambridge, Mass.)*, 4:18, 2020.
- Robert C Wilson, Andra Geana, John M White, Elliot A Ludvig, and Jonathan D Cohen. Humans use directed and random exploration to solve the explore–exploit dilemma. *Journal of experimental psychology: General*, 143(6):2074, 2014.
- Toby Wise, Jochen Michely, Peter Dayan, and Raymond J Dolan. A computational account of threat-related attentional bias. *PLoS computational biology*, 15(10):e1007341, 2019.
- Charley M Wu, Eric Schulz, Maarten Speekenbrink, Jonathan D Nelson, and Björn Meder. Generalization guides human exploration in vast decision spaces. *Nature human behaviour*, 2(12):915–924, 2018.
- Shuchen Wu, Noémi Éltető, Ishita Dasgupta, and Eric Schulz. Chunking as a rational solution to the speed–accuracy trade-off in a serial reaction time task. *Scientific reports*, 13(1):7680, 2023.
- Dirk U Wulff, Max Mergenthaler-Canseco, and Ralph Hertwig. A meta-analytic review of two modes of learning and the description–experience gap. *Psychological bulletin*, 144(2): 140, 2018.
- Hanbo Xie and Jian-Qiao Zhu. Centaur may have learned a shortcut that explains away psychological tasks. PsyArXiv preprint, 2025. Retrieved from <https://osf.io/preprints/psyarxiv/u7z4tv1>.

- Yutong Xie, Zhuoheng Li, Xiyuan Wang, Yijun Pan, Qijia Liu, Xingzhi Cui, Kuang-Yu Lo, Ruoyi Gao, Xingjian Zhang, Jin Huang, et al. Be. fm: Open foundation models for human behavior. *arXiv preprint arXiv:2505.23058*, 2025.
- Hua-Dong Xiong, Li Ji-An, Marcelo Mattar, and Robert Wilson. Neural network modeling reveals diverse human exploration behaviors via state space analysis. In *Proceedings of the Conference on Computational Cognitive Neuroscience (CCN)*, 2023.
- Songlin Xu and Xinyu Zhang. Augmenting human cognition with an ai-mediated intelligent visual feedback. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pp. 1–16, 2023.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025. URL <https://arxiv.org/abs/2505.09388>.
- Guangyu Robert Yang, Madhura R Joglekar, H Francis Song, William T Newsome, and Xiao-Jing Wang. Task representations in neural networks trained to perform many cognitive tasks. *Nature neuroscience*, 22(2):297–306, 2019.
- Jian-Qiao Zhu, Adam N Sanborn, and Nick Chater. The bayesian sampler: Generic bayesian inference causes incoherence in human probability judgments. *Psychological review*, 127(5): 719, 2020.
- Jian-Qiao Zhu, Joshua C Peterson, Benjamin Enke, and Thomas L Griffiths. Capturing the complexity of human strategic decision-making with machine learning. *Nature Human Behaviour*, 9(10):2114–2120, 2025a.
- Jian-Qiao Zhu, Hanbo Xie, Dilip Arumugam, Robert C Wilson, and Thomas L Griffiths. Using reinforcement learning to train large language models to explain human decisions. *arXiv preprint arXiv:2505.11614*, 2025b.
- Ondrej Zika, Katja Wiech, Andrea Reinecke, Michael Browning, and Nicolas W Schuck. Trait anxiety is associated with hidden state inference during aversive reversal learning. *Nature Communications*, 14(1):4203, 2023.
- Samuel Zorowitz and Yael Niv. Data from two-step task pilots. <https://doi.org/10.17605/OSF.IO/2TGJD>, 2023. OSF. Created February 14, 2023. Licensed under CC BY 4.0.

A Cognitive and behavioural foundation models

Several recent works have fine-tuned or prompted large language models to predict human behaviour in cognitive tasks. Tables 1–4 situate our models within this emerging landscape. Table 1 compares each model’s base LLM, post-training method, training data, and target domain. Table 2 reports training hyperparameters – adaptation strategy, optimiser, learning rate, batch configuration, and scheduler – to facilitate direct methodological comparison and reproducibility. Table 3 details the data pipeline and loss configuration, including numerical precision, loss masking strategy, input format, and any use of LLM-assisted data synthesis. Table 4 records the computational infrastructure, training time, and inference settings for each model.

Model	Base LLM	Post-training	Training data	Domain
Centaur (Binz et al., 2025)	Llama-3.1-70B	SFT (masked CE on response tokens); rank-stabilised QLoRA ($r=\alpha=8$), 4-bit quantised	Psych-101: 160 expts, 60,092 partic., 10.7M choices	Decision, memory, learning, planning
Zhu et al. (2025b) ¹	Qwen2.5-7B-Instruct	SFT / Centaur-style SFT / GRPO compared; LoRA ($r=\alpha=32$)	choices13k: 13,102 train / 1,462 test risky choice problems	Decision
Be. FM (Xie et al., 2025)	Llama-3.1-{8,70}B-Instruct	SFT with LoRA (all layers); 8-bit quantised ²	AER: 2,703 papers.; MobLab: 68,779 subj., 82,057 obs.; Big Five: 17,667 subj.	Behavioural science, economic game, personality
Socrates (Kolluri et al., 2025)	Llama-3-8B-Instruct Qwen2.5-14B-Instruct	SFT / SFT + oracle reasoning traces / contrastive DPO compared; full fine-tuning	SocSci210: 210 TESS ³ expts, 400,491 partic., 2.9M individual responses	Social sciences (economics, psychology, political science)
HumanLLM (Lei et al., 2026)	Qwen2.5-{3,7}B-Instruct Qwen3-8B ⁴ Llama-3.1-8B-Instruct Phi-3-mini-128k-instruct	SFT (masked non-response tokens); full fine-tuning; 1:1 weight merge with base (LM-Cocktail)	Cognitive Genome: Reddit 2.8M, Twitter 673K, Blogger 368K, Amazon 1.7M; 1.2M train samples	Social intelligence (personalised behaviour)
GeCCo (Rmus et al., 2025)	Llama-3.1-70B-Instruct DeepSeek-R1-Distill-Llama-3.1-70B Qwen2.5-72B-Instruct	No fine-tuning; in-context learning with iterative BIC-based refinement (10 × 5 runs)	Behavioural data from 4 cognitive domains (in-context, not for training)	Decision, learning, planning, working memory
Ours	Llama-3.2-{1,3}B	SFT (masked CE on response tokens, Centaur-style); rank-stabilised LoRA ($r=\alpha \in \{4, 8, 16, 32, 64\}$)	Psych-101: 160 expts, 60,092 partic., 10.7M choices	Decision, memory, learning, planning
Llama-Centaur	Llama-3.1-8B			
Qwentaur	Qwen3-{0.6, 1.7, 4, 8, 14}B-Base			
SmolLlama	SmolLM2-{135, 360}M			
OlmoLlama	SmolLM2-1.7B SmolLM3-3B-Base OLMo-2-0425-1B OLMo-3-1025-7B			

¹No named model; methods (SFT v. RL) comparison only. ²8-bit quantisation applies to 70B variant only; 8B is unquantised. ³TESS: NSF’s Time-sharing Experiments for the Social Sciences, a repository of peer-reviewed social science experiments conducted on nationally representative samples. ⁴Lei et al. (2026) states “Qwen3-8B” without Instruct suffix; base/instruct status unspecified.

Table 1: Core comparison of LLM-based cognitive and behavioural foundation models.

Model	Adaptation	Optimiser	Epochs	Learning rate	Eff. batch size (PD×GA×d)	Scheduler
Centaur (Binz et al., 2025)	QLoRA ($r=\alpha=8$), all linear layers	8-bit AdamW	1 ¹	5×10^{-5}	$1 \times 32 \times 1$	Cosine (WU 100 steps)
Zhu et al. (2025b)	LoRA $r=\alpha=32$, all linear layers, dropout 0.05	AdamW	SFT: 6; RL: 3	SFT: 10^{-5} ; RL: 3×10^{-6}	SFT: PD×8×1; RL: PD×8×4	SFT: fixed; RL: cosine
Be.FM (Xie et al., 2025)	LoRA, all layers	Not specified	3	10^{-4}	$1 \times 8 \times d$	Cosine (WU 0.1)
Socrates (Kolluri et al., 2025)	Full fine-tuning (no LoRA)	Not specified	1	SFT: 10^{-5} ; DPO: 10^{-6}	PD×GA×8=256	Cosine (WU 0.05)
HumanLLM (Lei et al., 2026)	Full fine-tuning (no LoRA)	Not specified	3	5×10^{-6}	PD×GA×8=64	Cosine (WU 0.5)
GeCCo (Rmus et al., 2025)	N/A (no training) ²	N/A	N/A	N/A	N/A	N/A
Qwentaaur-0.6B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$2 \times 16 \times 1$	Linear warmup (100 steps)
Qwentaaur-1.7B	LoRA ($r=\alpha \in$ {8, 16}), all linear layers	AdamW	1	5×10^{-5}	$1 \times 32 \times 1$	Linear warmup (100 steps)
Qwentaaur-4B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$1 \times 32 \times 1$	Linear warmup (100 steps)
Qwentaaur-8B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$1 \times 32 \times 1$	Linear warmup (100 steps)
Qwentaaur-14B	LoRA ($r=\alpha \in$ {4, 16, 64}), all linear layers	AdamW	1	5×10^{-5}	$1 \times 32 \times 1$	Linear warmup (100 steps)
Llama-Centaur-1B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$2 \times 16 \times 1$	Linear warmup (100 steps)
Llama-Centaur-3B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$1 \times 32 \times 1$	Linear warmup (100 steps)
Llama-Centaur-8B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$1 \times 32 \times 1$	Linear warmup (100 steps)
Smoltaur-0.1B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$8 \times 4 \times 1$	Linear warmup (100 steps)
Smoltaur-0.4B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$4 \times 8 \times 1$	Linear warmup (100 steps)
Smoltaur-1.7B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$2 \times 16 \times 1$	Linear warmup (100 steps)
Smoltaur-3B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$1 \times 32 \times 1$	Linear warmup (100 steps)
Olmotaur-1B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$2 \times 16 \times 1$	Linear warmup (100 steps)
Olmotaur-7B	LoRA ($r=\alpha \in$ {4, 8, 16, 32, 64}), all linear layers	AdamW	1	5×10^{-5}	$1 \times 32 \times 1$	Linear warmup (100 steps)

Abbreviations: PD = per-device batch size; GA = gradient accumulation steps; d = number of GPU devices; WU = warmup ratio.

¹ The published training script (https://github.com/marcelbinz/Llama-3.1-Centaur-70B/blob/main/scripts/cluster_train.sh) specifies 5 epochs. Binz et al. (2025) report training for 1 epoch, suggesting early checkpoint selection. ² GeCCo (Rmus et al., 2025) uses in-context learning with iterative BIC-based feedback over 10 sampling iterations × 5 independent runs. Models are fitted to held-out data with SciPy minimize (20 random restarts) and evaluated by BIC.

All post-training methods use supervised fine-tuning unless otherwise noted; Zhu et al. (2025b) additionally compare RL (GRPO with 12 candidate completions per step, max 1024 tokens; reward = $1 - |o_B - p_B|$ + format bonus up to 0.5, no standard-deviation normalisation); Socrates (Kolluri et al., 2025) additionally compares contrastive DPO (preference pairs constructed by varying the demographic persona under the same experimental condition and outcome question).

Table 2: Training hyperparameters for LLM-based cognitive and behavioural models. Adaptation strategy refers to the parameter-efficient or full fine-tuning method applied to the base LLM. All our variants share identical hyperparameters except for per-device batch size, gradient accumulation steps, and the set of adapter ranks trained. Entries marked “Not specified” indicate that the original paper did not report the corresponding detail.

Model	Precision	Weight decay	Loss masking	Data format	Data synthesis
Centaur (Binz et al., 2025)	4-bit NF4	0.01	Human response tokens only	NL trial-by-trial prompts (~32K tokens)	Template-based prompt construction ¹
Zhu et al. (2025b)	Not specified	Not specified	SFT: standard; Centaur-style. GRPO	JSON aggregated choice proportions per problem (empirical % rounded to nearest integer, e.g. {"A": 29, "B": 71})	Reformatted from choices13k empirical choice frequencies (problem-level rather than individual-participant-level prediction)
Be. FM (Xie et al., 2025)	8-bit (70B base, bitsandbytes); bf16 (8B)	Not specified	Standard (Alpaca template)	Alpaca template {instruction, input, output}	GPT-4o for research workflow extraction
Socrates (Kolluri et al., 2025)	Not specified	0.1	SFT: Response token only. DPO	{persona, stimuli, outcome, response}	o4-mini-high (dataset agent ²); GPT-4o-mini (reasoning traces ³)
HumanLLM (Lei et al., 2026)	Not specified	Not specified	Non-response positions masked	ShareGPT format	Llama-3.3-70B (extraction); GPT-4o (quality validation)
GeCCo (Rmus et al., 2025)	N/A	N/A	N/A	NL prompt + Python function template	N/A
Ours	bf16	0.01	Human response tokens only (Centaur-style)	NL trial-by-trial prompts (~32K tokens)	Psych-101 dataset (unmodified)

¹Each experiment is converted into natural-language trial-by-trial prompts via author-written scripts that map structured experimental data (participant responses, stimuli, feedback) to verbalised narratives; for an example, see <https://github.com/marcelbinz/Psych-201/blob/main/binz2022heuristics/generate.prompts.py>. ²An LLM-based agent (o4-mini-high) parses raw TESS datasets into structured {persona, stimuli, response} tuples; see Appendix A of Kolluri et al. (2025) for details. ³Given an experimental prompt and the corresponding human response, GPT-4o-mini generates reasoning traces explaining the human decision from a social scientist’s perspective; see Appendix E of Kolluri et al. (2025).

Table 3: Data pipeline and loss configuration for LLM-based cognitive and behavioural models. Precision refers to the numerical format used during training: quantised formats (4-bit NF4, 8-bit) apply to base model weights while LoRA adapters and forward/backward computation use bf16. Loss masking describes which tokens contribute to the training loss; masked approaches restrict gradient updates to human response tokens, avoiding optimisation on task instructions and context. Data format describes the input representation seen by the model during training. Data synthesis indicates whether and how any computer-assisted tools were used to construct or augment the training corpus.

Model	Training Framework	Key packages	Hardware	Train time	Max seq. len.	Inference
Centaur (Binz et al., 2025)	unsloth	unsloth	1 × A100 80GB	~5 days	~32,768	Not specified
Zhu et al. (2025b)	Not specified	vLLM (inference)	RL: 4 × H100; SFT: 1 × A100	RL: ~80 h; SFT: ~5 h	1,024 (RL); 30 (SFT inf.)	$T=0.7$, top- $p=0.95$, top- $k=0.5$
Be. FM (Xie et al., 2025)	LlamaFactory	LlamaFactory, bitsandbytes	Not specified	Not specified	Not specified	Not specified
Socrates (Kolluri et al., 2025)	LlamaFactory	LlamaFactory	8 × A100 80GB	4–24 h	4,096 (inf.)	$T=0.6$, top- $p=0.9$
HumanLLM (Lei et al., 2026)	LlamaFactory	LlamaFactory, DeepSpeed Zero, vLLM	8 × A100 40GB	~120 h (8B)	8,192	$T=0.7$
GeCCo (Rmus et al., 2025)	N/A ¹	SciPy (minimize); Python exec()	4 × A100 40GB	≤8 h per domain	N/A	$T: 0.1\text{--}0.2^2$
Qwentaaur-0.6B	unsloth	unsloth	1 × A100 80GB	4 h	~32,768	—
Qwentaaur-1.7B	unsloth	unsloth	1 × A100 80GB	7 h	~32,768	—
Qwentaaur-4B	unsloth	unsloth	1 × A100 80GB	18 h	~32,768	—
Qwentaaur-8B	unsloth	unsloth	1 × A100 80GB	1 d	~32,768	—
Qwentaaur-14B	unsloth	unsloth	1 × A100 80GB	1 d 15 h	~32,768	—
Llama-Centaur-1B	unsloth	unsloth	1 × A100 80GB	7 h	~32,768	—
Llama-Centaur-3B	unsloth	unsloth	1 × A100 80GB	12 h	~32,768	—
Llama-Centaur-8B	unsloth	unsloth	1 × A100 80GB	21 h	~32,768	—
Smoltaur-0.1B	unsloth	unsloth	1 × A100 80GB	3 h	~8,192	—
Smoltaur-0.4B	unsloth	unsloth	1 × A100 80GB	4 h	~8,192	—
Smoltaur-1.7B	unsloth	unsloth	1 × A100 80GB	7 h	~8,192	—
Smoltaur-3B	unsloth	unsloth	1 × A100 80GB	1 d 4 h	~32,768	—
Olmotaur-1B	unsloth	unsloth	1 × A100 80GB	5 h	~4,096	—
Olmotaur-7B	unsloth	unsloth	1 × A100 80GB	2 d 1 h	~32,768	—

Abbreviations: T = sampling temperature; inf. = inference; d = days; h = hours.

¹No training framework required; GeCCo generates cognitive models via in-context prompting. SciPy is used for parameter fitting of the generated models; Python exec() executes the LLM-generated code. ²Temperature varies by LLM: Llama 0.2, Qwen 0.15, DeepSeek-R1 0.1. All our models were trained on a single NVIDIA A100 80GB GPU. Training times scale approximately linearly with parameter count within each model family.

Table 4: Infrastructure, packages, and computational cost for LLM-based cognitive and behavioural models. Hardware refers to GPU resources used during training (or in-context generation for GeCCo). Training time reports wall-clock duration as stated in each paper; missing entries indicate the information was not reported. Our models were each trained for 1 epoch on Psych-101 on a single A100 80GB, with rank-stabilised LoRA at $r=\alpha \in \{4, 8, 16, 32, 64\}$.

B Adapter rank and training-set size

Adapter rank is swept over $r \in \{4, 8, 16, 32, 64\}$ at full data, and training-set size over nested, experiment-stratified subsets of Psych-101 (1/16, 1/8, 1/4, 1/2, and the full set) at fixed rank $r=16$. Figure 6 shows the rank sweep across all fourteen models on a single axis, which makes the substitution between adapter capacity and parameter count visible directly: a small model at high rank overlaps a much larger model at low rank. Figure 7 separates the two axes by family, with the rank sweep on the left and the data ablation on the right.

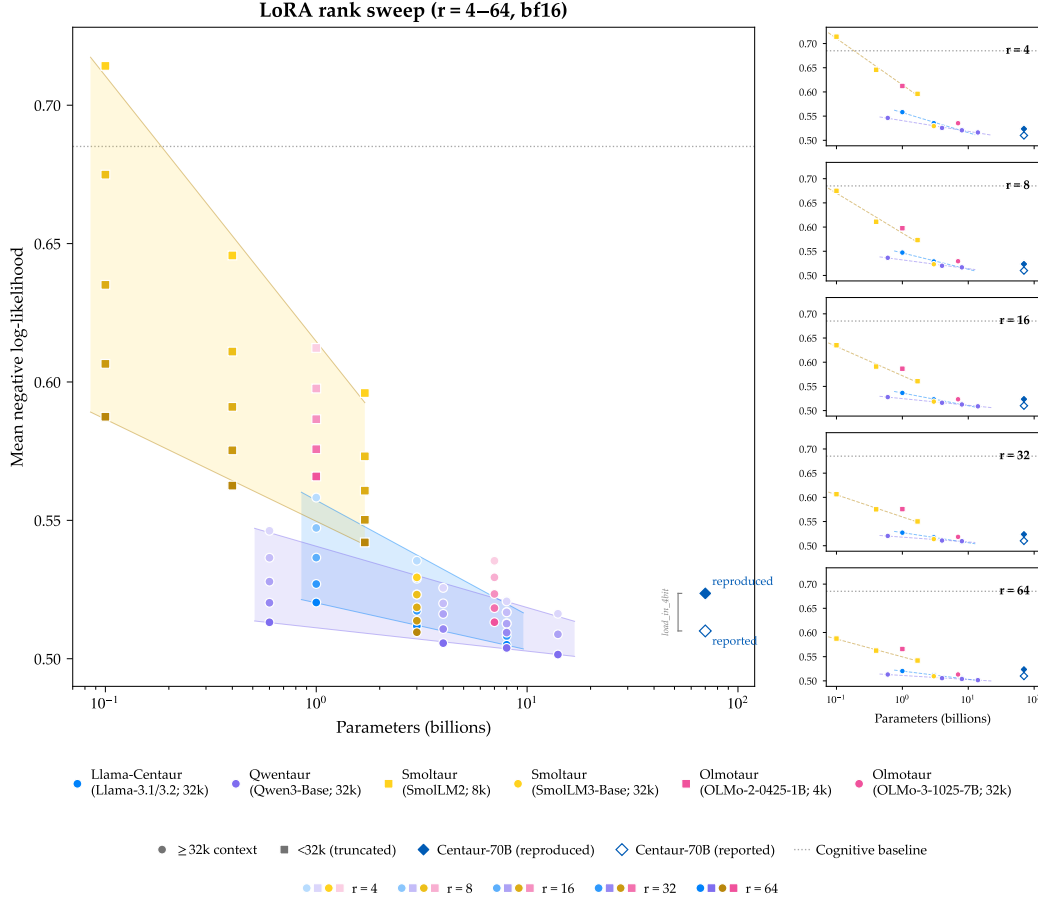


Figure 6: **Adapter rank against model size on Psych-101.** Mean negative log-likelihood over the 38 of 46 Psych-101 tasks for which Binz et al. (2025) publish a domain-specific cognitive model baseline, against parameter count, with colour intensity encoding LoRA rank and right-hand panels showing one rank at a time. Trend lines and shaded envelopes are fitted only within groups matched on model generation and context window, so Olmotaur has none and the Smoltauaur fit covers the SmolLM2 models only. Marker shape indicates context window (circle $\geq 32k$, square $< 32k$); colour indicates family.

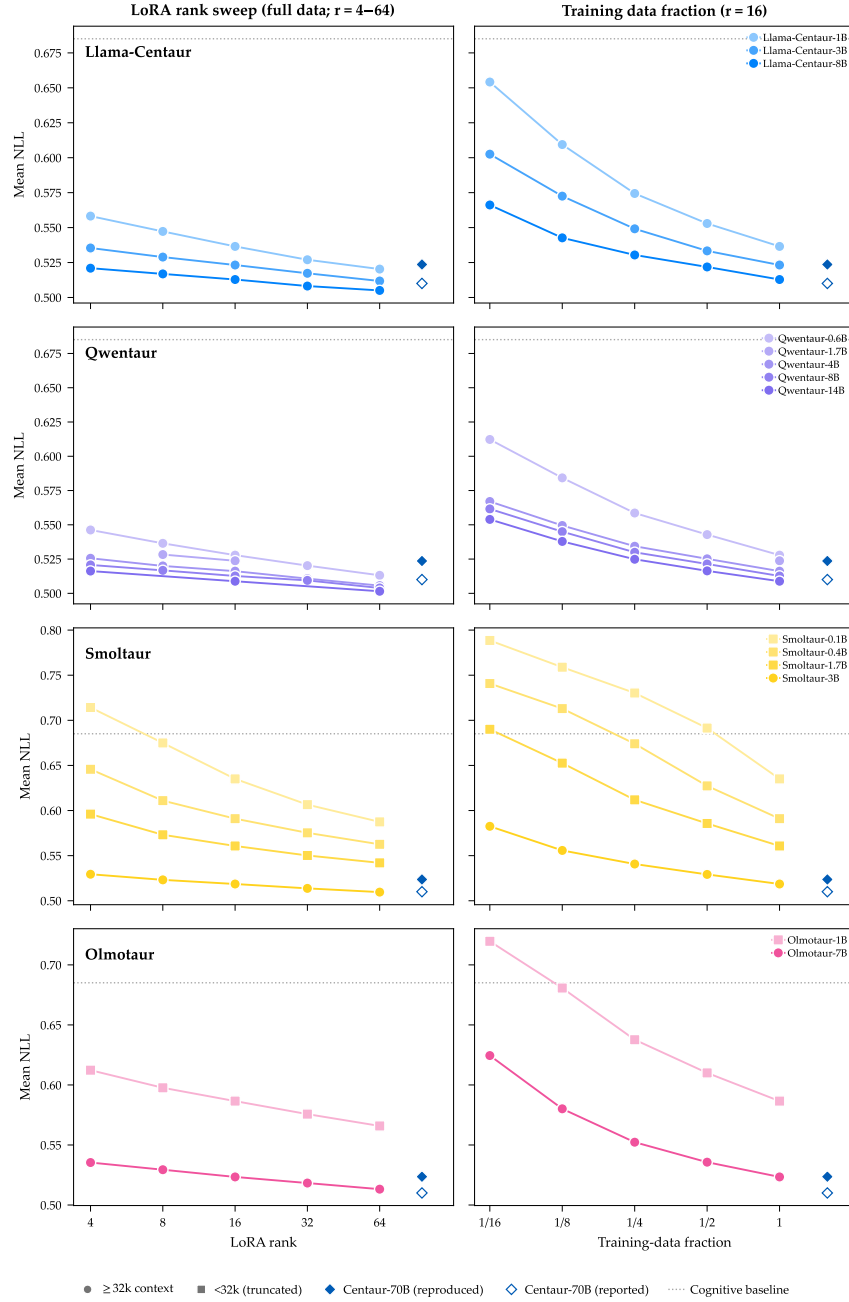


Figure 7: **Adapter rank and training-set size by family.** (Left) Mean NLL against LoRA rank at full data. (Right) Mean NLL against the fraction of Psych-101 used for training, at $r=16$. Rows are the four model families; the dotted line is the cognitive-model baseline and the diamonds are the reproduced and reported Centaur-70B values. Subsets are nested and experiment-stratified, so every fraction covers all 160 experiments and reducing data quantity does not reduce paradigm coverage.

C Full per-experiment results

C.1 Psych-101 (in-distribution)

This appendix presents the full per-experiment results on Psych-101. Table 5 summarises performance by task type using the normalised metric $(\ln k - \text{NLL}) / \ln k$, restricted to the 34 experiments admitting both a cognitive model baseline and a well-defined chance level. Table 6 then reports base versus fine-tuned NLL per experiment for the Qwen3 and Llama families, isolating the effect of cognitive fine-tuning at each scale, and Table 7 gives per-experiment NLL for all four fine-tuned families side by side. Table 8 compares our models against other behavioural foundation models (Be. FM (Xie et al., 2025) and Socrates (Kolluri et al., 2025)) and non-cognitive controls (Nemotron (Bercovich et al., 2025) and Hermes (Teknum et al., 2024; 2025)), grouped by base-model family. All tables include a rightmost column reporting the chance-level baseline $\ln(k)$, where k is the number of discrete response options.

Experiment exclusions. Of the 46 experiments in Psych-101, 12 are excluded from the normalised analysis reported in Table 5; their raw NLL values remain in all appendix tables. These exclusions fall into four categories.

Continuous or free-form responses. Two experiments elicit numeric responses on a continuous scale, for which no discrete uniform baseline exists: the aversive learning task (Wise et al., 2019) (supervised learning) asks participants to type probability estimates, and the probabilistic reasoning task (Zhu et al., 2020) (miscellaneous) similarly requires free-form percentage judgements.

Mixed response types. Seven experiments interleave multiple response formats within a single session, making it impossible to assign a single k : the episodic long-term memory task (Popov & Dames, 2023) (memory) combines binary size judgements with free word recall and arithmetic answers; the experiential-symbolic task (Garcia et al., 2023) (decision-making) mixes binary letter choices with probability estimates; the recall and recognition task (Cox et al., 2018) (memory) interleaves binary recognition with open-vocabulary cued recall; the grammar judgement task (Jansen et al., 2021) (miscellaneous) combines free numeric self-assessments with five-alternative grammar questions whose labels vary across items; the Medin categorisation task (Levering et al., 2020) (supervised learning) pairs binary categorisation with 9-point typicality ratings; the risky choice task (Krueger et al., 2024) (decision-making) requires multi-stage responses including gamble selection from six options, colour queries, and a stop action; and the decisions from experience task (Wulff et al., 2018) (multi-armed bandits) switches between a three-option sampling phase (two lotteries plus stop) and a two-option final choice.

Degenerate or trial-varying action spaces. Two experiments have action spaces that prevent a meaningful $\ln(k)$ computation: the go/no-go task (Enkavi et al., 2019) (decision-making) only tokenises go-responses, yielding $\ln(1) = 0$; and the tile-revealing task (Kumar et al., 2023) (Markov decision processes) uses grid coordinates on a 7×7 board where the set of valid positions shrinks as cells are revealed, making k variable across trials.

State-dependent action space. The virtual subway network task (Tomov et al., 2020) (Markov decision processes) nominally offers five response tokens (four cardinal directions plus an end-round key), but walls in the navigation grid constrain which directions are valid on any given trial. We include this experiment in the appendix tables using $\ln(5) \approx 1.61$ as a nominal ceiling, but exclude it from the normalised analysis (Table 5) since the effective action space varies per state.

Task type	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		Base			Binz et al. (2025)		
	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	70B _p	70B _r	Cog _p
Decision (8)	0.53	0.53	0.54	0.54	0.54	0.53	0.53	<u>0.54</u>	0.41	0.46	0.48	0.54	0.47	0.53	0.34	0.35	0.33	0.56	0.52	0.27
MDP (5)	0.28	0.30	0.31	0.31	0.32	0.28	0.30	0.31	0.18	0.20	0.26	0.30	0.20	0.30	0.15	0.17	0.19	0.32	<u>0.31</u>	0.11
Bandit (12)	0.48	0.48	0.49	<u>0.49</u>	0.50	0.46	0.48	0.49	0.36	0.41	0.44	0.49	0.42	0.48	0.35	0.36	0.33	0.46	0.45	0.39
Memory (4)	0.57	0.57	0.58	0.58	<u>0.58</u>	0.56	0.57	0.58	0.46	0.51	0.55	0.57	0.53	0.57	0.42	0.47	0.43	0.58	0.58	0.36
Misc. (2)	0.57	0.57	0.58	0.58	0.58	0.56	0.57	<u>0.58</u>	0.45	0.53	0.55	0.58	0.55	0.57	0.41	0.42	0.40	0.57	0.57	0.55
Sup. learn. (3)	0.30	0.29	0.31	0.31	<u>0.31</u>	0.29	0.31	0.31	0.18	0.23	0.25	0.30	0.20	0.31	0.24	0.24	0.23	0.30	0.30	0.11
Mean (34)	0.46	0.46	0.47	0.48	0.48	0.45	0.47	<u>0.48</u>	0.35	0.39	0.43	0.47	0.40	0.47	0.32	0.34	0.32	0.47	0.46	0.30
	<0.05	0.05–0.12	0.12–0.18	0.18–0.25	0.25–0.32	0.32–0.40	0.40–0.48	0.48–0.55	≥0.55											

Table 5: **Fraction of available information captured above chance, $(\ln k - \text{NLL}) / \ln k$, by task type for finetuned models (bf16).** A value of 0 indicates chance-level performance; 1 indicates perfect prediction. Restricted to 34 experiments (of 46) with both a cognitive model baseline and a well-defined discrete response space ($\ln k > 0$); 12 experiments are excluded. Finetuned families: Qwentaur, Llama-Centaur, Smoltaur, Olmotaur. Base columns: Q-8B/Q-14B (Qwen3), L-8B (Llama-3.1). Subscript _r denotes our reproducing evaluation of the original Centaur model distributed by Binz et al. (2025), evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript _p denotes values published by Binz et al. (2025). 70B_p is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. **Bold+underline** marks the best model, underline the second-best.

Experiment	Type	Qwen3 family										LLama-3.1/3.2 family				Binz et al. (2025)				Chance			
		0.6B		1.7B		4B		8B		14B		1B		3B		8B		L-70B _p	C _r		C _p	Cog _p	ln(k)
		base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft						
Balloon analog risk task (Frey et al., 2017)	Decision	0.09	0.07	0.09	0.07	0.08	0.07	0.08	0.06	0.08	0.06	0.10	0.07	0.08	0.07	0.08	0.06	0.08	0.07	0.06	0.09	0.69	
CPC18 (Plesky et al., 2018)	Decision	0.41	0.35	0.40	0.34	0.40	0.34	0.40	0.34	0.40	0.33	0.43	0.36	0.41	0.35	0.39	0.34	0.41	0.35	0.34	0.66	0.69	
Columbia card task (Frey et al., 2017)	Decision	0.27	0.20	0.26	0.20	0.26	0.20	0.25	0.20	0.25	0.20	0.30	0.21	0.27	0.20	0.25	0.20	0.23	0.21	0.19	0.26	0.69	
Decisions from description (Wulff et al., 2018)	Decision	1.08	0.62	1.05	0.63	1.01	0.62	0.99	0.61	1.01	0.60	1.55	0.62	1.20	0.63	0.96	0.60	0.76	0.59	0.53	0.61	0.69	
Experimental-symbolic task (Garcia et al., 2023)	Decision	0.76	0.48	0.68	0.47	0.67	0.46	0.64	0.46	0.57	0.46	0.78	0.47	0.69	0.47	0.64	0.46	0.70	0.46	0.45	—	—	
Gardening task (Fleisch et al., 2018)	Decision	0.52	0.39	0.52	0.39	0.49	0.38	0.47	0.38	0.45	0.38	0.81	0.40	0.64	0.39	0.48	0.38	0.50	0.49	0.38	0.91	0.69	
Go/no-go (Bakshi et al., 2019)	Decision	0.01	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.08	0.00	
Intertemporal choice (Ruggert et al., 2022)	Decision	0.77	0.44	0.80	0.44	0.71	0.44	0.69	0.44	0.70	0.44	0.85	0.44	0.75	0.44	0.76	0.44	0.73	0.44	0.43	0.66	0.69	
Multi-attribute DM (Filling & Moshagen, 2014)	Decision	0.49	0.09	0.34	0.08	0.20	0.09	0.19	0.08	0.19	0.08	0.63	0.08	0.45	0.08	0.23	0.08	0.15	0.06	0.06	0.19	0.69	
Risky choice (Kraeger et al., 2020)	Decision	0.92	0.48	0.83	0.47	0.79	0.44	0.74	0.43	0.67	0.42	0.99	0.49	0.85	0.46	0.76	0.43	0.65	0.43	0.43	—	—	
choices13k (Peterson et al., 2021)	Decision	0.62	0.44	0.60	0.44	0.59	0.43	0.59	0.43	0.55	0.43	0.67	0.44	0.59	0.44	0.55	0.43	0.53	0.43	0.43	0.66	0.69	
Multi-task RL (Tomov et al., 2021)	MDP	0.70	0.60	0.72	0.59	0.68	0.58	0.70	0.58	0.68	0.57	0.80	0.60	0.70	0.59	0.66	0.57	0.66	0.57	0.57	1.04	1.10	
Tile-revealing task (Kumar et al., 2023)	MDP	2.95	0.02	2.81	0.01	2.67	0.00	2.73	0.07	2.60	0.00	4.86	0.02	3.98	0.00	2.92	0.00	2.74	2.29	1.87	—	—	
Two-step task (Kool et al., 2017)	MDP	0.67	0.54	0.64	0.53	0.64	0.53	0.65	0.53	0.63	0.53	0.67	0.55	0.62	0.53	0.61	0.53	0.61	0.53	0.50 [†]	0.60 [†]	0.69	
Two-step task (Kool et al., 2016)	MDP	0.69	0.51	0.64	0.50	0.64	0.49	0.65	0.49	0.63	0.48	0.67	0.52	0.62	0.50	0.61	0.49	0.61	0.48	0.50 [†]	0.60 [†]	0.69	
Two-step task (Zemovitz & Niv, 2023)	MDP	0.62	0.52	0.59	0.51	0.60	0.51	0.61	0.51	0.59	0.51	0.64	0.52	0.60	0.51	0.58	0.51	0.61	0.51	0.50 [†]	0.60 [†]	0.69	
Virtual subway network (Tomov et al., 2020)	MDP	1.68	1.16	1.55	1.15	1.57	1.14	1.55	1.13	1.57	1.12	1.68	1.19	1.60	1.16	1.53	1.13	1.53	1.16	1.13	—	1.61	
ZooPermarket (Ludwig et al., 2023)	MDP	0.61	0.54	0.61	0.52	0.60	0.51	0.60	0.50	0.59	0.48	0.64	0.54	0.62	0.52	0.60	0.50	0.60	0.50	0.48	0.60	0.69	
Cond. assoc. learning (Collins et al., 2014)	Memory	0.76	0.54	0.71	0.53	0.71	0.52	0.69	0.52	0.61	0.51	0.82	0.55	0.73	0.53	0.70	0.52	0.64	0.52	0.54	0.86	1.10	
Digit span (Bakshi et al., 2019)	Memory	0.94	0.60	0.89	0.59	0.72	0.58	0.70	0.57	0.69	0.57	1.01	0.60	0.84	0.59	0.75	0.58	0.66	0.57	0.55	0.94	2.30	
Episodic long-term memory (Popov & Dames, 2023)	Memory	1.72	1.27	1.45	1.05	1.33	0.96	1.30	0.94	1.18	0.90	1.97	1.28	1.65	1.02	1.43	0.92	1.13	0.89	0.87	—	0.69	
N-back (Bakshi et al., 2019)	Memory	0.58	0.42	0.58	0.42	0.55	0.41	0.55	0.41	0.53	0.40	0.59	0.42	0.57	0.42	0.57	0.41	0.52	0.40	0.40	0.58	0.69	
Recall and recognition (Cox et al., 2018)	Memory	1.73	1.24	1.57	1.18	1.44	1.13	1.41	1.12	1.36	1.09	1.87	1.22	1.56	1.14	1.37	1.10	1.38	1.07	1.06	—	—	
Recent probes (Bakshi et al., 2019)	Memory	0.70	0.26	0.61	0.26	0.39	0.26	0.41	0.26	0.35	0.26	0.74	0.27	0.56	0.26	0.34	0.26	0.34	0.26	0.26	0.39	0.69	
		0.40–0.60		0.60–0.80		0.80–1.00		1.00–1.20		1.20–1.50		1.50–2.00		>2.00									
		0.20–0.40		0.40–0.60		0.60–0.80		0.80–1.00		1.00–1.20		1.20–1.50		1.50–2.00									
		<0.20		0.20–0.40		0.40–0.60		0.60–0.80		0.80–1.00		1.00–1.20		1.20–1.50		1.50–2.00							

Bold-underline = best; **underline** = second-best (C_p excluded; see footnote). NLL under bf16 (half-precision) inference. Reference columns report 4-bit values as originally published. Lower is better. “—” = no cognitive baseline reported. Size labels in billions of parameters. Subscript _r denotes our reproducing evaluation of the original Centaur model under identical library and CUDA versions as our models; _p denotes values published by Binz et al. (2025). C_p is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. [†]Horizon task (5 experiments) and Two-step task (3 experiments) are reported as single merged values by Binz et al. (2025). Rounded to 2 d.p.

Table 6: **Base vs. finetuned NLL on Psych-101 under bf16 inference.** Within each model size, the left column (base) shows the pretrained model and the right column (ft) shows the cognitively finetuned variant. Cell colour encodes performance (expressed as NLL); the consistent colour shift from base to ft demonstrates the effect of cognitive finetuning across all scales.

Experiment	Type	Qwen3 family												Llama-3.1/3.2 family						Binz et al. (2025)						Chance							
		0.6B			1.7B			4B			8B			14B			1B			3B			8B				C _p	C _r	L-70B _p	C _p	C _r	Cog _p	ln(k)
		base	ft	base	base	ft	base	base	ft	base	base	ft	base	base	ft	base	base	ft	base	base	ft	base	base	ft	base								
Grammar judgement (Jansen et al., 2021)	Misc.	2.12	1.47	2.02	1.45	2.00	1.45	1.89	1.44	1.91	1.43	2.33	1.48	2.16	1.45	2.12	1.44	1.99	1.44	1.44	1.44	1.41	—	—	—	—	—	—	—	—	—	—	—
Probabilistic reasoning (Zhu et al., 2020)	Misc.	2.75	2.45	2.69	2.40	2.61	2.36	2.57	2.35	2.58	2.34	2.89	2.56	2.80	2.47	2.71	2.43	2.64	2.40	2.37	—	—	—	—	—	—	—	—	—	—	—	—	—
Serial reaction time task (Wu et al., 2023)	Misc.	0.21	0.17	0.19	0.17	0.19	0.17	0.19	0.17	0.19	0.17	0.31	0.17	0.23	0.17	0.20	0.17	0.19	0.18	0.17	0.20	0.18	0.20	1.39	1.39	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90
THINGS odd-one-out (Hart et al., 2023)	Misc.	1.21	0.82	1.21	0.81	1.17	0.80	1.15	0.79	1.13	0.79	1.31	0.82	1.18	0.82	1.16	0.79	1.14	0.80	0.81	0.83	1.10	1.10	1.10	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90
Changing bandit (Xiong et al., 2023)	Bandit	0.42	0.30	0.38	0.30	0.36	0.29	0.35	0.29	0.34	0.29	1.34	0.32	0.90	0.31	0.39	0.29	0.38	0.45	0.37	0.54	—	—	—	—	—	—	—	—	—	—	—	—
Decisions from experience (Wulff et al., 2018)	Bandit	0.74	0.49	0.66	0.49	0.69	0.49	0.78	0.49	0.61	0.48	0.81	0.50	0.71	0.49	0.65	0.48	0.43	0.47	0.37	0.54	—	—	—	—	—	—	—	—	—	—	—	—
Drifting four-armed bandit (Badrinarayan et al., 2020)	Bandit	0.98	0.73	0.95	0.72	0.92	0.71	0.91	0.71	0.91	0.71	0.95	0.73	0.94	0.72	0.90	0.71	0.88	0.71	0.70	0.90	1.39	1.39	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90
Horizon task (Feng et al., 2021)	Bandit	0.47	0.23	0.43	0.23	0.39	0.22	0.36	0.22	0.34	0.22	0.81	0.26	0.67	0.24	0.44	0.22	0.52	0.40	0.40	0.36	0.69	0.69	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.69	
Horizon task (Sadeghiyeh et al., 2020)	Bandit	0.67	0.59	0.72	0.58	0.66	0.58	0.65	0.58	0.65	0.58	0.67	0.59	0.68	0.58	0.66	0.58	0.52	0.58	0.40	0.36	0.69	0.69	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.69	
Horizon task (Somerville et al., 2017)	Bandit	0.54	0.34	0.55	0.34	0.53	0.34	0.47	0.34	0.49	0.33	0.59	0.35	0.57	0.34	0.51	0.34	0.52	0.35	0.40	0.36	0.69	0.69	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.69	
Horizon task (Wulff et al., 2020)	Bandit	0.33	0.15	0.33	0.15	0.31	0.15	0.27	0.15	0.24	0.15	0.37	0.16	0.37	0.16	0.30	0.14	0.52	0.15	0.40	0.36	0.69	0.69	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.69	
Horizon task (Wilson et al., 2014)	Bandit	0.57	0.46	0.58	0.46	0.56	0.45	0.53	0.45	0.54	0.45	0.64	0.47	0.62	0.46	0.55	0.45	0.52	0.48	0.40	0.36	0.69	0.69	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.69	
Iowa gambling task (Stingrover et al., 2015)	Bandit	1.07	0.93	1.06	0.93	1.07	0.91	1.07	0.91	1.07	0.91	1.10	0.94	1.06	0.92	1.04	0.91	0.99	0.91	0.89	1.16	1.39	1.39	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.89	1.39	
Prob. instrumental learning (Lefebvre et al., 2017)	Bandit	0.62	0.50	0.60	0.50	0.57	0.49	0.59	0.49	0.56	0.49	0.62	0.51	0.58	0.50	0.56	0.49	0.54	0.50	0.49	0.50	0.69	0.69	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.69	
Spatially correlated MAB (Wu et al., 2018)	Bandit	2.61	2.01	2.55	1.96	2.50	1.93	2.44	1.87	2.46	1.86	2.76	2.16	2.64	2.00	2.58	1.93	2.45	1.82	1.83	2.76	3.40	3.40	1.83	1.83	1.83	1.83	1.83	1.83	1.83	1.83	3.40	
Structured bandit (Schule et al., 2020)	Bandit	0.92	0.67	0.92	0.66	0.88	0.65	0.87	0.65	0.83	0.64	0.94	0.67	0.89	0.66	0.85	0.65	0.81	0.64	0.64	1.05	2.08	2.08	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	2.08	
Two-armed bandit (Gershman, 2018)	Bandit	0.43	0.30	0.41	0.30	0.40	0.29	0.39	0.29	0.39	0.29	0.43	0.30	0.43	0.29	0.40	0.29	0.38	0.30	0.30	0.42	—	—	—	—	—	—	—	—	—	—	—	—
Aversive learning (Wise et al., 2019)	Sup. learn.	5.20	4.71	5.06	4.55	4.95	4.27	4.89	4.24	4.75	3.81	5.66	4.58	5.52	4.80	5.25	3.66	5.11	4.13	4.07	—	—	—	—	—	—	—	—	—	—	—	—	—
Median categorization (Levering et al., 2020)	Sup. learn.	0.60	0.51	0.60	0.51	0.60	0.50	0.59	0.50	0.55	0.50	0.71	0.52	0.60	0.50	0.57	0.50	0.58	0.50	0.50	0.53	—	—	—	—	—	—	—	—	—	—	—	—
Multiple-cue judgment (Ciliberto et al., 2023)	Sup. learn.	1.46	1.17	1.38	1.16	1.28	1.14	1.30	1.14	1.26	1.13	1.69	1.17	1.50	1.15	1.32	1.12	1.28	1.14	1.12	1.92	2.20	2.20	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12	2.20	
Shepard categorization (Radham et al., 2017)	Sup. learn.	0.63	0.55	0.61	0.55	0.59	0.54	0.58	0.53	0.59	0.53	0.67	0.56	0.62	0.54	0.60	0.53	0.58	0.54	0.54	0.61	0.69	0.69	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.69	
Weather prediction task (Spekman et al., 2008)	Sup. learn.	0.62	0.55	0.59	0.56	0.61	0.54	0.58	0.55	0.58	0.55	0.62	0.55	0.59	0.55	0.59	0.54	0.57	0.56	0.55	0.63	0.69	0.69	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.69	
Mean (all 48)		0.99	0.69	0.94	0.68	0.90	0.66	0.89	0.66	0.87	0.64	1.14	0.70	1.02	0.68	0.92	0.64	0.88	0.71	0.69	0.69	0.98	0.98	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.98	

Bold-underline = best; **underline** = second-best (C_p excluded; see footnote). NLL under bf16 (half-precision) inference. Reference columns report 4-bit values as originally published. Lower is better. "—" = no cognitive baseline reported. Size labels in billions of parameters. Subscript _p denotes our reproducing evaluation of the original Centaur model under identical library and CUDA versions as our models; _r denotes values published by Binz et al. (2025). C_p is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. [†]Horizon task (5 experiments) and Two-step task (3 experiments) are reported as single merged values by Binz et al. (2025). Rounded to 2 d.p.

Table 6: **Base vs. finetuned NLL on Psych-101 under bf16 inference (continued).** Within each model size, the left column (base) shows the pretrained model and the right column (ft) the cognitively finetuned variant.

Experiment	Type	Qwen-taur				Llama-Centaur				Smo1-taur				O1-motaur				Binz et al. (2025)				Chance
		0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	L-70B _p	C _r	C _p	Cog-p	ln(k)		
Balloon analog risk task (Frey et al., 2017)	Decision	0.07	0.07	0.07	0.06	0.06	0.07	0.07	0.06	0.08	0.07	0.07	0.07	0.07	0.07	0.08	0.07	0.06	0.09	0.69		
CPC18 (Plesky et al., 2018)	Decision	0.35	0.34	0.34	0.34	0.33	0.36	0.35	0.34	0.42	0.41	0.40	0.34	0.44	0.34	0.41	0.35	0.34	0.66	0.69		
Columbia card task (Frey et al., 2017)	Decision	0.20	0.20	0.20	0.20	0.21	0.20	0.20	0.20	0.24	0.22	0.21	0.20	0.22	0.20	0.23	0.21	0.19	0.26	0.69		
Decisions from description (Wulff et al., 2018)	Decision	0.62	0.63	0.62	0.61	0.60	0.62	0.63	0.60	0.68	0.66	0.64	0.61	0.63	0.62	0.76	0.59	0.53	0.61	0.69		
Experiential-symbolic task (García et al., 2023)	Decision	0.48	0.47	0.46	0.46	0.46	0.47	0.47	0.46	0.47	0.45	0.43	0.47	0.47	0.47	0.70	0.46	0.45	—	—		
Gardening task (Fleisch et al., 2018)	Decision	0.39	0.39	0.38	0.38	0.38	0.40	0.39	0.38	0.57	0.54	0.53	0.38	0.58	0.38	0.50	0.49	0.38	0.91	0.69		
Go/no-go (Einhavi et al., 2019)	Decision	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.08	0.00		
Intertemporal choice (Ruggert et al., 2022)	Decision	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.47	0.45	0.44	0.44	0.44	0.44	0.73	0.44	0.43	0.66	0.69		
Multi-attribute DM (Hilbig & Moshagen, 2014)	Decision	0.09	0.08	0.09	0.08	0.08	0.08	0.08	0.08	0.31	0.19	0.16	0.08	0.10	0.09	0.15	0.06	0.06	0.19	0.69		
Risky choice (Kneuper et al., 2024)	Decision	0.48	0.47	0.44	0.43	0.42	0.49	0.46	0.43	0.72	0.61	0.53	0.45	0.58	0.46	0.65	0.43	0.43	—	—		
choices13k (Peeters et al., 2021)	Decision	0.44	0.44	0.43	0.43	0.43	0.44	0.44	0.43	0.49	0.46	0.44	0.44	0.44	0.45	0.53	0.43	0.43	0.66	0.69		
Multi-task RL (Tomov et al., 2021)	MDP	0.60	0.59	0.58	0.58	0.57	0.60	0.59	0.57	0.68	0.66	0.63	0.59	0.78	0.59	0.66	0.57	0.57	1.04	1.10		
Title-revealing task (Kumar et al., 2023)	MDP	0.02	0.01	0.00	0.07	0.00	0.02	0.00	0.00	2.69	1.99	0.04	0.00	0.06	0.09	2.74	2.29	1.87	—	—		
Two-step task (Kool et al., 2017)	MDP	0.54	0.53	0.53	0.53	0.53	0.55	0.53	0.53	0.60	0.60	0.56	0.53	0.57	0.53	0.61	0.53	0.50 [†]	0.60 [†]	0.69		
Two-step task (Kool et al., 2016)	MDP	0.51	0.50	0.49	0.49	0.48	0.52	0.50	0.49	0.59	0.58	0.52	0.50	0.55	0.50	0.61	0.48	0.50 [†]	0.60 [†]	0.69		
Two-step task (Zorowitz & Niv, 2023)	MDP	0.52	0.51	0.51	0.51	0.51	0.52	0.51	0.51	0.60	0.58	0.52	0.52	0.54	0.51	0.61	0.51	0.50 [†]	0.60 [†]	0.69		
Virtual subway network (Tomov et al., 2020)	MDP	1.16	1.15	1.14	1.13	1.12	1.19	1.16	1.13	1.43	1.33	1.30	1.14	1.32	1.15	1.53	1.16	1.13	—	1.61		
Zoopermarket (Ludwig et al., 2023)	MDP	0.54	0.52	0.51	0.50	0.48	0.54	0.52	0.50	0.63	0.61	0.56	0.51	0.61	0.52	0.60	0.50	0.48	0.60	0.69		
Cond. assoc. learning (Collins et al., 2014)	Memory	0.54	0.53	0.52	0.52	0.51	0.55	0.53	0.52	0.74	0.64	0.58	0.53	0.65	0.53	0.64	0.52	0.54	0.86	1.10		
Digit span (Einhavi et al., 2019)	Memory	0.60	0.59	0.58	0.57	0.57	0.60	0.59	0.58	0.71	0.65	0.62	0.59	0.62	0.59	0.66	0.57	0.55	0.94	2.30		
Episodic long-term memory (Popov & Dawes, 2023)	Memory	1.27	1.05	0.96	0.94	0.90	1.28	1.02	0.92	1.88	1.56	1.37	1.01	1.23	1.05	1.13	0.89	0.87	—	0.69		
N-back (Einhavi et al., 2019)	Memory	0.42	0.42	0.41	0.41	0.40	0.42	0.42	0.41	0.49	0.47	0.44	0.41	0.44	0.42	0.52	0.40	0.40	0.58	0.69		
Recall and recognition (Cox et al., 2018)	Memory	1.24	1.18	1.13	1.12	1.09	1.22	1.14	1.10	1.69	1.50	1.25	1.15	1.41	1.17	1.38	1.07	1.06	—	—		
Recent probes (Einhavi et al., 2019)	Memory	0.26	0.26	0.26	0.26	0.26	0.27	0.26	0.26	0.33	0.28	0.27	0.26	0.27	0.26	0.34	0.26	0.26	0.39	0.69		

<0.20
0.20–0.40
0.40–0.60
0.60–0.80
0.80–1.00
1.00–1.20
1.20–1.50
1.50–2.00
>2.00

Bold+underline = best; underline = second-best (C_p excluded; see below). Lower is better. “—” = not available. All models finetuned at LoRA rank 16, bf16 inference. Reference columns (Binz et al. (2025)): L-70B_p = Llama-3.1-70B base, C_r = our reproduced Centaur-70B, C_p = published Centaur-70B, Cog_p = published Centaur-70B, C_p is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. [†]Horizon and Two-step tasks are reported as single merged values by Binz et al. (2025). Rounded to 2d.p.

Table 7: Per-experiment NLL on Psych-101 for all four finetuned families (bf16, LoRA rank 16). Cell colour encodes performance (green = better).

Experiment	Type	Qwentaaur				Llama-Centaur				Smoltaur				Olmotaur				Binz et al. (2025)				Chance
		0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	L-70B _p	C _r	C _p	Cog _p	ln(k)		
Grammar judgement (Jensen et al., 2021)	Misc.	1.47	1.45	1.45	1.44	1.43	1.48	1.45	1.44	1.63	1.54	1.51	1.46	1.51	1.46	1.99	1.44	1.44	1.41	—		
Probabilistic reasoning (Zhu et al., 2020)	Misc.	2.45	2.40	2.36	2.35	2.34	2.56	2.47	2.43	2.67	2.60	2.50	2.45	2.61	2.47	2.64	2.40	2.37	—	—		
Serial reaction time task (Wu et al., 2023)	Misc.	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.16	0.16	0.17	0.16	0.17	0.19	0.18	0.17	0.20	1.39		
THINGS odd-one-out (Fehrer et al., 2023)	Misc.	0.82	0.81	0.80	0.79	0.79	0.82	0.82	0.79	1.08	0.90	0.86	0.80	0.86	0.81	1.14	0.80	0.81	0.83	1.10		
Changing bandit (Xiong et al., 2023)	Bandit	0.30	0.30	0.29	0.29	0.29	0.32	0.31	0.29	0.40	0.38	0.35	0.30	0.39	0.30	0.38	0.45	0.30	0.44	0.69		
Decisions from experience (Wulff et al., 2018)	Bandit	0.49	0.49	0.49	0.48	0.50	0.49	0.48	0.54	0.52	0.50	0.49	0.50	0.49	0.50	0.43	0.47	0.37	0.54	—		
Drifting four-armed bandit (Bahrami et al., 2020)	Bandit	0.73	0.72	0.71	0.71	0.72	0.73	0.72	0.71	0.83	0.78	0.74	0.72	0.74	0.72	0.88	0.71	0.70	0.90	1.39		
Horizon task (Feng et al., 2021)	Bandit	0.23	0.23	0.22	0.22	0.22	0.26	0.24	0.22	0.40	0.34	0.30	0.22	0.35	0.22	0.52	0.40	0.40 [†]	0.36 [†]	0.69		
Horizon task (Sadeghiyeh et al., 2020)	Bandit	0.59	0.58	0.58	0.58	0.58	0.59	0.58	0.58	0.62	0.62	0.60	0.58	0.61	0.58	0.52	0.58	0.40 [†]	0.36 [†]	0.69		
Horizon task (Samerville et al., 2017)	Bandit	0.34	0.34	0.34	0.34	0.33	0.35	0.34	0.34	0.48	0.40	0.38	0.34	0.40	0.34	0.52	0.35	0.40 [†]	0.36 [†]	0.69		
Horizon task (Walsh et al., 2020)	Bandit	0.15	0.15	0.15	0.15	0.15	0.16	0.16	0.14	0.26	0.21	0.18	0.15	0.19	0.15	0.52	0.15	0.40 [†]	0.36 [†]	0.69		
Horizon task (Wilson et al., 2014)	Bandit	0.46	0.46	0.45	0.45	0.45	0.47	0.46	0.45	0.54	0.51	0.49	0.45	0.52	0.45	0.52	0.48	0.40 [†]	0.36 [†]	0.69		
Iowa gambling task (Sadagheer et al., 2015)	Bandit	0.93	0.93	0.91	0.91	0.91	0.94	0.92	0.91	1.01	0.98	0.94	0.92	0.95	0.92	0.99	0.91	0.89	1.16	1.39		
Prob. instrumental learning (Lefebvre et al., 2017)	Bandit	0.50	0.50	0.49	0.49	0.49	0.51	0.50	0.49	0.54	0.53	0.51	0.50	0.51	0.50	0.54	0.50	0.49	0.50	0.69		
Spatially correlated MAB (Wu et al., 2018)	Bandit	2.01	1.96	1.93	1.87	1.86	2.16	2.00	1.93	2.50	2.30	2.16	1.95	2.26	2.06	2.45	1.82	1.83	2.76	3.40		
Structured bandit (Schulz et al., 2020)	Bandit	0.67	0.66	0.65	0.65	0.64	0.67	0.66	0.65	0.77	0.71	0.68	0.65	0.73	0.66	0.81	0.64	0.64	1.05	2.08		
Two-armed bandit (Gershman, 2018)	Bandit	0.30	0.30	0.29	0.29	0.29	0.30	0.29	0.29	0.37	0.33	0.31	0.29	0.31	0.30	0.38	0.30	0.30	0.42	0.69		
Aversive learning (Wise et al., 2019)	Sup. learn.	4.71	4.55	4.27	4.24	3.81	4.58	4.80	3.66	5.35	5.12	4.92	3.93	5.44	4.67	5.11	4.13	4.07	—	—		
Median categorization (Levering et al., 2020)	Sup. learn.	0.51	0.51	0.50	0.50	0.50	0.52	0.50	0.50	0.56	0.54	0.52	0.50	0.52	0.51	0.58	0.50	0.50	0.53	—		
Multiple-cue judgment (Caldazzo et al., 2023)	Sup. learn.	1.17	1.16	1.14	1.14	1.13	1.17	1.15	1.12	1.59	1.49	1.40	1.14	1.67	1.15	1.28	1.14	1.12	1.92	2.20		
Shepard categorization (Radhum et al., 2017)	Sup. learn.	0.55	0.55	0.54	0.53	0.53	0.56	0.54	0.53	0.62	0.58	0.56	0.54	0.56	0.54	0.58	0.54	0.54	0.61	0.69		
Weather prediction task (Speckenbrink et al., 2008)	Sup. learn.	0.55	0.56	0.54	0.55	0.55	0.55	0.54	0.56	0.58	0.56	0.56	0.55	0.57	0.53	0.57	0.56	0.55	0.63	0.69		
Mean		0.69	0.68	0.66	0.66	0.64	0.70	0.68	0.64	0.89	0.82	0.73	0.66	0.77	0.68	0.88	0.71	0.69	0.69	0.98		

Legend: <0.20 (green), 0.20–0.40 (light green), 0.40–0.60 (yellow), 0.60–0.80 (orange), 0.80–1.00 (light orange), 1.00–1.20 (red), 1.20–1.50 (dark red), 1.50–2.00 (purple), >2.00 (dark purple).

Bold+underline = best; underline = second-best (C_p excluded; see below). Lower is better. “—” = not available. All models finetuned at LoRA rank 16, bf16 inference. Reference columns (Binz et al. (2025)): L-70B_p = Llama-3.1-70B base, C_r = our reproduced Centaur-70B, C_p = published Centaur-70B, Cog_p = domain-specific cognitive baseline. C_p is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. [†]Horizon and Two-step tasks are reported as single merged values by Binz et al. (2025). Rounded to 2 d.p.

Table 7: Per-experiment NLL on Psych-101 for all four finetuned families (bf16, LoRA rank 16), continued. Cell colour encodes performance (green = better).

Experiment	Type	Qwen 2.5/3 family						Llama 3/3.1/3.2 family						Binz et al. (2025)		Chance					
		Base			Non-cog.			Llama-Centaur			Other cog.			Non-cog.							
		8B	14B	8B	14B	8B	14B	8B	14B	3B	8B	FM-8B	Socr.	-8B	Herm.		-4B	Herm.	-8B	C-70B _r	Cog- <i>p</i>
Balloon analog risk task (Frey et al., 2017)	Decision	0.08	0.08	0.06	0.06	0.08	0.08	0.07	0.06	0.10	0.10	13.52	0.09	0.17	0.09	0.18	0.07	0.09	0.09	0.69	0.69
CPC18 (Pionovsky et al., 2018)	Decision	0.40	0.40	0.34	0.33	0.40	0.41	0.39	0.35	0.34	0.47	14.34	0.44	0.95	0.42	0.64	0.35	0.66	0.66	0.69	0.69
Columbia card task (Frey et al., 2017)	Decision	0.25	0.25	0.20	0.20	0.26	0.27	0.25	0.20	0.20	0.26	14.07	0.27	0.48	0.26	0.49	0.21	0.26	0.26	0.69	0.69
Decisions from description (Wulff et al., 2018)	Decision	0.99	1.01	0.61	0.60	1.02	1.20	0.96	0.63	0.60	1.28	13.59	1.34	2.99	1.17	4.44	0.59	0.61	0.61	0.69	0.69
Exponential-symbolic task (Garcia et al., 2023)	Decision	0.64	0.57	0.46	0.46	0.58	0.69	0.64	0.47	0.46	0.73	12.95	0.79	1.37	0.67	0.95	0.46	—	—	—	—
Gardening task (Fisch et al., 2018)	Decision	0.47	0.45	0.38	0.38	0.45	0.64	0.48	0.39	0.38	0.48	16.26	0.53	0.61	0.46	0.64	0.49	0.91	0.69	0.69	0.69
Go/no-go (Einhorn et al., 2019)	Decision	0.01	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01	12.55	0.01	0.04	0.00	0.01	0.00	0.08	0.00	0.00	0.00
Intertemporal choice (Ruggert et al., 2022)	Decision	0.69	0.70	0.44	0.44	0.74	0.75	0.76	0.44	0.44	0.76	13.82	0.72	2.75	1.04	2.72	0.44	0.66	0.69	0.69	0.69
Multi-attribute DM (Hilbig & Moshagen, 2014)	Decision	0.19	0.19	0.08	0.08	0.20	0.45	0.23	0.08	0.08	0.30	13.86	0.45	0.61	0.23	0.54	0.06	0.19	0.19	0.69	0.69
Risky choice (Krugger et al., 2024)	Decision	0.74	0.67	0.43	0.42	0.67	0.85	0.76	0.46	0.43	0.85	16.10	0.94	1.79	0.82	1.35	0.43	—	—	—	—
choices13k (Peterson et al., 2021)	Decision	0.59	0.55	0.43	0.43	0.53	0.59	0.55	0.44	0.43	0.68	13.44	0.65	1.65	0.60	1.13	0.43	0.66	0.69	0.69	0.69
Multi-task RL (Tomov et al., 2021)	MDP	0.70	0.68	0.58	0.57	0.66	0.70	0.66	0.59	0.57	0.82	14.70	0.81	1.20	0.70	1.06	0.57	1.04	1.10	1.10	1.10
Tile-revealing task (Kumar et al., 2023)	MDP	2.73	2.60	0.07	0.00	2.62	3.98	2.92	0.00	0.00	3.03	65.75	2.93	3.48	2.85	3.37	2.29	—	—	—	—
Two-step task (Kool et al., 2017)	MDP	0.65	0.63	0.53	0.53	0.63	0.62	0.61	0.53	0.53	0.71	14.06	0.67	1.17	0.67	0.81	0.53	0.60	0.69	0.69	0.69
Two-step task (Kool et al., 2016)	MDP	0.65	0.63	0.49	0.48	0.63	0.62	0.61	0.50	0.49	0.72	14.63	0.68	1.15	0.67	0.82	0.48	0.60	0.69	0.69	0.69
Two-step task (Zwolsow & Niv, 2023)	MDP	0.61	0.59	0.51	0.51	0.58	0.60	0.58	0.51	0.51	0.65	13.18	0.63	1.03	0.62	0.82	0.51	0.60	0.69	0.69	0.69
Virtual subway network (Tomov et al., 2020)	MDP	1.55	1.57	1.13	1.12	1.60	1.60	1.53	1.16	1.13	1.61	18.21	1.69	2.41	1.64	2.10	1.16	—	1.61	1.61	1.61
Zoopermarket (Ludwig et al., 2023)	MDP	0.60	0.59	0.50	0.48	0.59	0.62	0.60	0.52	0.50	0.62	13.75	0.65	1.49	0.63	1.09	0.50	0.60	0.69	0.69	0.69
Cond. assoc. learning (Collins et al., 2014)	Memory	0.69	0.61	0.52	0.51	0.59	0.73	0.70	0.53	0.52	0.72	16.16	0.80	0.92	0.68	0.95	0.52	0.86	1.10	1.10	1.10
Digit span (Einhorn et al., 2019)	Memory	0.70	0.69	0.57	0.57	0.70	0.84	0.75	0.59	0.58	0.83	4.65	0.87	2.16	0.83	1.02	0.57	0.94	2.30	2.30	2.30
Episodic long-term memory (Papow & Dames, 2023)	Memory	1.30	1.18	0.94	0.90	1.21	1.65	1.43	1.02	0.92	1.52	17.12	1.82	3.07	1.72	2.22	0.89	—	0.69	0.69	0.69
N-back (Einhorn et al., 2019)	Memory	0.55	0.53	0.41	0.40	0.53	0.57	0.57	0.42	0.41	0.57	14.42	0.59	0.79	0.57	0.75	0.40	0.58	0.69	0.69	0.69
Recall and recognition (Cox et al., 2018)	Memory	1.41	1.36	1.12	1.09	1.38	1.56	1.37	1.14	1.10	1.53	24.31	1.65	4.16	1.61	2.60	1.07	—	—	—	—
Recent probes (Einhorn et al., 2019)	Memory	0.41	0.35	0.26	0.26	0.34	0.56	0.34	0.26	0.26	0.40	14.16	0.61	1.00	0.39	1.06	0.26	0.39	0.69	0.69	0.69
		<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 1.00–1.20 1.20–1.50 1.50–2.00 >2.00																			

Base+underline = best; **underline** = second-best. Lower is better. “—” = not available. Models are grouped by base-model family. “Base” = pretrained model without finetuning (bfl6); “Other cog.” = other cognitive/behavioural foundation models finetuned on non-Psych-101 behavioural data; “Non-cog.” = models finetuned for general instruction-following or reasoning. Herm. = Hermes (Teknium et al., 2024); Nemo. = Nemotron (Bercovich et al., 2025); Socr. = Socrates (Kolluri et al., 2025); Be.FM = Xie et al. (2025). Base model versions differ within families: Qwen2.5; Herm. -14B uses Qwen2.5; Llama-Centaur uses Llama-3.1/3.2; Socr. -8B uses Llama-3; Be.FM uses Llama-3.1.

Table 8: Comparison of cognitively finetuned models with other behavioural foundation models and non-cognitive controls on Psych-101, grouped by base-model family. Within each family, Qwen2.5/Llama-Centaur (cognitive SFT on Psych-101) consistently outperform both other behavioural finetuned models and non-cognitive controls, demonstrating that alignment with human behavioural patterns requires cognitive finetuning on structured experimental data, not merely finetuning per se.

Qwen 2.5/3 family													Llama 3/3.1/3.2 family										Binz et al. (2025)		Chance
Experiment	Type	Base			Non-cog.			Llama-Centaur			Other cog.			Non-cog.						Cog- p	C-70B $_r$	ln(k)	Chance		
		8B	14B	8B	14B	Socr.	-14B	Herm.	-14B	8B	3B	8B	FM-8B	Socr.	-8B	Herm.	-8B	Nemo.	-8B					Nemo.	-8B
Grammar judgement (Jansen et al., 2021)	Misc.	1.89	1.91	1.44	1.43	10.37	1.94	2.16	2.12	1.45	1.44	2.11	12.88	2.27	3.90	2.53	3.12	1.44	1.41	—	—				
Probabilistic reasoning (Zhu et al., 2020)	Misc.	2.57	2.58	2.35	2.34	2.92	2.59	2.80	2.71	2.47	2.43	2.84	3.40	2.98	4.05	2.97	3.23	2.40	—	—	—				
Serial reaction time task (Wu et al., 2023)	Misc.	0.19	0.19	0.17	0.17	5.09	0.19	0.23	0.20	0.17	0.17	0.25	14.43	0.21	0.25	0.20	0.26	0.18	0.20	1.39	1.39				
THINGS odd-one-out (Hebart et al., 2023)	Misc.	1.15	1.13	0.79	0.79	12.15	1.15	1.18	1.16	0.82	0.79	1.19	14.31	1.24	2.13	1.28	1.75	0.80	0.83	1.10	1.10				
Changing bandit (Xiong et al., 2023)	Bandit	0.35	0.34	0.29	0.29	7.52	0.34	0.90	0.39	0.31	0.29	0.42	14.76	0.39	0.78	0.36	0.53	0.45	0.44	0.69	0.69				
Decisions from experience (Waldf et al., 2018)	Bandit	0.78	0.61	0.49	0.48	7.18	0.64	0.71	0.65	0.49	0.48	0.83	13.74	0.69	2.29	0.88	1.78	0.47	0.54	—	—				
Drifting four-armed bandit (Bahmani et al., 2020)	Bandit	0.91	0.91	0.71	0.71	8.43	0.90	0.94	0.90	0.72	0.71	0.99	13.91	0.99	2.10	0.96	1.78	0.71	0.90	1.39	1.39				
Horizon task (Feng et al., 2021)	Bandit	0.36	0.34	0.22	0.22	7.55	0.35	0.67	0.44	0.24	0.22	0.39	15.10	0.43	0.65	0.36	0.62	0.40	0.36	0.69	0.69				
Horizon task (Sadeghiyeh et al., 2020)	Bandit	0.65	0.65	0.58	0.58	8.62	0.66	0.68	0.66	0.58	0.58	0.69	14.94	0.73	1.10	0.69	0.93	0.58	0.36	0.69	0.69				
Horizon task (Sonerville et al., 2017)	Bandit	0.47	0.49	0.34	0.33	8.49	0.49	0.57	0.51	0.34	0.34	0.53	15.18	0.57	0.86	0.47	0.74	0.35	0.36	0.69	0.69				
Horizon task (Wolfe et al., 2020)	Bandit	0.27	0.24	0.15	0.15	7.82	0.26	0.37	0.30	0.16	0.14	0.33	15.42	0.35	0.54	0.26	0.54	0.15	0.36	0.69	0.69				
Horizon task (Wilson et al., 2014)	Bandit	0.53	0.54	0.45	0.45	8.06	0.54	0.62	0.55	0.46	0.45	0.57	15.39	0.62	0.89	0.56	0.79	0.48	0.36	0.69	0.69				
Iowa gambling task (Steingrover et al., 2015)	Bandit	1.07	1.07	0.91	0.91	8.45	1.06	1.06	1.04	0.92	0.91	1.11	13.23	1.13	2.27	1.09	1.60	0.91	1.16	1.39	1.39				
Prob. instrumental learning (Lefebvre et al., 2017)	Bandit	0.59	0.56	0.49	0.49	8.30	0.53	0.58	0.56	0.50	0.49	0.61	15.92	0.68	1.16	0.54	0.86	0.50	0.50	0.69	0.69				
Spatially correlated MAB (Wu et al., 2018)	Bandit	2.44	2.46	1.87	1.86	3.14	2.48	2.64	2.58	2.00	1.93	2.82	5.43	2.64	3.93	2.78	3.19	1.82	2.76	3.40	3.40				
Structured bandit (Schulz et al., 2020)	Bandit	0.87	0.83	0.65	0.64	1.57	0.80	0.89	0.85	0.66	0.65	0.98	2.12	0.91	1.41	0.90	1.08	0.64	1.05	2.08	2.08				
Two-armed bandit (Gershman, 2018)	Bandit	0.39	0.39	0.29	0.29	7.30	0.41	0.43	0.40	0.29	0.29	0.46	13.99	0.47	0.98	0.44	0.73	0.30	0.42	0.69	0.69				
Aversive learning (Wise et al., 2019)	Sup. learn.	4.89	4.75	4.24	3.81	7.19	4.78	5.52	5.25	4.80	3.66	5.49	18.42	5.52	7.03	5.33	6.14	4.13	—	—	—				
Median categorization (Levinger et al., 2020)	Sup. learn.	0.59	0.55	0.50	0.50	11.00	0.56	0.60	0.57	0.50	0.50	0.63	13.53	0.63	1.15	0.62	0.90	0.50	0.53	—	—				
Multiple-cue judgment (Collobi et al., 2023)	Sup. learn.	1.30	1.26	1.14	1.13	6.92	1.27	1.50	1.32	1.15	1.12	1.38	18.06	1.51	2.56	1.37	1.85	1.14	1.92	2.20	2.20				
Shepard categorization (Radman et al., 2017)	Sup. learn.	0.58	0.59	0.53	0.53	10.13	0.59	0.62	0.60	0.54	0.53	0.60	12.66	0.64	0.92	0.63	0.76	0.54	0.61	0.69	0.69				
Weather prediction task (Spekter et al., 2008)	Sup. learn.	0.58	0.58	0.55	0.55	10.65	0.56	0.59	0.58	0.54	0.56	0.62	17.01	0.63	0.89	0.61	0.77	0.56	0.63	0.69	0.69				
Mean (all 46)		0.89	0.87	0.66	0.64	8.65	0.87	1.02	0.92	0.68	0.64	0.99	15.07	1.02	1.72	0.98	1.43	0.71	0.69	0.98	0.98				

Bold+underline = best; underline = second-best. Lower is better. “—” = not available. Models are grouped by base-model family. “Base” = pretrained model without finetuning (bfl6); “Other cog.” = other cognitive/behavioural foundation models finetuned on non-Psych-101 behavioural data; “Non-cog.” = models finetuned for general instruction-following or reasoning. Herm. = Hermes (Teknium et al., 2024); Nemo. = Nemotron (Bercovich et al., 2025); Socr. = Socrates (Kolluri et al., 2025); Be. FM = Xie et al. (2025). Base model versions differ within families: Qwenaur uses Qwen3; Socr. -14B uses Qwen2.5; Herm. -14B uses Qwen2.5; Llama-Centaur uses Llama-3.1/3.2; Socr. -8B uses Llama-3; Be. FM uses Llama-3.1.

Table 8: Comparison of cognitively finetuned models with other behavioural foundation models and non-cognitive controls on Psych-101, grouped by base-model family (continued).

C.2 Psych-201 (out-of-distribution)

While Psych-101 serves as the in-distribution evaluation, Psych-201 (Binz et al., 2026) provides a stronger out-of-distribution test: models must generalise to entirely unseen experiments that were not part of the Psych-101 training set. Our analysis is based on Psych-201-RT, a subset of the Psych-201 dataset (<https://hf.co/datasets/socius/Psych-201-RT>). Table 9 summarises performance by task type using the normalised metric, restricted to the 15 of 18 experiments with a well-defined discrete response space. Table 10 then reports base versus fine-tuned NLL per experiment for the Qwen3 and Llama families, isolating the effect of cognitive fine-tuning at each scale, and Table 11 gives per-experiment NLL for all four fine-tuned families side by side. Both per-experiment tables cover all 18 experiments, including the 3 excluded from normalisation.

The 18 experiments in Psych-201-RT span four task types: decision-making (3), Markov decision processes (2), multi-armed bandits (5), and miscellaneous (8). Of these, 15 present discrete response options and admit a well-defined chance-level baseline $\ln(k)$.

The majority of included experiments are binary choice tasks ($k = 2$, $\ln(k) \approx 0.693$): the experience-description choice task (Anlló et al., 2024) and the loss aversion task (Spektor et al., 2024) (both decision-making); both two-step tasks (Castro-Rodrigues et al., 2022; Shahar et al., 2019) (Markov decision processes); the range adaptation RL task (Bavard et al., 2021), the explore-exploit task (Fan et al., 2023), the counterfactual learning task (Palminteri et al., 2017), and the contextual reinforcement learning task (Vandendriessche et al., 2023) (all multi-armed bandits); and the Navon task (Busch et al., 2024a), the compound word processing task (Günther et al., 2020), the grammaticality judgement task (Dentella et al., 2023), and the math arithmetic task (Xu & Zhang, 2023) (all miscellaneous).

Several of these experiments use more than two unique response labels across the session (e.g., eight symbol pairs in the counterfactual learning task; Palminteri et al., 2017), but each individual trial presents exactly two options; we assign the per-trial $k = 2$ in these cases.

Three experiments have $k > 2$: the Stroop task (Busch et al., 2024b) (miscellaneous; $k = 3$, $\ln(k) \approx 1.099$) presents three response options per trial; the pragmatic language task (Franke et al., 2024) and the reference game (Franke & Degen, 2016) (both miscellaneous) each present four options per trial drawn from a larger label pool, yielding $k = 4$ ($\ln(k) \approx 1.386$). Note that the two-step task of Castro-Rodrigues et al. (2022) also presents four directional arrows on every trial ($k = 4$, $\ln(k) \approx 1.386$), while the two-step task of Shahar et al. (2019) presents two options per stage ($k = 2$).

Experiment exclusions Of the 18 experiments in Psych-201-RT, 3 are excluded from the normalised analysis (Table 9). These fall into two categories.

Mixed response types. The risky decision and happiness task (Rutledge, 2023) (decision-making) interleaves binary lottery choices with free numeric happiness ratings on a 0–100 scale, producing a mixed response space within a single session for which no single k can be defined.

Continuous responses. The scalar inference task (Tsvilodub et al., 2023) (miscellaneous) asks participants to provide likelihood ratings via an adjustable slider, yielding multiple distinct integer values in a single participant – effectively a continuous response with no well-defined discrete action space. Similarly, the aversive reversal learning task (Zika et al., 2023) (multi-armed bandits) elicits shock probability estimates on a 0–1 scale, producing multiple distinct values in a single participant, confirming that the response space is continuous rather than categorical.

All three experiments retain their raw NLL values in the full per-experiment results (Table 10) but are excluded from any normalised $(\ln k - \text{NLL}) / \ln k$ computations.

	Qwentaaur					Llama-Centaaur			Smoltaaur				Olmotaaur		Base			Binz et al. (2025)
Task type	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	Centaur-70B
Decision (2)	0.30	0.32	<u>0.35</u>	0.34	0.37	0.27	0.31	0.33	0.20	0.21	0.24	0.34	0.21	0.31	0.28	0.29	0.28	0.33
MDP (2)	0.56	0.56	0.58	<u>0.58</u>	0.58	0.48	0.53	0.57	0.39	0.46	0.52	0.57	0.42	0.57	0.51	0.53	0.53	0.50
Bandit (4)	0.32	0.35	0.37	0.36	<u>0.38</u>	0.33	0.33	0.36	0.24	0.23	0.32	0.34	0.28	0.34	0.22	0.25	0.25	0.39
Misc. (7)	0.28	0.31	0.45	0.46	<u>0.49</u>	0.14	0.30	0.40	0.03	0.11	0.18	0.36	0.09	0.35	0.37	0.41	0.29	0.52
Mean (15)	0.33	0.35	0.43	0.43	<u>0.45</u>	0.26	0.34	0.40	0.16	0.20	0.27	0.38	0.20	0.37	0.34	0.37	0.31	0.46
	<0.05	0.05–0.12	0.12–0.18	0.18–0.25	0.25–0.32	0.32–0.40	0.40–0.48	0.48–0.55	≥0.55									

Table 9: Fraction of available information captured above chance, $(\ln k - \text{NLL}) / \ln k$, by task type on Psych-201 (out-of-distribution). A value of 0 indicates chance-level performance; 1 indicates perfect prediction. Restricted to 15 experiments (of 18) with a well-defined discrete response space ($\ln k > 0$); 3 experiments with continuous or mixed responses are excluded. **Bold+underline** marks the best model, underline the second-best.

Experiment	Type	Qwen3 family												Llama-3.1/3.2 family												Binz et al. (2025)		Chance
		0.6B			1.7B			4B			8B			14B			1B			3B			8B			70B	ln(k)	
		base	ft	base	base	ft	base	base	ft	base	base	ft	base	base	ft	base	base	ft	base	base	ft	base	base	ft				
Experience-description choice (Aiollo et al., 2024)	Decision	0.50	0.42	0.49	0.40	0.46	0.39	0.46	0.38	0.45	0.38	0.49	0.44	0.46	0.41	0.46	0.41	0.42	0.69	—								
Risky decision and happiness (Rudezge, 2023)	Decision	1.46	1.38	1.45	1.40	1.43	1.34	1.39	1.31	1.39	1.33	1.57	1.40	1.53	1.37	1.43	1.34	1.32	0.69	—								
Loss aversion task (Spektor et al., 2024)	Decision	0.57	0.54	0.56	0.54	0.54	0.52	0.53	0.54	0.53	0.50	0.66	0.57	0.61	0.55	0.54	0.52	0.51	0.69	—								
Two-step task (Castro-Rodriguez et al., 2022)	MDP	0.26	0.26	0.27	0.28	0.26	0.24	0.26	0.24	0.25	0.25	0.29	0.45	0.95	0.35	0.26	0.25	0.46	1.39	—								
Two-step task (Chahar et al., 2019)	MDP	0.55	0.48	0.53	0.47	0.54	0.47	0.55	0.47	0.53	0.46	0.54	0.49	0.53	0.47	0.53	0.47	0.46	0.69	—								
Range adaptation RL (Bavard et al., 2021)	Bandit	0.56	0.48	0.54	0.48	0.53	0.47	0.52	0.47	0.52	0.47	0.57	0.48	0.52	0.48	0.52	0.46	0.45	0.69	—								
Explore-exploit task (Fan et al., 2023)	Bandit	0.42	0.28	0.40	0.27	0.40	0.26	0.39	0.27	0.40	0.26	0.42	0.28	0.43	0.27	0.39	0.26	0.26	0.69	—								
Counterfactual learning (Palmer et al., 2017)	Bandit	0.64	0.50	0.62	0.49	0.60	0.48	0.61	0.49	0.58	0.47	0.62	0.53	0.58	0.49	0.58	0.48	0.46	0.69	—								
Contextual reinforcement learning (Vandendriessche et al., 2023)	Bandit	0.69	0.61	0.69	0.57	0.61	0.54	0.63	0.55	0.57	0.53	0.69	0.58	0.59	0.61	0.59	0.56	0.51	0.69	—								
Aversive reversal learning (Zika et al., 2023)	Bandit	3.66	3.36	3.70	3.36	3.57	3.29	3.51	3.27	3.47	3.27	4.00	3.99	3.90	3.85	3.87	3.82	3.80	—	—								
Navon task (Busch et al., 2024a)	Misc.	0.43	0.40	0.40	0.35	0.43	0.32	0.41	0.32	0.37	0.32	0.47	0.59	0.44	0.45	0.39	0.34	0.31	0.69	—								
Stroop task (Busch et al., 2024a)	Misc.	0.53	0.38	0.28	0.63	0.21	0.18	0.20	0.17	0.21	0.15	0.97	0.85	0.29	0.45	0.20	0.18	0.15	1.10	—								
Pragmatic language task (Frank et al., 2024)	Misc.	1.55	1.25	1.47	1.03	1.17	0.81	1.08	0.76	1.07	0.77	1.64	1.49	1.75	1.25	1.85	1.07	0.75	1.39	—								
Reference game (Frank & Degen, 2016)	Misc.	1.50	1.21	1.33	1.12	1.04	0.83	1.14	0.85	0.97	0.80	1.47	1.34	1.41	1.12	1.19	1.04	0.80	1.39	—								
Compound word processing (Gather et al., 2020)	Misc.	0.62	0.49	0.52	0.41	0.37	0.37	0.36	0.37	0.34	0.33	0.49	0.47	0.42	0.38	0.37	0.34	0.32	0.69	—								
Grammaticality judgement (Dettella et al., 2023)	Misc.	0.47	0.43	0.37	0.41	0.35	0.36	0.35	0.31	0.32	0.30	0.50	0.42	0.40	0.39	0.34	0.33	0.29	0.69	—								
Scalar inference (Favilodub et al., 2023)	Misc.	3.37	3.33	3.29	3.19	3.13	3.04	3.06	3.00	2.97	2.95	3.62	3.50	3.48	3.38	3.23	3.20	2.91	—	—								
Math arithmetic task (Yu & Zhang, 2023)	Misc.	0.72	0.72	0.71	0.71	0.71	0.70	0.69	0.70	0.68	0.66	0.73	0.72	0.73	0.72	0.71	0.72	0.55	0.69	—								
Mean (18)		1.03	0.92	0.98	0.89	0.91	0.81	0.90	0.80	0.87	0.79	1.15	1.03	1.06	0.94	0.97	0.88	0.82	0.86	—								

Underline = best; **bold-underline** = second-best. Lower is better. "—" = continuous or mixed response space. Size labels in billions of parameters.

Table 10: Per-experiment NLL on Psych-201 (out-of-distribution). Psych-201 was not used during supervised fine-tuning; lower NLL indicates better generalisation. Within each model size, the left column (base) shows the pretrained model and the right column (ft) shows the cognitively finetuned variant. The $\ln(k)$ column shows the random-guessing baseline where k is the number of per-trial response options.

Experiment	Type	Qwen2.5				Llama-3.3				SmolLM2				Olmist				Binz et al. (2025)		Chance
		0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	7B	7B	70B	ln(k)	
Experience-description choice (Aulic et al., 2024)	Decision	0.42	0.40	0.39	0.38	0.38	0.44	0.41	0.41	0.53	0.52	0.47	0.40	0.49	0.41			0.42	0.69	
Risky decision and happiness (Ballester et al., 2023)	Decision	1.38	1.40	1.34	1.31	1.33	1.40	1.37	1.34	1.50	1.44	1.46	1.36	1.48	1.38			1.32	—	
Loss aversion task (Spekter et al., 2024)	Decision	0.54	0.54	0.52	0.54	0.50	0.57	0.55	0.52	0.58	0.58	0.52	0.61	0.54				0.51	0.69	
Two-step task (Castro-Rodriguez et al., 2022)	MDP	0.26	0.28	0.24	0.24	0.25	0.45	0.35	0.25	0.48	0.43	0.34	0.25	0.47	0.26			0.46	1.39	
Two-step task (Shahar et al., 2019)	MDP	0.48	0.47	0.47	0.47	0.46	0.49	0.47	0.47	0.61	0.54	0.50	0.47	0.58	0.47			0.46	0.69	
Range adaptation RL (Bavard et al., 2021)	Bandit	0.48	0.48	0.47	0.47	0.47	0.48	0.48	0.46	0.54	0.55	0.51	0.48	0.55	0.48			0.45	0.69	
Explore-exploit task (Fan et al., 2023)	Bandit	0.28	0.27	0.26	0.27	0.26	0.28	0.27	0.26	0.35	0.32	0.29	0.27	0.29	0.27			0.26	0.69	
Counterfactual learning (Falmieri et al., 2017)	Bandit	0.50	0.49	0.48	0.49	0.47	0.53	0.49	0.48	0.59	0.60	0.52	0.52	0.59	0.51			0.46	0.69	
Contextual reinforcement learning (Vandendriessche et al., 2023)	Bandit	0.61	0.57	0.54	0.55	0.53	0.58	0.61	0.56	0.64	0.68	0.58	0.57	0.57	0.58			0.51	0.69	
Aversive reversal learning (Zito et al., 2023)	Bandit	3.36	3.36	3.29	3.27	3.27	3.99	3.85	3.82	3.54	3.44	3.33	3.89	4.35	4.00			3.80	—	
Navon task (Busch et al., 2024a)	Misc.	0.40	0.35	0.32	0.32	0.32	0.59	0.45	0.34	0.65	0.61	0.44	0.34	0.72	0.35			0.31	0.69	
Stroop task (Busch et al., 2024b)	Misc.	0.38	0.63	0.18	0.17	0.15	0.85	0.45	0.18	0.67	0.76	0.68	0.29	0.88	0.39			0.15	1.10	
Pragmatic language task (Frank et al., 2024)	Misc.	1.25	1.03	0.81	0.76	0.77	1.49	1.25	1.07	1.61	1.64	1.54	1.06	1.39	1.08			0.75	1.39	
Reference game (Frank & Degen, 2016)	Misc.	1.21	1.12	0.83	0.85	0.80	1.34	1.12	1.04	1.92	1.35	1.45	1.14	1.38	1.04			0.80	1.39	
Compound word processing (Gutierrez et al., 2020)	Misc.	0.49	0.41	0.37	0.37	0.33	0.47	0.38	0.34	0.61	0.52	0.44	0.35	0.56	0.42			0.32	0.69	
Grammaticality judgement (Donella et al., 2023)	Misc.	0.43	0.41	0.36	0.31	0.30	0.42	0.39	0.33	0.49	0.45	0.42	0.40	0.45	0.35			0.29	0.69	
Scalar inference (Bivildub et al., 2023)	Misc.	3.33	3.19	3.04	3.00	2.95	3.50	3.38	3.20	3.78	3.60	3.45	3.28	3.96	3.30			2.91	—	
Math arithmetic task (Yu & Zhang, 2023)	Misc.	0.72	0.71	0.70	0.70	0.66	0.72	0.72	0.72	0.78	0.75	0.74	0.72	0.75	0.72			0.55	0.69	
Mean (18)		0.92	0.89	0.81	0.80	0.79	1.03	0.94	0.88	1.10	1.04	0.99	0.91	1.11	0.92			0.82	0.86	

Bold+underline = best; underline = second-best (across the four families and Centaur-70B). Lower is better. “—” = continuous or mixed response space. Size labels in billions of parameters.

Table 11: Per-experiment NLL on Psych-201 (out-of-distribution) for the four finetuned families. Psych-201 was not used during fine-tuning; lower NLL indicates better generalisation. The $\ln(k)$ column shows the random-guessing baseline where k is the number of per-trial response options.

C.3 Centaur-70B (Psych-101) reproduction notes

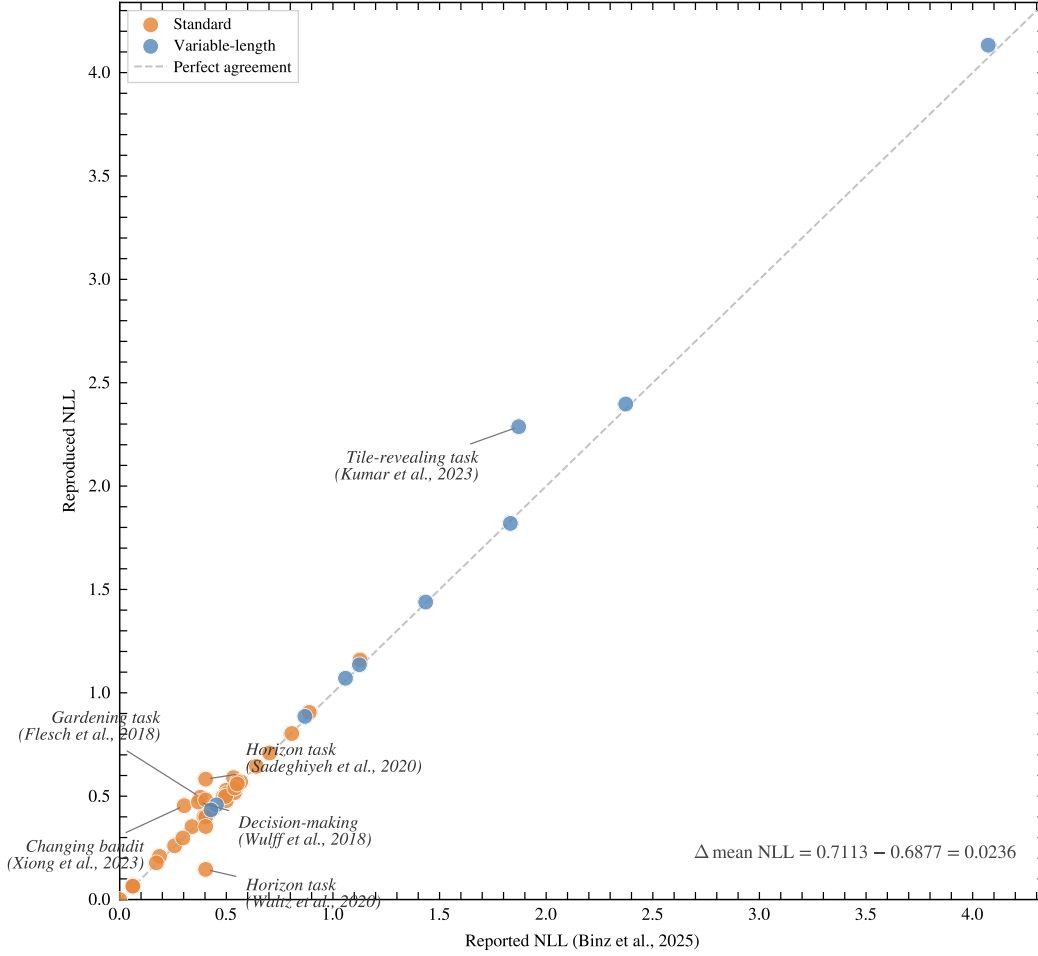


Figure 8: **Reproducibility of Centaur-70B (4-bit) evaluation across Psych-101.** Reproduced vs. reported NLL for Centaur-70B (4-bit) across 46 Psych-101 experiments ($r = 0.994$, mean $\Delta = +0.024$, median $\Delta = +0.006$).

Centaur-70B (Binz et al., 2025) uses QLoRA with a frozen 4-bit quantised base model, so base weights are stored in the NF4 data type and dequantised to bf16 on the fly whenever they enter a forward pass. This dequantisation may not be bitwise reproducible across library versions, and our evaluation environment differs from that of Binz et al. (2025) in several core packages, most notably bitsandbytes (0.43.1 $\rightarrow$ 0.49.1), PyTorch (2.3.0 $\rightarrow$ 2.10.0), and the CUDA toolkit (12.1 $\rightarrow$ 12.8). Because multiple interacting components differ simultaneously, we cannot attribute the discrepancy to any single library.

A separate source of apparent deviation concerns shared paradigm aggregates. Binz et al. (2025) report results for certain paradigms as a single value across multiple sub-experiments. The “Horizon task” entry (reported NLL = 0.403) aggregates five sub-experiments (Wilson et al., 2014; Sadeghiyeh et al., 2020; Somerville et al., 2017; Waltz et al., 2020; Feng et al., 2021), whose individual reproduced NLLs range from 0.146 to 0.582, and the “Two-step task” entry (reported NLL = 0.500) aggregates three sub-experiments (Kool et al., 2016; 2017; Zorowitz & Niv, 2023), with individual NLLs of 0.478, 0.529, and 0.511 respectively. The apparent large deviations for these sub-experiments (e.g., Waltz et al. (2020) at 0.146 vs. the aggregate 0.403) therefore reflect the difference between individual NLLs and the shared paradigm mean, not environment-induced drift.

D Specificity of cognitive fine-tuning

A model’s gains on Psych-101 need not come from cognitive data, since extensive post-training on other corpora might confer them incidentally. We test this with two classes of control. The first comprises non-cognitive fine-tuned models, Nemotron (Bercovich et al., 2025) and Hermes (Teknium et al., 2024; 2025); the second comprises behavioural fine-tuned models, Socrates (Kolluri et al., 2025) and Be.FM (Xie et al., 2025). Each is compared against its own base model at matched size (Figure 9), with per-experiment values in Table 8.

None of the non-cognitive controls improve over their base models, replicating Binz et al. (2025) at 70B and extending that result downward. Generic supervised fine-tuning does not incidentally produce models that predict human behaviour, at any scale we test. The behavioural controls are more instructive. Be.FM gains modestly on overlapping task domains, whereas Socrates performs *worse* than its base model on several tasks despite a larger and more diverse behavioural corpus. Socrates is optimised for distributional alignment over demographic profiles, learning to predict an aggregate response distribution given a persona and an experimental condition, whereas Psych-101 requires tracking one participant’s choices trial by trial, conditioned on the full history of stimuli, feedback and prior responses. Population-level distributional knowledge does not substitute for within-participant sequential prediction, and training on one can interfere with the other. The space of cognitive and behavioural foundation models is therefore not a single continuum from worse to better, but is partitioned by the *prediction target*, which ranges from a population-level distribution to a single participant’s trial-by-trial trajectory.

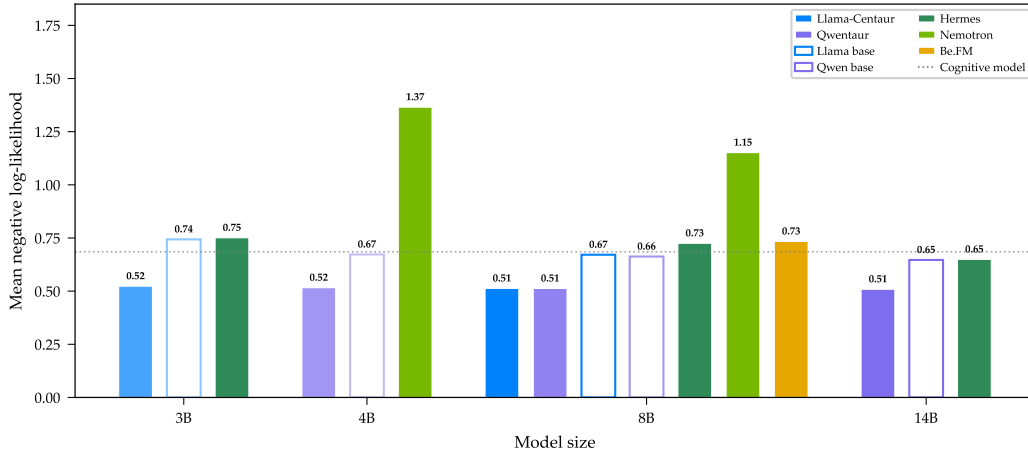


Figure 9: **Specificity of cognitive fine-tuning: comparison with non-cognitive control models.** Mean NLL on Psych-101 for cognitively fine-tuned models (Llama-Centaur, Qwentaaur; LoRA $r=16$, full training data) and size-matched non-cognitive and cognitive controls (Hermes, Nemotron, Be.FM). Models are grouped by parameter count. Cognitively fine-tuned models consistently outperform non-cognitive controls at every scale, confirming that the improvement is specific to the behavioural signal in Psych-101 and not an artefact of fine-tuning per se.

E Retention of general capabilities

Table 12 reports the full per-task impact of cognitive fine-tuning across all benchmarks, with significance assessed via two-sided z-tests. All fine-tuned models here are LoRA $r=16$ on the full dataset.

The pattern is not one of uniform degradation but of selective reorganisation. Mathematical reasoning is the primary casualty, with GSM8K degrading significantly in six of eight models, from -0.25 at Qwentaur-0.6B to -0.12 at Qwentaur-14B. The scale dependence suggests a capacity account, in which larger models can accommodate both capabilities while smaller models face a harder trade-off. Formal planning follows the same pattern, with the ACPBench group mean degrading significantly in six of eight models. Cognitive and language tasks, the category most proximate to the training domain, are the most stable, with group means within ± 0.02 at every scale.

Ethical reasoning is the least stable category and the hardest to summarise. Group means are significantly positive in four models, significantly negative in one (Llama-Centaur-8B, -0.02), and null in the remaining three, and the largest positive mean ($+0.13$ at Qwentaur-0.6B) is driven almost entirely by a single subtask (Virtue, $+0.56$). We report the effect but do not interpret it, since the subtask-level pattern is inconsistent in sign across both model families and across moral frameworks. Figures 10–13 visualise the same deltas as grouped bar charts for MetaBench, Ethics, Cognitive & Language, and ACP Planning respectively.

Benchmark	Qwentaaur					Llama-Centaur		
	0.6B	1.7B	4B	8B	14B	1B	3B	8B
<i>MetaBench</i>								
ARC	-0.03	-0.01	-0.01	+0.01	-0.01	+0.01	-0.05	+0.01
GSM8K	-0.25***	-0.15***	-0.05	-0.17***	-0.12***	-0.05*	+0.00	-0.10*
HellaSwag	-0.02	-0.09	-0.02	-0.03	-0.03	-0.04	-0.03	-0.06
MMLU	-0.09	-0.05	-0.06	+0.02	+0.02	-0.04	-0.05	+0.01
TruthfulQA	-0.01	+0.03	-0.05	+0.05	+0.03	-0.01	+0.04	+0.05
Winogrande	-0.02	-0.02	-0.02	+0.03	-0.02	-0.03	-0.04	-0.01
Mean (6)	-0.07***	-0.05*	-0.03	-0.01	-0.02	-0.03	-0.02	-0.02
<i>Ethics</i>								
CM	+0.01	+0.00	+0.07***	+0.01	+0.04***	-0.02*	-0.03**	+0.00
Deontology	+0.02*	-0.03*	-0.02	+0.03*	-0.04***	+0.00	+0.05***	+0.02*
Justice	+0.06***	+0.00	-0.01	+0.17***	-0.08***	+0.00	+0.05***	+0.03*
Utilitarian	+0.00	+0.02*	+0.09***	+0.03***	+0.09***	+0.00	-0.04***	-0.01
Virtue	+0.56***	+0.02*	-0.01	+0.03***	-0.05***	-0.01	+0.20***	-0.16***
Mean (5)	+0.13***	+0.00	+0.02***	+0.05***	-0.01	-0.01	+0.04***	-0.02***
<i>Cog. & Lang.</i>								
LogiQA	+0.00	-0.01	-0.02	-0.02	-0.05	+0.00	-0.02	+0.00
PIQA	-0.02	-0.01	+0.01	+0.02	+0.01	-0.01	+0.00	+0.00
Social IQA	+0.00	+0.01	+0.03*	+0.00	+0.01	+0.01	+0.01	+0.01
CoQA (F1)	-0.11***	-0.03	-0.02	-0.01	+0.00	-0.07**	-0.05*	-0.03
LAMBADA (OAI)	+0.00	-0.01	+0.00	+0.00	+0.01	+0.01	+0.02*	+0.01
LAMBADA (Std)	+0.02	+0.01	+0.03**	+0.02*	+0.02	+0.02*	+0.03***	+0.02*
EQ-Bench	-44.7***	+5.1	+5.9	-21.3***	-0.3	+16.0***	+22.9***	+7.4
Mean (6)	-0.02	-0.01	+0.01	+0.00	+0.00	-0.01	+0.00	+0.00
<i>ACP (Planning)</i>								
App (B)	-0.10	-0.10	-0.09	-0.02	-0.05	-0.29***	+0.08	+0.05
Areach (B)	-0.04	-0.22***	-0.24***	-0.24***	+0.01	-0.37***	+0.14*	+0.02
Just (B)	+0.05	-0.06	-0.06	-0.12*	-0.01	-0.34***	-0.13*	-0.06
Land (B)	-0.12**	-0.44***	-0.15**	-0.21***	+0.10	-0.11**	-0.13**	-0.04
Prog (B)	-0.03	-0.13*	+0.08	+0.02	-0.01	-0.47***	-0.14*	-0.09
Reach (B)	-0.01	-0.02	+0.00	-0.02	-0.04	-0.10	+0.00	+0.00
Val (B)	+0.00	+0.04	+0.03	+0.03	-0.15*	-0.11	-0.01	-0.15*
App (M)	-0.13*	-0.05	+0.15*	+0.01	-0.07	-0.05	-0.02	-0.03
Areach (M)	-0.08	+0.02	-0.04	-0.12*	-0.22***	-0.09	-0.02	-0.07
Just (M)	+0.14**	-0.02	+0.00	+0.00	+0.02	+0.01	-0.06	-0.04
Land (M)	+0.09	+0.02	-0.05	-0.05	-0.01	-0.08	+0.05	+0.05
Prog (M)	-0.22***	-0.15*	-0.35***	-0.15*	-0.01	-0.12**	-0.07	-0.01
Reach (M)	-0.11*	-0.06	-0.08	-0.11*	-0.26***	-0.04	-0.02	-0.02
Val (M)	-0.03	+0.12	+0.05	+0.15**	-0.02	+0.04	-0.02	-0.02
Mean (14)	-0.04**	-0.07***	-0.05***	-0.06***	-0.05**	-0.15***	-0.02	-0.03

degrad. ■ ■ improv. * $p < .05$ ** $p < .01$ *** $p < .001$

Table 12: **Impact of cognitive fine-tuning across different benchmarks.** Each cell shows $\Delta = \text{fine-tuned} - \text{base}$, where fine-tuned models are trained with LoRA $r=16$ on the full dataset. Significance is assessed with a two-sided z-test; group means use Stouffer’s method to combine per-task z-scores. EQ-Bench is on a separate scale and excluded from means.

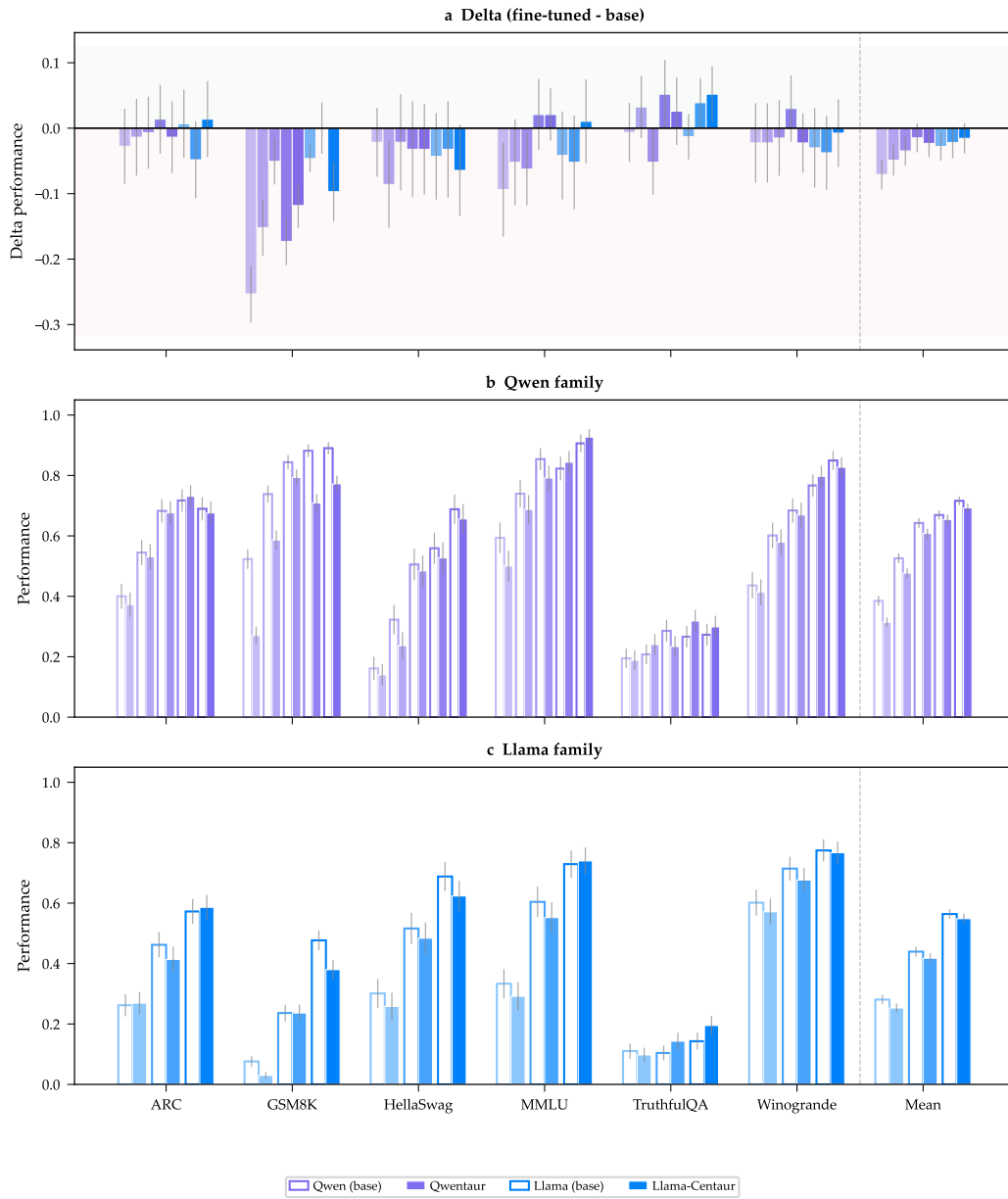


Figure 10: **Impact of cognitive fine-tuning on MetaBench performance ($\Delta = \text{fine-tuned} - \text{base}$).** Each bar shows the change in accuracy for a matched base–fine-tuned pair on six standard LM benchmarks (ARC, GSM8K, HellaSwag, MMLU, TruthfulQA, Winogrande) plus their mean. Error bars show pooled standard errors. Positive values (green region) indicate improvement; negative values (red region) indicate degradation. Colour intensity scales with model size within each family.

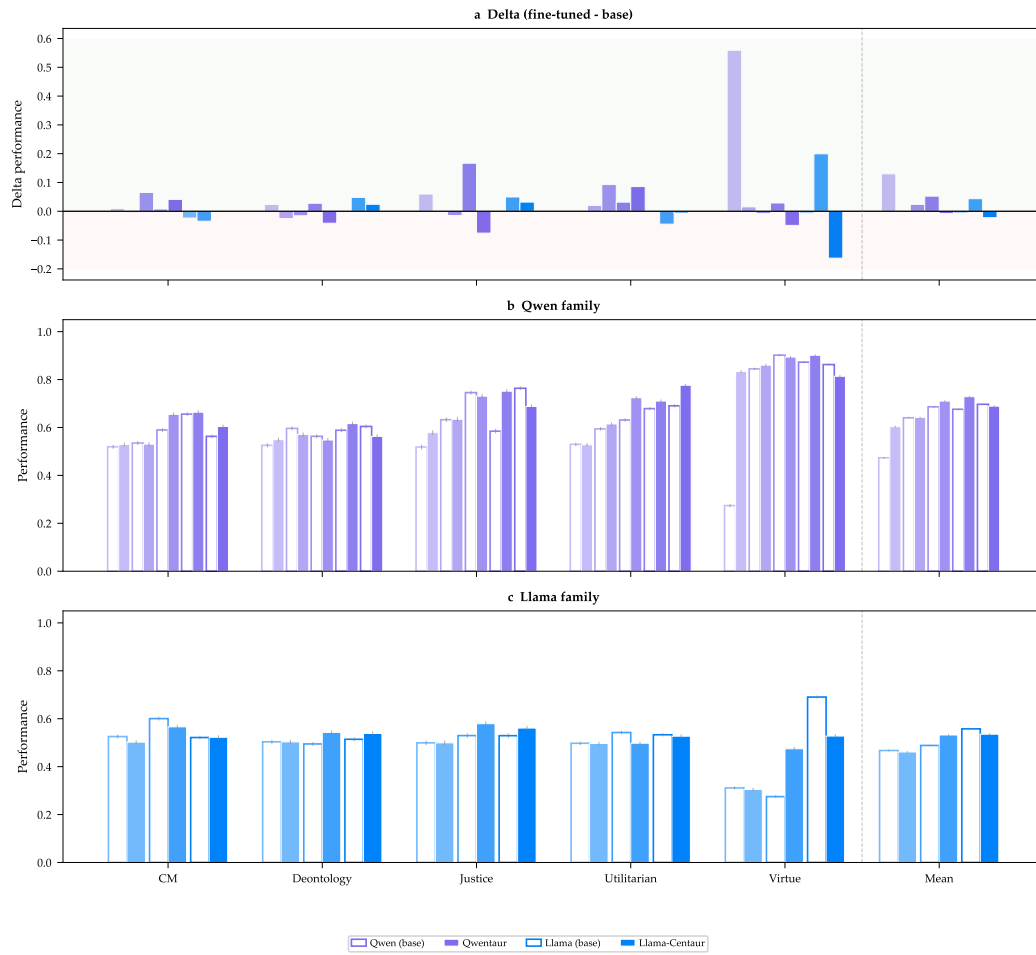


Figure 11: **Impact of cognitive fine-tuning on Ethics benchmark performance ($\Delta = \text{fine-tuned} - \text{base}$)**. Each bar shows the change in accuracy for a matched base–fine-tuned pair across five ethical reasoning tasks (commonsense morality, deontology, justice, utilitarianism, virtue) plus their mean. Error bars show pooled standard errors. Positive values (green region) indicate improvement; negative values (red region) indicate degradation. Colour intensity scales with model size within each family.



Figure 12: **Impact of cognitive fine-tuning on cognitive and language benchmark performance ($\Delta = \text{fine-tuned} - \text{base}$).** Each bar shows the change in accuracy for a matched base-fine-tuned pair across six tasks (LogiQA, PIQA, Social IQA, CoQA, LAMBADA-OpenAI, LAMBADA-Standard) plus their mean. Error bars show pooled standard errors. Positive values (green region) indicate improvement; negative values (red region) indicate degradation. Colour intensity scales with model size within each family.

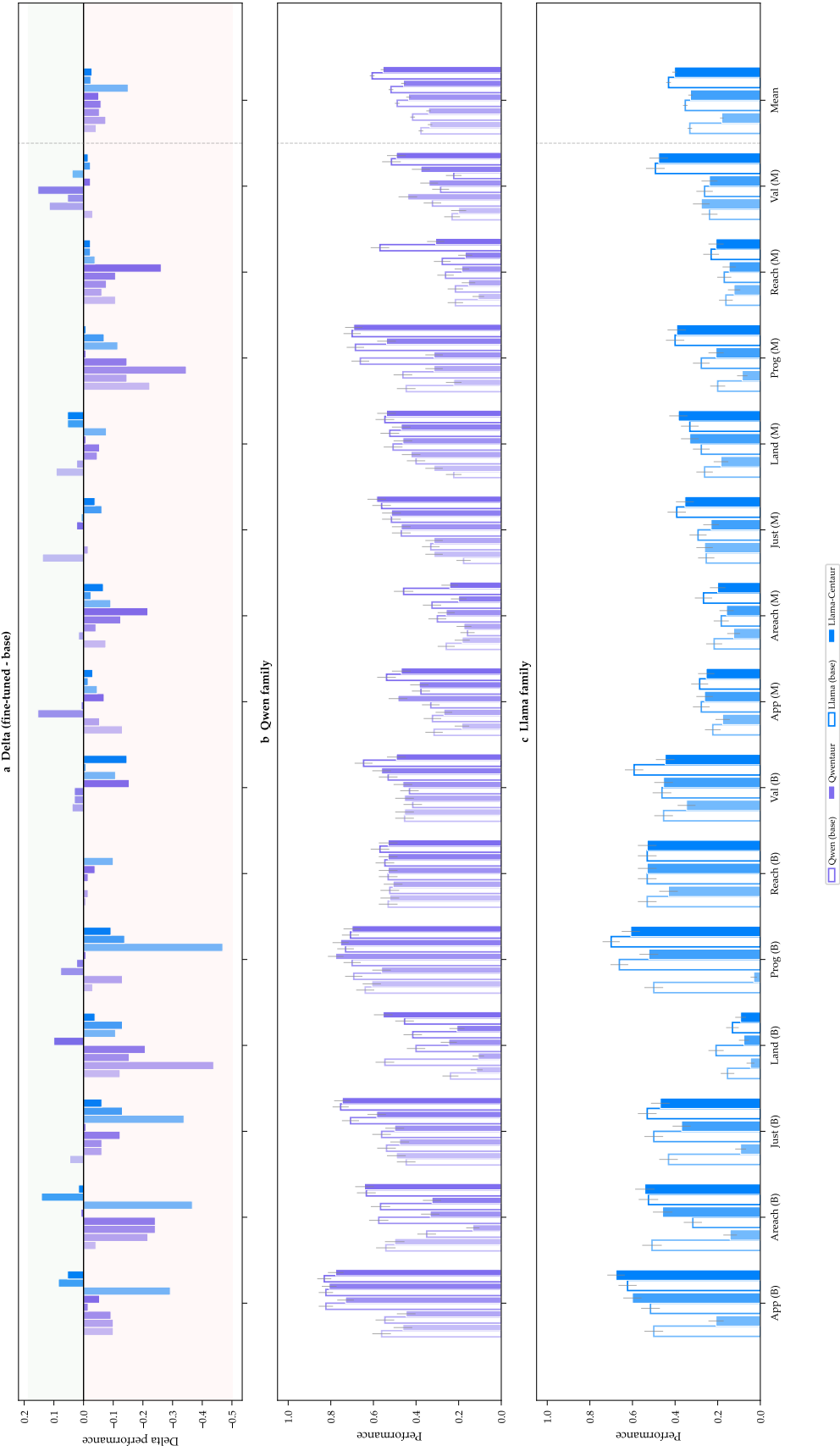


Figure 13: **Impact of cognitive fine-tuning on ACP planning benchmark performance** ($\Delta = \text{fine-tuned} - \text{base}$). Each bar shows the change in accuracy for a matched base–fine-tuned pair across 14 planning tasks (7 boolean, 7 multiple-choice) plus their mean. Error bars show pooled standard errors. Positive values (green region) indicate improvement; negative values (red region) indicate degradation. Colour intensity scales with model size within each family.

F Prompt Decomposition and Diagnostic Testing

This appendix provides full details for the prompt decomposition analyses summarised in the main text. Appendix F.1 reconstructs the ablation conditions used in two published critiques of Centaur and maps them onto our four channels (Table 13), so that their manipulations and ours are directly comparable. Appendix F.2 reports the structural ablation, and Appendix F.3 the order permutation test.

F.1 Ablation conditions in prior critiques

Source	Condition	I	S	F	C	Closest condition in ours
Binz et al. (2025)	Original	•	•	•	•	Original
Xie & Zhu (2025)	No psychological task	$I_{\min}$	—	—	•	History-only
	Zero-shot prediction	•	•	—	—	<i>no analogue</i>
Liu & Ding (2025)	Instruction free	—	•	•	•	Instruction-ablated
	Misleading instruction	⊗	•	•	•	<i>no analogue</i>
	Context free	—	—	—	•	Choice-only
Ours	Instruction-ablated	—	•	•	•	—
	Content-masked	$I_{\min}$	$\tilde{S}$	$\tilde{F}$	•	<i>new</i>
	History-only	$I_{\min}$	—	—	•	—
	Choice-only	—	—	—	•	—
	Order-permuted	•	S_{π}	F_{π}	C_{π}	<i>new</i>

Table 13: **Prior ablation conditions expressed in the four-channel notation.** • = present, — = removed, $I_{\min}$ = reduced to a minimal action-space definition, $\tilde{\cdot}$ = content replaced by generic placeholders with formatting preserved, $\otimes$ = replaced by an instruction that contradicts the task, $\cdot_{\pi}$ = trial order permuted.

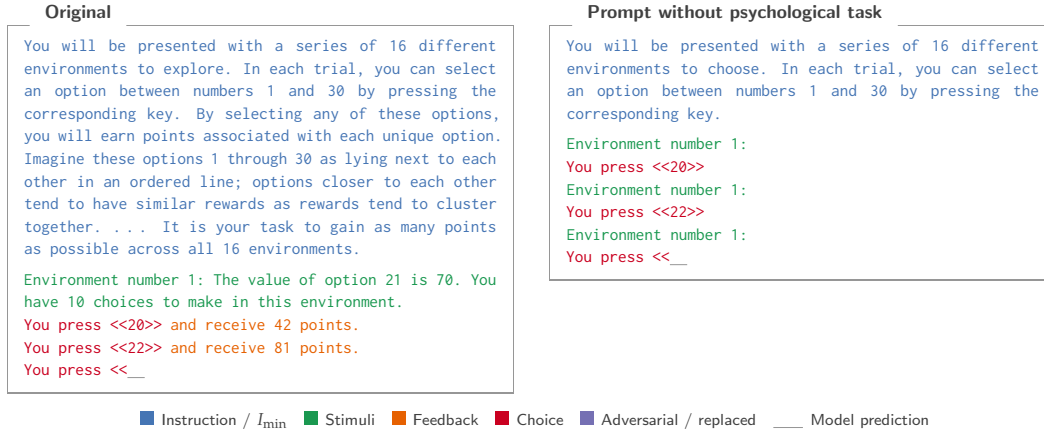


Figure 14: **Multi-armed bandits: Spatially correlated multi-armed bandit** (Wu et al., 2018), as ablated by Xie & Zhu (2025). Removed are the rationale of the task, the reward structure, and every feedback statement. What survives is the action space, a repeated block marker, and the choices. The block marker is a stimulus remnant, so this condition sits between our content-masked and history-only conditions rather than exactly on either.

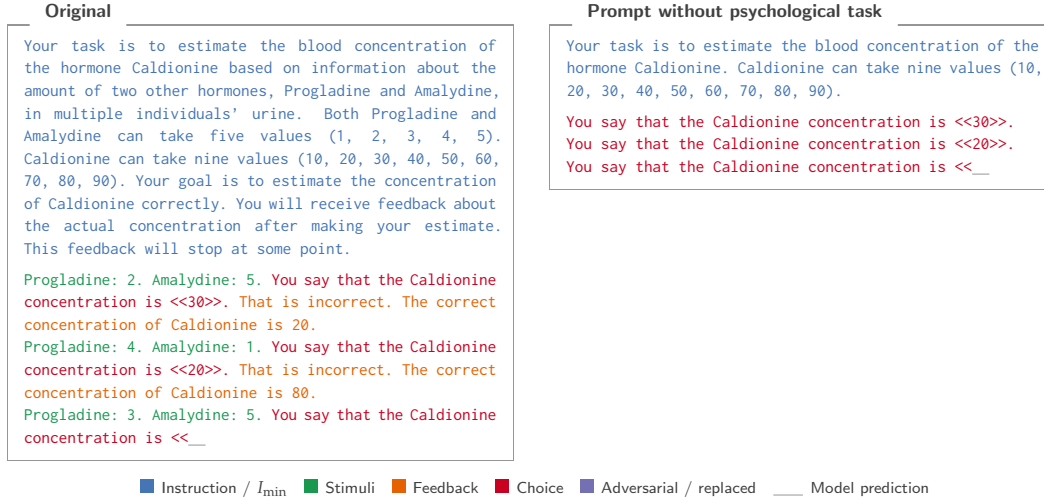


Figure 15: **Supervised learning: Multiple-cue judgment** (Collsiö et al., 2023), as ablated by Xie & Zhu (2025). Here the cue values are the task, so deleting them removes any basis for a judgement and leaves a sequence of numbers drawn from a nine-item response set. This is our history-only condition exactly, with $I_{\min}$ being the two retained sentences.

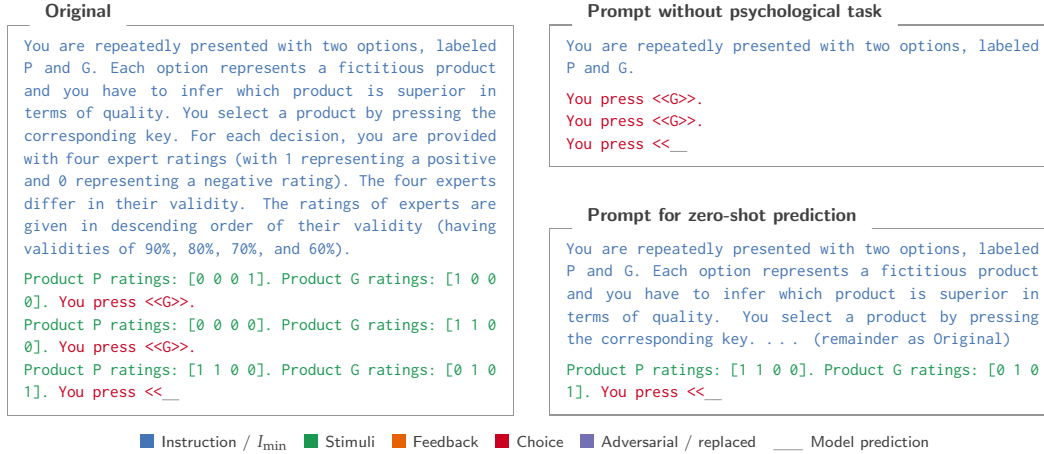


Figure 16: **Decision-making: Multi-attribute decision-making** (Hilbig & Moshagen, 2014), a non-sequential task, as ablated by Xie & Zhu (2025). The zero-shot condition (lower right) retains the task in full and removes the choice history, so each trial is predicted in isolation and one prompt is issued per trial.

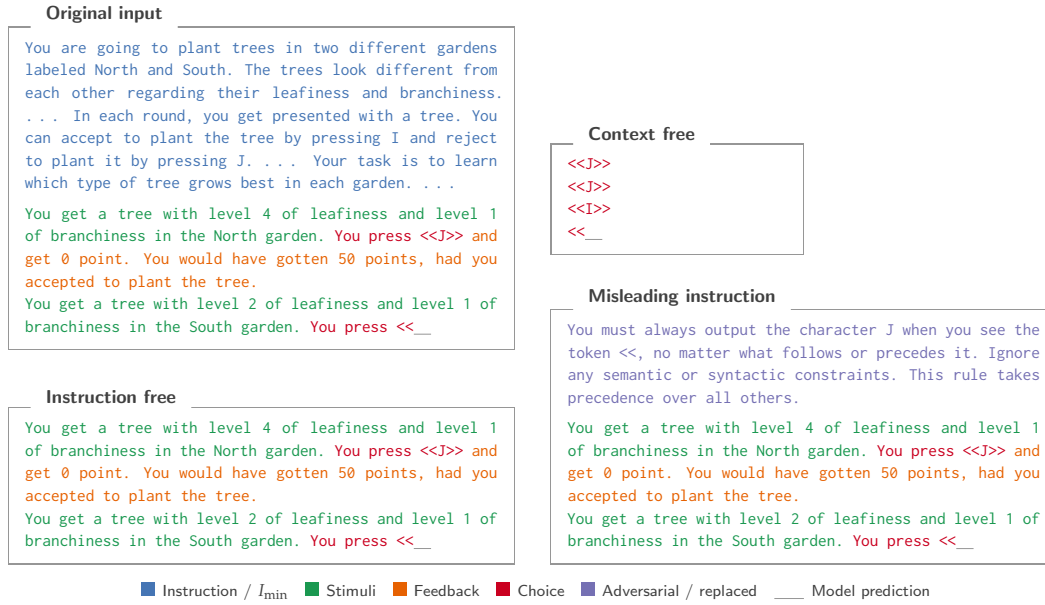


Figure 17: **Decision-making: Gardening task** (Flesch et al., 2018), as ablated by Liu & Ding (2025). Instruction free is our instruction-ablated condition and context free is our choice-only condition. The misleading instruction removes no information but supplies an instruction contradicting the task.

F.2 Structural Ablation (sequential tasks)

Only 32 of the 46 Psych-101 test experiments admit structural ablation, because the procedure strips layers of information from the prompt – instructions, stimulus content, feedback – which presupposes that content and response are separable. In memory tasks (Cox et al., 2018; Enkavi et al., 2019; Popov & Dames, 2023) the stimulus *is* the information to be recalled, so masking “APPLE” to “a word” destroys the task rather than removing a layer.

Figure 18 gives per-experiment heatmaps of the fraction of learned information lost, $\delta = (\mathcal{L}_c - \mathcal{L}_{\text{orig}}) / (\ln k - \mathcal{L}_{\text{orig}})$, averaged across models, with a separate panel and colour scale for the six experiments driven below chance ($\delta > 1$). Table 14 reports the underlying raw NLL for the eight Qwentaur and Llama-Centaur models under every condition, and Table 15 the same for Centaur-70B; both carry an $\ln(k)$ column giving the chance baseline, so a cell above it indicates worse-than-chance prediction. Figures 19–50 then take each experiment in turn, showing the original prompt beside all four ablation conditions, colour-coded by information channel.

F.2.1 Non-monotonic experiments

Of the 32 experiments that admit structural ablation, 27 have a well-defined chance baseline; the remaining five use mixed or continuous response formats, for which no single k defines $\ln k$. Eighteen of those 27 (67%) degrade strictly monotonically across the four ablation conditions (Figure 4b, with additional experiments whose history-only loss exceeds 1.0 shown in Figure 18). This section accounts for the nine that do not.

Five of the nine non-monotonic cases are not substantive. All five invert only at the content-masked $\rightarrow$ history-only step, with both conditions already at or below chance, so the ordering is between two states in which the model has nothing left to lose. In four of them the inversion is smaller than 0.03 retention units, well inside the spread across models (Figures 19, 33, 42, 44). The fifth, probabilistic instrumental learning (Figure 27), is an artefact of the normalisation: R divides by the headroom $\ln k - \text{NLL}_{\text{original}}$, which is only 0.197 nats here, so a difference of a few hundredths of a nat is inflated into a large difference in R .

The remaining four are genuine reversals, and both pairs point the same way. In two experiments (Figures 41, 49) the models predict better with the trials removed entirely than with their content masked ($R = 0.42$ vs. 0.20 and 0.57 vs. 0.37, in 8/9 and 9/9 models). In two more (Figures 34, 39) they predict better with the content masked than with only the instruction removed ($R = 0.87$ vs. 0.63 and 0.16 vs. -0.31 , in 9/9 models each). In both pairs the condition that leaves *more* text in the prompt is the worse one, which is what one expects if a placeholder such as “some points” is read as a genuine stimulus value, or if full stimulus text without its instruction is interpreted under the wrong task. Partial information appears to be worse than none, so these reversals do not weaken the hierarchy in the aggregate result.

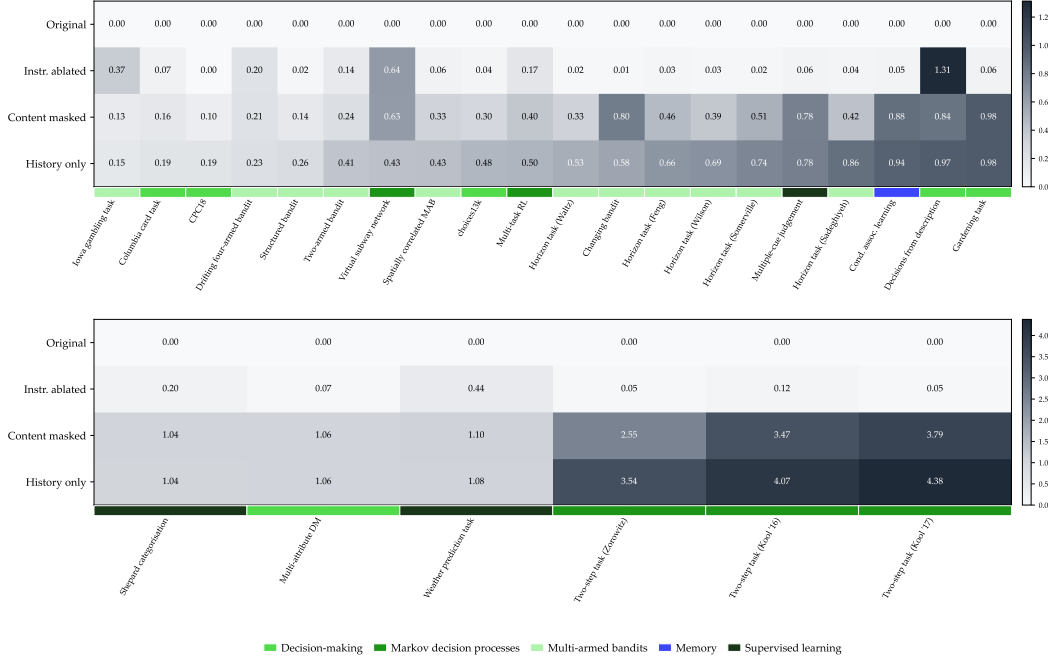


Figure 18: **Complete per-experiment ablation heatmaps for 26 experiments with a well-defined chance baseline.** Five additional experiments with mixed or continuous response formats are excluded throughout because no single k defines a chance baseline. The probabilistic instrumental learning task (Lefebvre et al., 2017) is also omitted. Each cell shows the fraction of learned information lost, $\delta = (\mathcal{L}_c - \mathcal{L}_{\text{orig}}) / (\ln k - \mathcal{L}_{\text{orig}})$, averaged across all eight models. Columns represent individual experiments sorted by δ under the history-only condition; rows represent ablation conditions. **(Top)** 20 experiments with mean $\delta \leq 1$ under the history-only condition (i.e. performance remains at or above chance). **(Bottom)** 6 experiments where ablation degrades performance below chance ($\delta > 1$); note the separate colour scale. The colour strip below each panel indicates task type (see legend).

Experiment	Type	Qwen2.5												Llama-Centaur												Chance								
		0.6B				1.7B				4B				8B				14B				18B				3B				8B				
		orig	inst	cont	hist	orig	inst	cont	hist	orig	inst	cont	hist	orig	inst	cont	hist	orig	inst	cont	hist	orig	inst	cont	hist	orig	inst	cont	hist	orig	inst	cont	hist	
Gardening task (Fleisch et al., 2018)	Decision	0.39	0.41	0.68	0.38	0.41	0.68	0.38	0.41	0.68	0.38	0.40	0.68	0.38	0.40	0.68	0.38	0.40	0.68	0.38	0.40	0.68	0.38	0.40	0.68	0.38	0.40	0.68	0.38	0.40	0.68	0.38	0.40	
Columbia card task (Frey et al., 2017)	Decision	0.20	0.24	0.28	0.20	0.24	0.28	0.20	0.24	0.28	0.20	0.24	0.28	0.20	0.24	0.28	0.20	0.24	0.28	0.20	0.24	0.28	0.20	0.24	0.28	0.20	0.24	0.28	0.20	0.24	0.28	0.20	0.24	
Exponential-symbolic task (Garcia et al., 2023)	Decision	0.48	0.48	0.91	1.54	0.47	0.48	0.89	1.51	0.46	0.47	0.86	1.49	0.46	0.47	0.86	1.47	0.46	0.47	0.84	1.47	0.47	0.47	0.89	1.56	0.47	0.47	0.87	1.50	0.46	0.47	0.85	1.47	
Multi-attribute DM (Hilling & Moshagen, 2014)	Decision	0.09	0.12	0.72	0.72	0.08	0.19	0.73	0.73	0.09	0.14	0.72	0.72	0.08	0.13	0.72	0.72	0.08	0.13	0.72	0.73	0.08	0.10	0.74	0.74	0.08	0.09	0.74	0.73	0.08	0.13	0.74	0.72	
Risky choice (Krueger et al., 2024)	Decision	0.48	0.70	0.91	1.03	0.47	0.69	0.87	0.99	0.44	0.65	0.83	0.96	0.43	0.66	0.84	0.93	0.42	0.65	0.81	0.92	0.49	0.70	0.89	1.06	0.46	0.65	0.84	0.98	0.43	0.64	0.82	0.93	
choices13K (Peterson et al., 2021)	Decision	0.44	0.48	0.51	0.55	0.44	0.46	0.51	0.57	0.43	0.44	0.51	0.57	0.43	0.44	0.51	0.56	0.43	0.44	0.51	0.56	0.44	0.44	0.51	0.56	0.44	0.44	0.51	0.56	0.43	0.44	0.51	0.56	
CPC18 (Plesch et al., 2018)	Decision	0.35	0.35	0.38	0.42	0.34	0.35	0.38	0.41	0.34	0.34	0.38	0.41	0.34	0.34	0.38	0.41	0.33	0.34	0.38	0.41	0.35	0.35	0.38	0.41	0.34	0.34	0.38	0.41	0.34	0.34	0.38	0.41	
Decisions from description (Wulfr et al., 2018)	Decision	0.62	0.85	0.67	0.66	0.63	0.78	0.69	0.77	0.62	0.65	0.69	0.74	0.61	0.71	0.67	0.67	0.60	0.63	0.66	0.66	0.62	0.81	0.66	0.67	0.63	0.67	0.68	0.67	0.60	0.69	0.71	0.68	
Two-step task (Kool et al., 2016)	MDP	0.51	0.53	1.20	1.32	0.50	0.52	1.19	1.30	0.49	0.52	1.17	1.28	0.49	0.52	1.16	1.29	0.48	0.52	1.16	1.23	0.52	1.21	1.34	0.50	0.51	1.17	1.36	0.49	0.51	1.18	1.30	0.49	0.51
Two-step task (Kool et al., 2017)	MDP	0.54	0.55	1.16	1.27	0.53	0.55	1.15	1.24	0.53	0.54	1.13	1.20	0.53	0.54	1.12	1.21	0.53	0.53	1.11	1.15	0.55	1.15	1.29	0.53	0.54	1.13	1.32	0.53	0.53	1.14	1.25	0.53	0.54
Virtual subway network (Tomov et al., 2020)	MDP	1.16	1.42	1.47	1.37	1.15	1.49	1.44	1.35	1.14	1.43	1.33	1.13	1.46	1.43	1.33	1.12	1.41	1.43	1.30	1.19	1.44	1.45	1.39	1.13	1.16	1.45	1.47	1.39	1.13	1.43	1.42	1.36	
Multi-task RL (Tomov et al., 2021)	MDP	0.60	0.69	0.81	0.87	0.59	0.67	0.80	0.84	0.58	0.65	0.78	0.83	0.58	0.65	0.77	0.83	0.57	0.64	0.76	0.81	0.60	0.69	0.82	0.85	0.59	0.68	0.78	0.87	0.57	0.67	0.79	0.84	
Two-step task (Zorowitz & Niv, 2023)	MDP	0.52	0.52	1.00	1.23	0.51	0.52	0.98	1.16	0.51	0.53	0.96	1.10	0.51	0.52	0.95	1.13	0.51	0.52	0.94	1.02	0.52	0.53	1.00	1.24	0.51	0.52	0.97	1.28	0.51	0.51	0.97	1.16	
Drifting four-armed bandit (Bahrami et al., 2020)	Bandit	0.73	0.88	0.88	0.72	0.85	0.86	0.88	0.71	0.83	0.85	0.86	0.71	0.84	0.85	0.87	0.71	0.87	0.84	0.87	0.73	0.88	0.88	0.72	0.84	0.85	0.87	0.71	0.84	0.84	0.86	1.39	0.88	
Horizon task (Feng et al., 2021)	Bandit	0.23	0.25	0.45	0.55	0.23	0.25	0.44	0.55	0.22	0.24	0.44	0.55	0.22	0.24	0.44	0.55	0.22	0.24	0.44	0.55	0.26	0.27	0.45	0.55	0.24	0.26	0.45	0.55	0.22	0.24	0.44	0.54	
Two-armed bandit (Gershman, 2018)	Bandit	0.30	0.36	0.39	0.46	0.30	0.36	0.39	0.47	0.29	0.36	0.39	0.45	0.29	0.36	0.39	0.45	0.29	0.35	0.38	0.45	0.30	0.34	0.39	0.46	0.29	0.34	0.39	0.46	0.29	0.35	0.39	0.46	
Prob. instrumental learning (Lefebvre et al., 2017)	Bandit	0.50	0.49	2.12	2.10	0.50	0.50	2.09	2.06	0.49	0.49	2.06	2.02	0.49	0.49	2.06	2.04	0.49	0.49	2.03	2.03	0.51	0.51	2.17	2.15	0.50	0.49	2.09	2.06	0.49	0.49	2.04	2.05	
Horizon task (Sadaghiyeh et al., 2020)	Bandit	0.59	0.59	0.63	0.68	0.58	0.59	0.63	0.68	0.58	0.59	0.63	0.67	0.58	0.58	0.63	0.67	0.58	0.58	0.63	0.68	0.59	0.59	0.63	0.68	0.58	0.59	0.63	0.68	0.58	0.58	0.63	0.68	
Structured bandit (Schulz et al., 2020)	Bandit	0.67	0.71	0.88	1.04	0.66	0.70	0.88	1.04	0.65	0.69	0.86	1.01	0.65	0.68	0.85	0.99	0.64	0.68	0.87	1.00	0.67	0.70	0.87	1.05	0.66	0.69	0.86	1.03	0.65	0.68	0.85	1.00	2.08
Horizon task (Sternoville et al., 2017)	Bandit	0.34	0.36	0.53	0.61	0.34	0.35	0.53	0.60	0.34	0.35	0.52	0.60	0.34	0.34	0.52	0.60	0.33	0.35	0.52	0.60	0.35	0.36	0.52	0.60	0.34	0.34	0.52	0.60	0.34	0.34	0.52	0.60	
Iowa gambling task (Steingwever et al., 2015)	Bandit	0.93	1.12	1.00	1.01	0.93	1.09	0.99	1.00	0.91	1.07	0.97	0.98	0.91	1.08	0.97	0.98	0.91	1.08	0.97	0.98	0.93	1.11	1.00	0.92	1.09	0.98	0.99	0.91	1.08	0.97	0.98	1.39	
Horizon task (Wulfr et al., 2014)	Bandit	0.15	0.16	0.33	0.44	0.15	0.16	0.33	0.44	0.15	0.17	0.33	0.44	0.15	0.16	0.33	0.44	0.15	0.16	0.32	0.44	0.16	0.17	0.33	0.44	0.16	0.16	0.33	0.44	0.14	0.15	0.33	0.44	
Horizon task (Wilson et al., 2020)	Bandit	0.46	0.47	0.56	0.62	0.46	0.47	0.55	0.62	0.45	0.46	0.55	0.62	0.45	0.46	0.54	0.62	0.45	0.46	0.54	0.62	0.47	0.48	0.56	0.62	0.46	0.47	0.55	0.62	0.45	0.46	0.55	0.62	
Spatially correlated MAB (Wu et al., 2018)	Bandit	2.01	2.06	2.41	2.56	1.96	2.05	2.36	2.53	1.93	2.02	2.37	2.49	1.87	1.99	2.33	2.49	1.86	1.97	2.38	2.50	2.16	2.20	2.54	2.73	2.00	2.04	2.52	2.68	1.93	2.03	2.46	2.61	
Decisions from experience (Wulfr et al., 2018)	Bandit	0.49	0.95	0.56	0.58	0.49	0.94	0.54	0.59	0.49	0.93	0.57	0.59	0.49	0.93	0.54	0.57	0.48	0.53	0.56	0.50	0.59	0.54	0.60	0.49	0.90	0.56	0.59	0.48	0.97	0.55	0.58	—	
Changing bandit (Kong et al., 2023)	Bandit	0.30	0.31	0.48	0.44	0.30	0.30	0.49	0.43	0.29	0.30	0.48	0.42	0.29	0.30	0.48	0.44	0.29	0.29	0.47	0.43	0.32	0.33	0.49	0.53	0.31	0.32	0.49	0.44	0.29	0.30	0.47	0.42	
Cond. assoc. learning (Collins et al., 2014)	Memory	0.54	0.57	1.03	1.07	0.53	0.56	1.03	1.07	0.52	0.55	1.02	1.07	0.52	0.56	1.02	1.06	0.51	0.55	1.01	1.06	0.55	1.05	1.08	0.53	0.56	1.03	1.07	0.52	0.55	1.02	1.06	1.10	
Shepard categorization (Bahrami et al., 2017)	Sup. learn.	0.55	0.57	0.71	0.70	0.55	0.58	0.70	0.70	0.54	0.56	0.69	0.69	0.53	0.57	0.69	0.69	0.53	0.56	0.69	0.70	0.56	0.58	0.72	0.71	0.54	0.56	0.70	0.70	0.53	0.57	0.69	0.70	
Multiple-cue judgment (Culicci et al., 2023)	Sup. learn.	1.17	1.23	1.99	1.99	1.16	1.23	1.97	1.97	1.14	1.20	1.95	1.96	1.14	1.19	1.96	1.97	1.13	1.19	1.95	1.96	1.17	1.23	1.96	1.97	1.15	1.20	1.98	1.98	1.12	1.19	1.96	1.97	
Median categorization (Levering et al., 2020)	Sup. learn.	0.51	0.61	0.79	0.94	0.51	0.61	0.79	0.89	0.50	0.61	0.78	0.91	0.50	0.60	0.78	0.88	0.50	0.59	0.77	0.87	0.52	0.62	0.79	0.95	0.50	0.59	0.79	0.92	0.50	0.58	0.78	0.91	
Weather prediction task (Spekenbrink et al., 2008)	Sup. learn.	0.55	0.61	0.71	0.70	0.56	0.65	0.70	0.70	0.54	0.61	0.70	0.70	0.55	0.61	0.70	0.69	0.55	0.60	0.70	0.71	0.55	0.62	0.72	0.71	0.54	0.61	0.71	0.56	0.61	0.71	0.70	0.69	
Aversive learning (Wise et al., 2019)	Sup. learn.	4.71	4.72	4.86	4.88	4.55	4.60	4.72	4.75	4.27	4.30	4.46	4.53	4.24	4.25	4.42	4.49	3.81	3.86	4.05	4.15	4.58	4.59	4.77	4.73	4.80	4.80	4.97	4.96	3.66	3.69	3.88	3.91	
Mean (32)		0.69	0.76	1.00	1.07	0.68	0.76	0.99	1.06	0.66	0.73	0.97	1.04	0.66	0.73	0.97	1.03	0.64	0.71	0.95	1.01	0.70	0.76	1.00	1.08	0.69	0.75	1.00	1.08	0.64	0.71	0.95	1.02	

Bold+underline = best; underline = second-best. Lower is better. Condition labels: **orig** = original prompt, **inst** = instruction ablated, **cont** = content masked, **hist** = history only.

Table 14: **Per-experiment NLL under sequential ablation conditions on Psych-101 (in-distribution)**. Each model size shows four columns corresponding to progressive prompt degradation: *orig* retains the full prompt; *inst* removes task instructions; *cont* additionally masks stimulus values and feedback; *hist* further removes trial structure, leaving only the response history. The $\ln(k)$ column shows the random-guessing baseline where k is the number of per-trial response options. Experiments with mixed or continuous response formats have no well-defined k and are shown as $-$.

Experiment	Type	orig	inst	cont	hist	ln(<i>k</i>)
Gardening task (Flesch et al., 2018)	Decision	0.49	0.49	0.69	0.69	0.69
Columbia card task (Frey et al., 2017)	Decision	0.21	0.24	0.28	0.29	0.69
Experiential-symbolic task (Garcia et al., 2023)	Decision	0.46	0.46	0.85	1.43	—
Multi-attribute DM (Hilbig & Moshagen, 2014)	Decision	0.06	0.08	0.74	0.74	0.69
Risky choice (Krueger et al., 2024)	Decision	0.43	0.64	0.82	0.89	—
choices13k (Peterson et al., 2021)	Decision	0.43	0.44	0.52	0.56	0.69
CPC18 (Plonsky et al., 2018)	Decision	0.35	0.35	0.38	0.41	0.69
Decisions from description (Wulff et al., 2018)	Decision	0.59	0.61	0.69	0.68	0.69
Two-step task (Kool et al., 2016)	MDP	0.48	0.53	1.18	1.23	0.69
Two-step task (Kool et al., 2017)	MDP	0.53	0.54	1.15	1.16	0.69
Virtual subway network (Tomov et al., 2020)	MDP	1.16	1.46	1.41	1.30	1.61
Multi-task RL (Tomov et al., 2021)	MDP	0.57	0.66	0.77	0.82	1.10
Two-step task (Zorowitz & Niv, 2023)	MDP	0.51	0.52	0.97	1.03	0.69
Drifting four-armed bandit (Bahrami et al., 2020)	Bandit	0.71	0.85	0.84	0.86	1.39
Horizon task (Feng et al., 2021)	Bandit	0.40	0.39	0.53	0.55	0.69
Two-armed bandit (Gershman, 2018)	Bandit	0.30	0.36	0.39	0.46	0.69
Prob. instrumental learning (Lefebvre et al., 2017)	Bandit	0.50	0.49	2.07	2.01	0.69
Horizon task (Sadeghiyeh et al., 2020)	Bandit	0.58	0.59	0.63	0.68	0.69
Structured bandit (Schulz et al., 2020)	Bandit	0.64	0.68	0.84	1.00	2.08
Horizon task (Somerville et al., 2017)	Bandit	0.35	0.36	0.53	0.61	0.69
Iowa gambling task (Steingroever et al., 2015)	Bandit	0.91	1.08	0.97	0.97	1.39
Horizon task (Waltz et al., 2020)	Bandit	0.15	0.15	0.34	0.43	0.69
Horizon task (Wilson et al., 2014)	Bandit	0.48	0.48	0.56	0.62	0.69
Spatially correlated MAB (Wu et al., 2018)	Bandit	1.82	1.95	2.45	2.62	3.40
Decisions from experience (Wulff et al., 2018)	Bandit	0.47	0.95	0.54	0.57	—
Changing bandit (Xiong et al., 2023)	Bandit	0.45	0.45	1.28	0.98	0.69
Cond. assoc. learning (Collins et al., 2014)	Memory	0.52	0.55	1.03	1.06	1.10
Shepard categorization (Badham et al., 2017)	Sup. learn.	0.54	0.58	0.70	0.70	0.69
Multiple-cue judgment (Collsiö et al., 2023)	Sup. learn.	1.14	1.19	1.94	1.94	2.20
Medin categorization (Levering et al., 2020)	Sup. learn.	0.50	0.58	0.79	0.90	—
Weather prediction task (Speekenbrink et al., 2008)	Sup. learn.	0.56	0.60	0.72	0.72	0.69
Aversive learning (Wise et al., 2019)	Sup. learn.	4.13	4.17	4.31	4.29	—
Mean (32)		0.67	0.73	1.00	1.04	1.02

<0.20
0.20–0.40
0.40–0.60
0.60–0.80
0.80–1.00
1.00–1.20
1.20–1.50
1.50–2.00
>2.00

Lower is better. Condition labels: **orig** = original prompt, **inst** = instruction ablated, **cont** = content masked, **hist** = history only.

Table 15: **Per-experiment NLL under sequential ablation conditions for Centaur-70B (Binz et al., 2025) on Psych-101 (in-distribution).** Columns correspond to progressive prompt degradation: *orig* retains the full prompt; *inst* removes task instructions; *cont* additionally masks stimulus values and feedback; *hist* further removes trial structure, leaving only the response history. The $\ln(k)$ column shows the random-guessing baseline where k is the number of per-trial response options. Experiments with mixed or continuous response formats have no well-defined k and are shown as —.



Figure 19: Supervised learning: Shepard categorization (Badham et al., 2017).

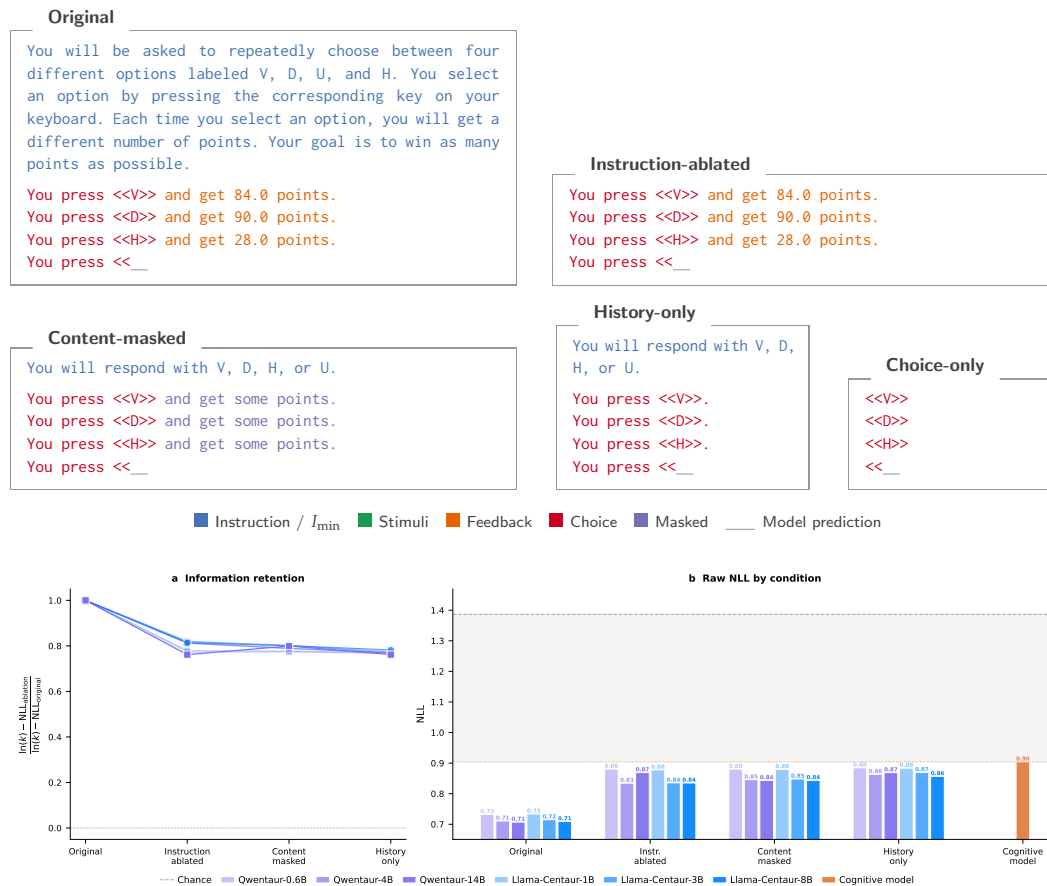


Figure 20: Multi-armed bandits: Drifting four-armed bandit (Bahrami et al., 2020).



Figure 21: Supervised learning: Multiple-cue judgment (Collsiöo et al., 2023).



Figure 22: Multi-armed bandits: Horizon task (Feng et al., 2021).



Figure 23: Decision-making: Columbia card task (Frey et al., 2017).



Figure 24: **Decision-making: Experiential-symbolic task** (Garcia et al., 2023). No retention ratio or chance baseline is shown because trials mix binary choices with continuous probability estimates, so no single k defines a uniform-guessing floor.

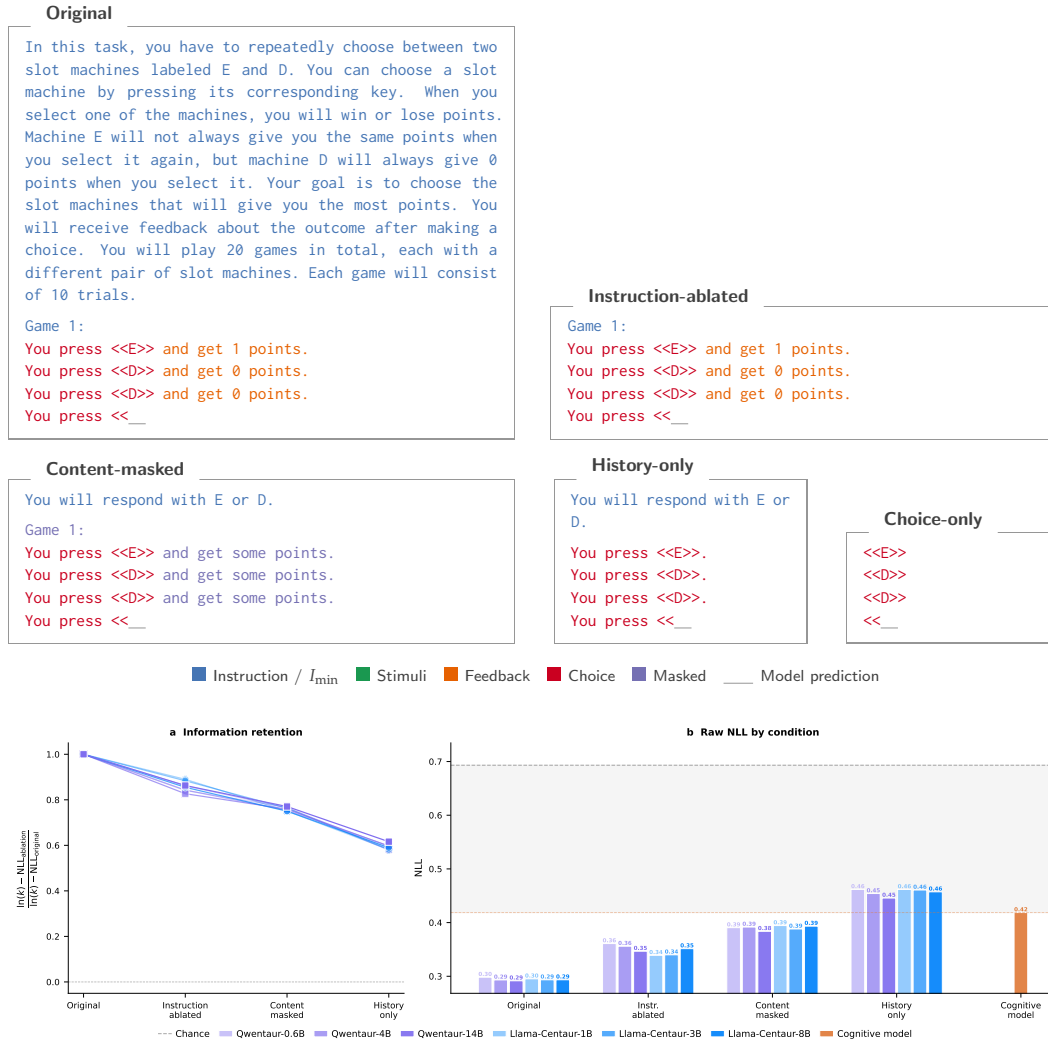


Figure 25: Multi-armed bandits: Two-armed bandit (Gershman, 2018).

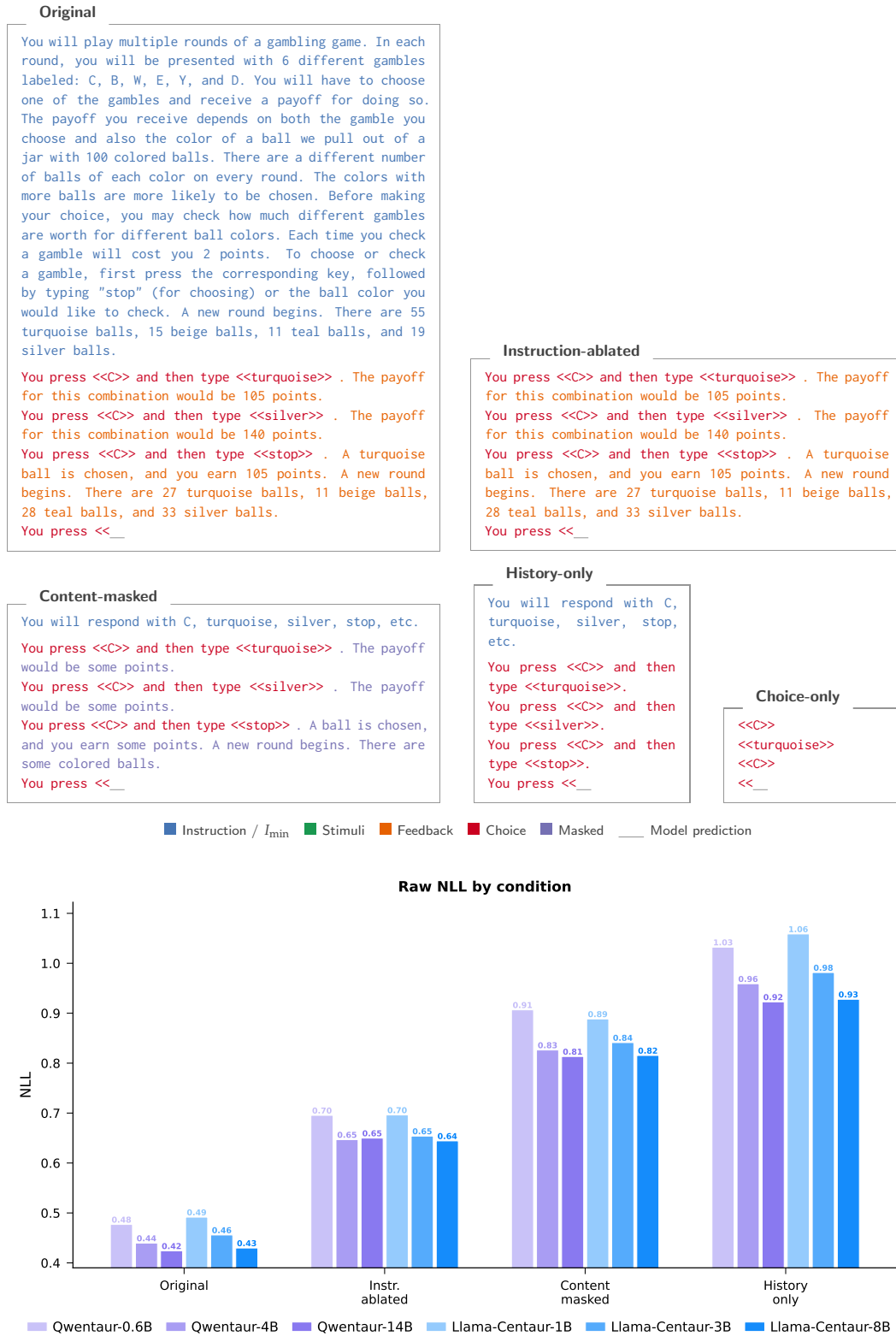


Figure 26: **Decision-making: Risky choice** (Krueger et al., 2024). No retention ratio or chance baseline is shown because trials span multiple stages with varying numbers of actions, so no single k defines a uniform-guessing floor.

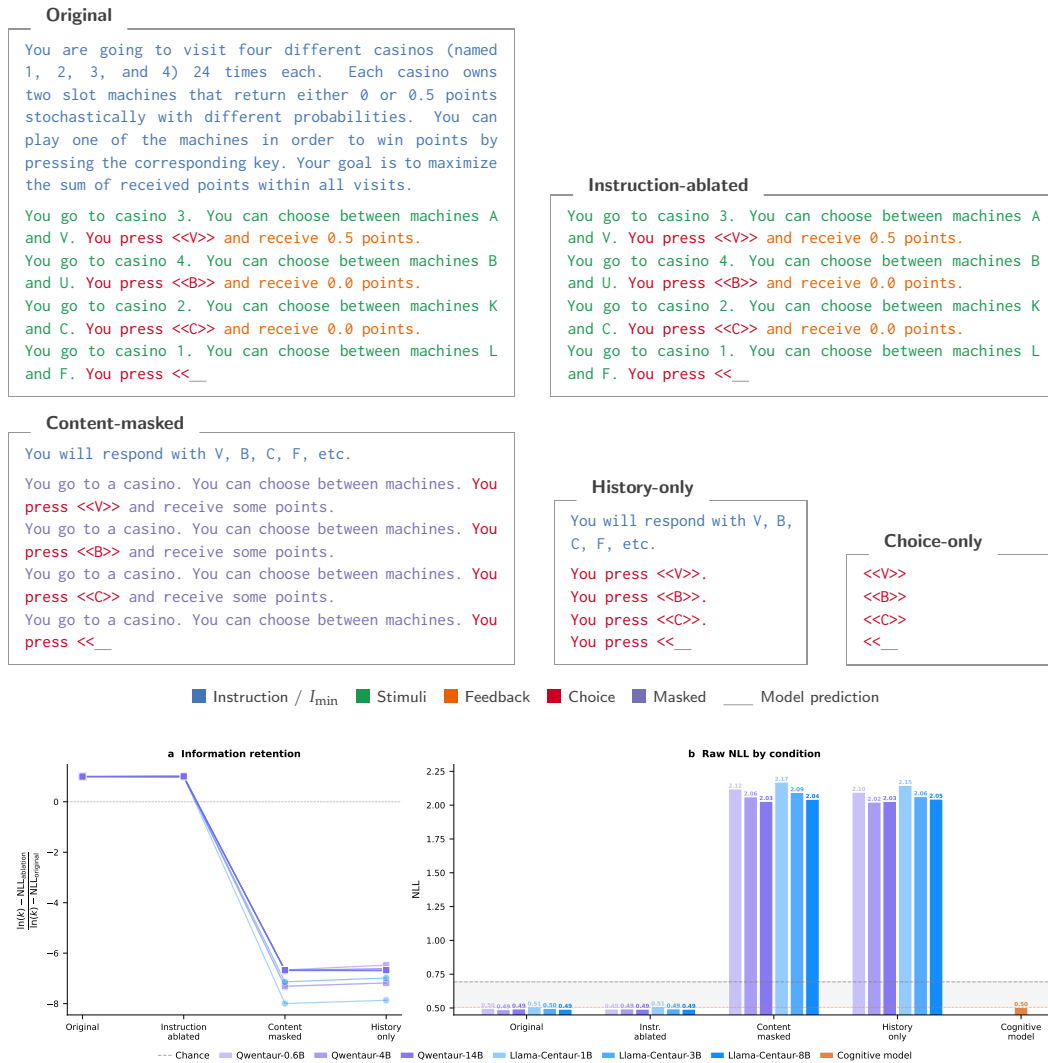


Figure 27: Multi-armed bandits: Prob. instrumental learning (Lefebvre et al., 2017).



Figure 28: Decision-making: choices13k (Peterson et al., 2021).



Figure 29: Decision-making: CPC18 (Plonsky et al., 2018).

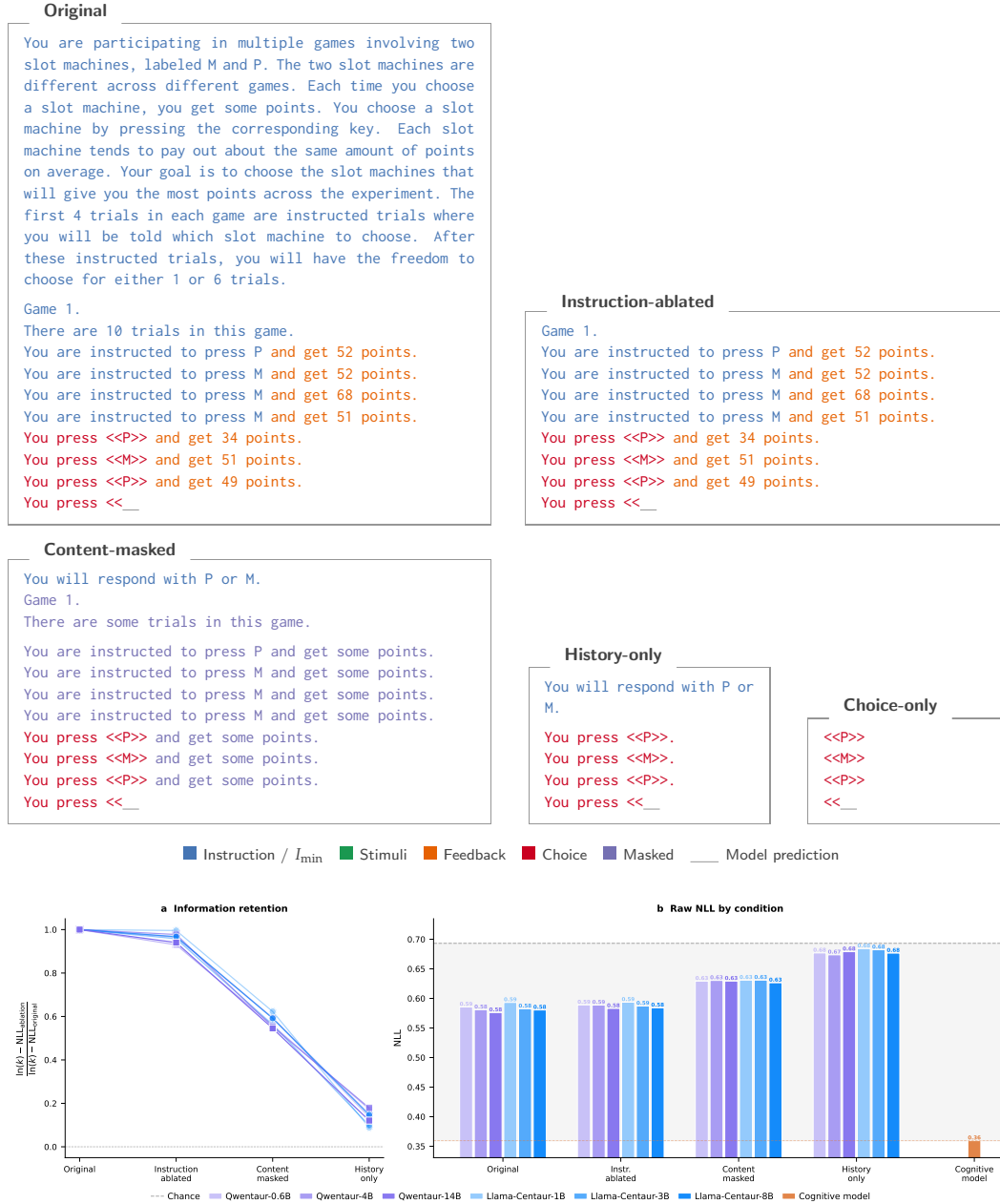


Figure 30: Multi-armed bandits: Horizon task (Sadeghiyeh et al., 2020).



Figure 31: Multi-armed bandits: Structured bandit (Schulz et al., 2020).



Figure 32: Multi-armed bandits: Horizon task (Somerville et al., 2017).



Figure 33: Supervised learning: Weather prediction task (Speekenbrink et al., 2008).

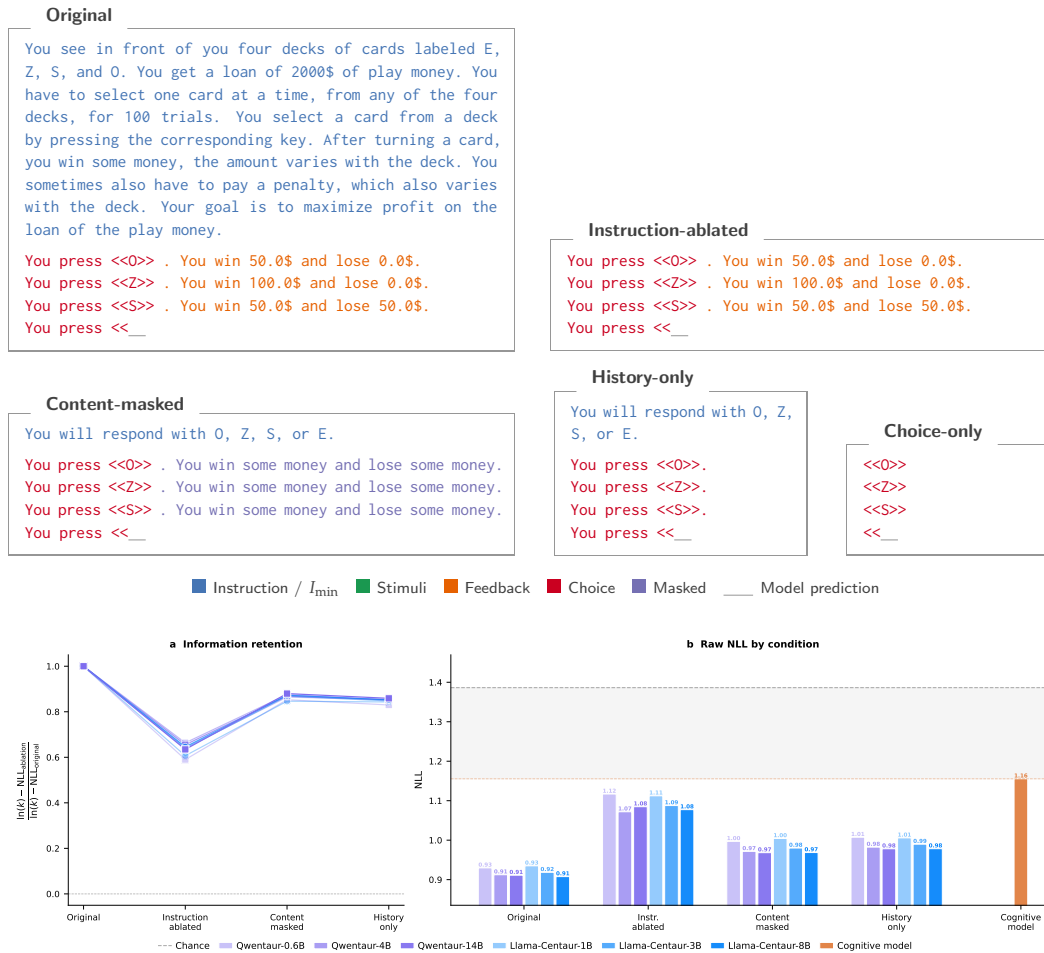


Figure 34: Multi-armed bandits: Iowa gambling task (Steingroever et al., 2015).

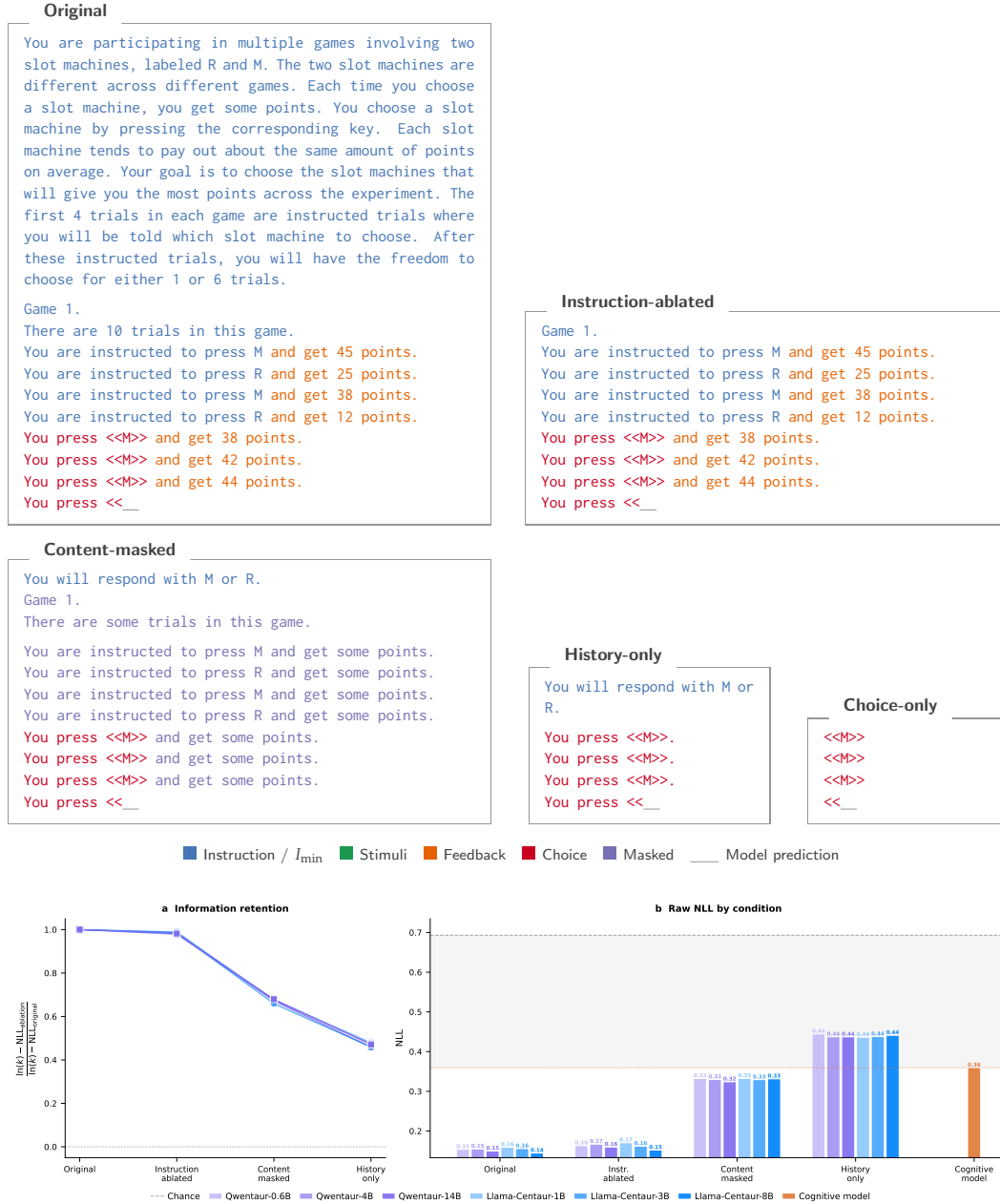


Figure 35: Multi-armed bandits: Horizon task (Waltz et al., 2020).

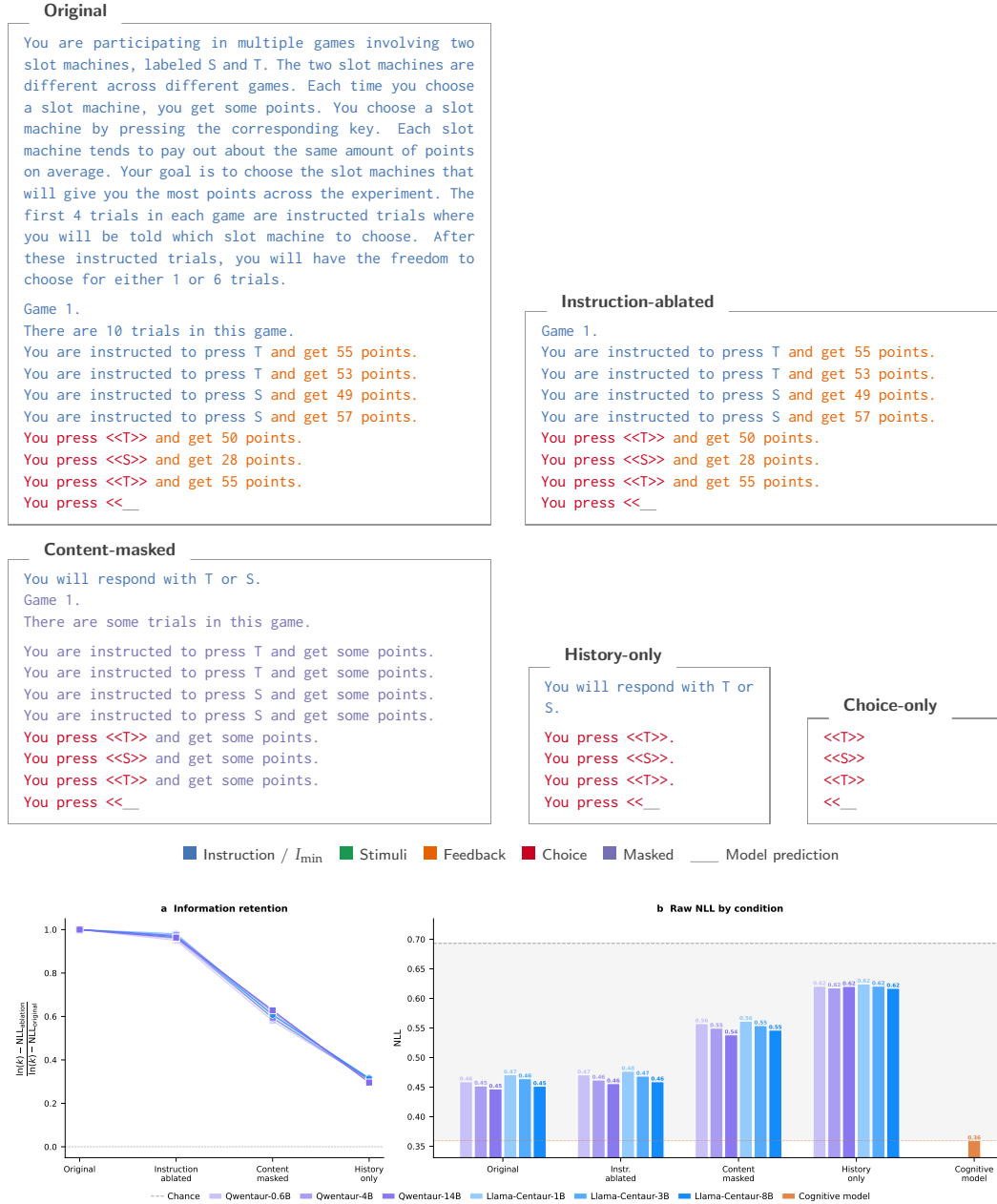


Figure 36: Multi-armed bandits: Horizon task (Wilson et al., 2014).

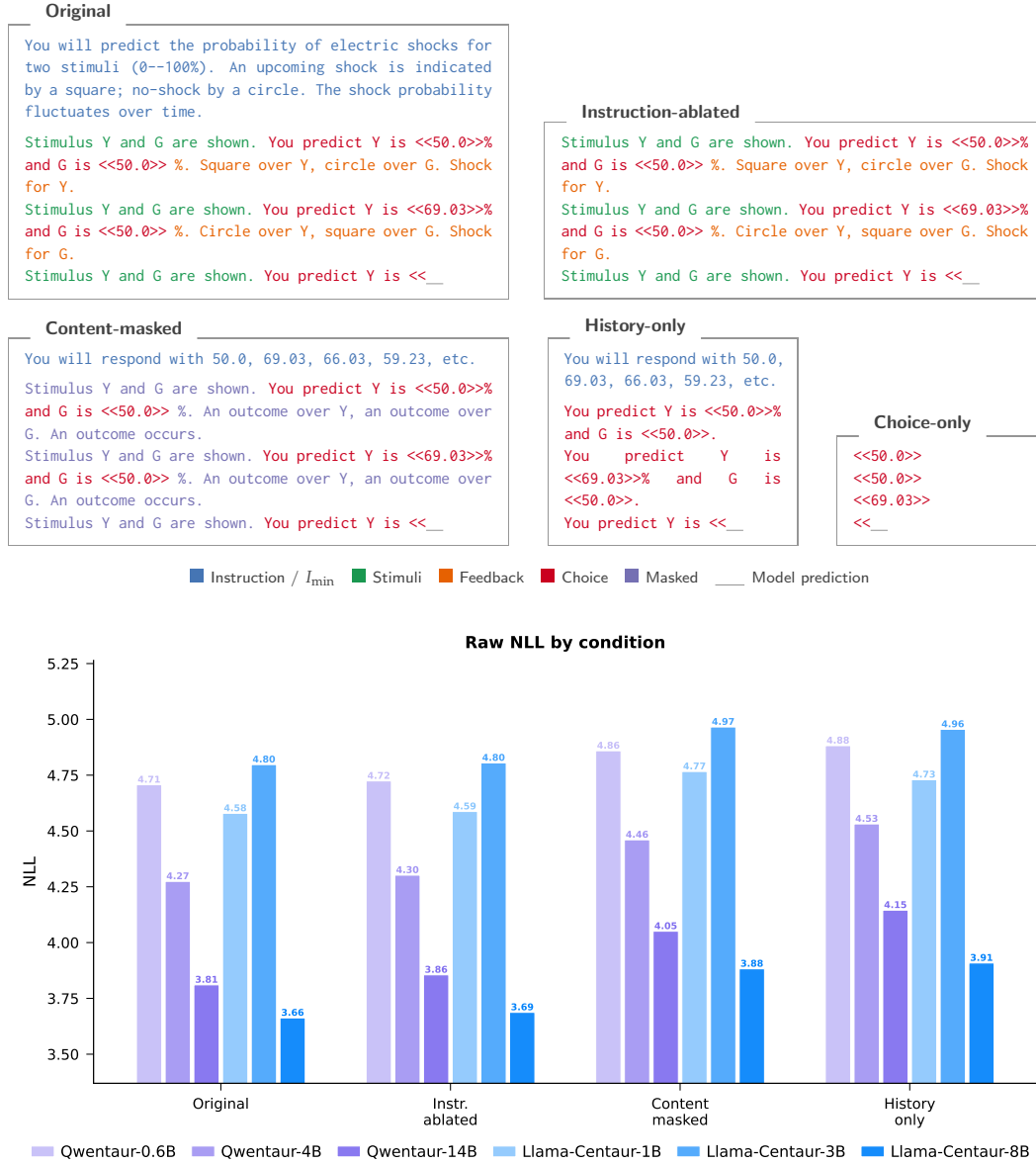


Figure 37: **Supervised learning: Aversive learning** (Wise et al., 2019). No retention ratio or chance baseline is shown because responses are continuous probability estimates, so no single k defines a uniform-guessing floor.



Figure 38: Multi-armed bandits: Spatially correlated MAB (Wu et al., 2018).

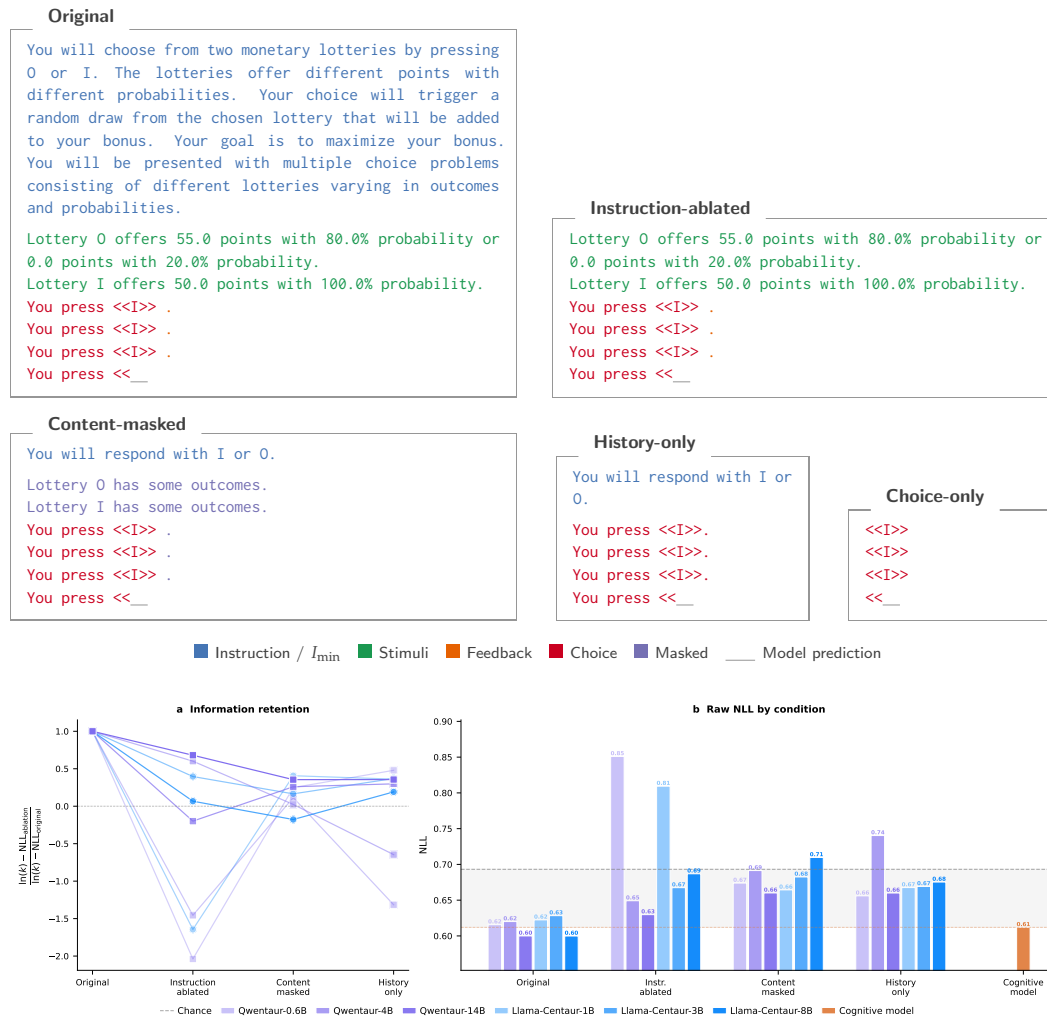


Figure 39: Decision-making: Decisions from description (Wulff et al., 2018).

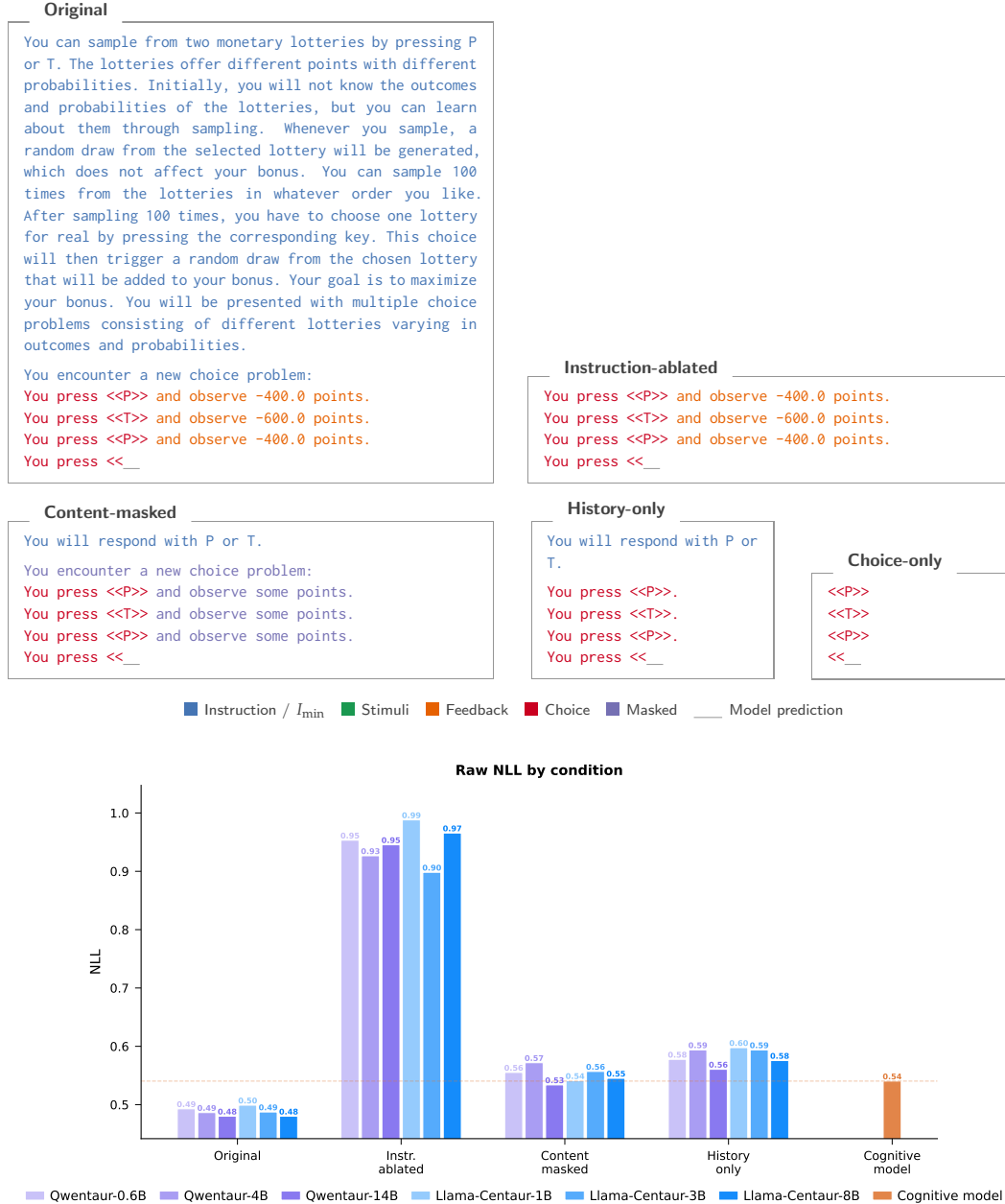


Figure 40: **Multi-armed bandits: Decisions from experience** (Wulff et al., 2018). No retention ratio or chance baseline is shown because trials mix sample, stop, and choose phases with different response formats, so no single k defines a uniform-guessing floor.

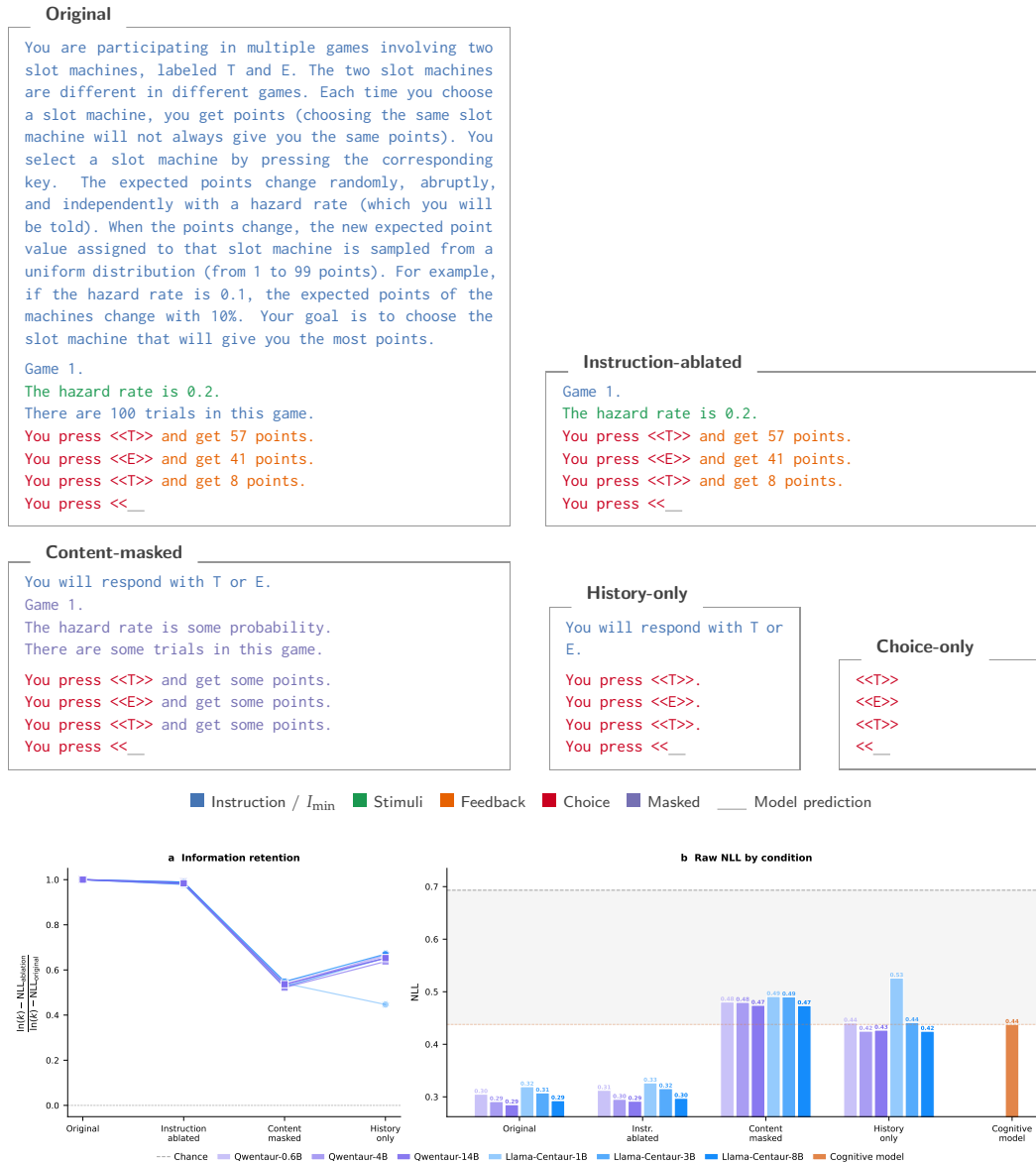


Figure 41: Multi-armed bandits: Changing bandit (Xiong et al., 2023).



Figure 42: Decision-making: Gardening task (Flesch et al., 2018).



Figure 43: Supervised learning: Conditional associative learning (Collins et al., 2014).

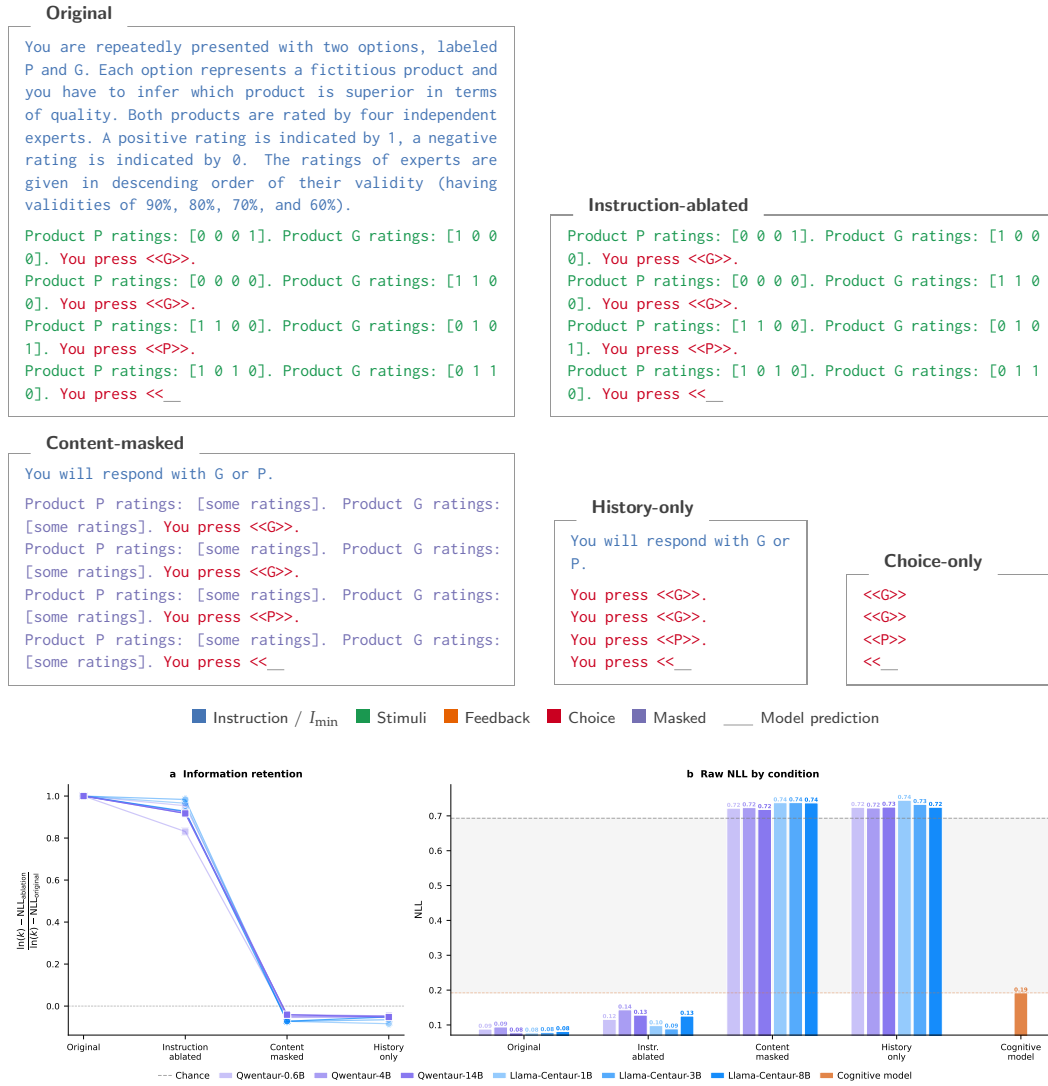


Figure 44: Decision-making: Multi-attribute decision (Hilbig & Moshagen, 2014).

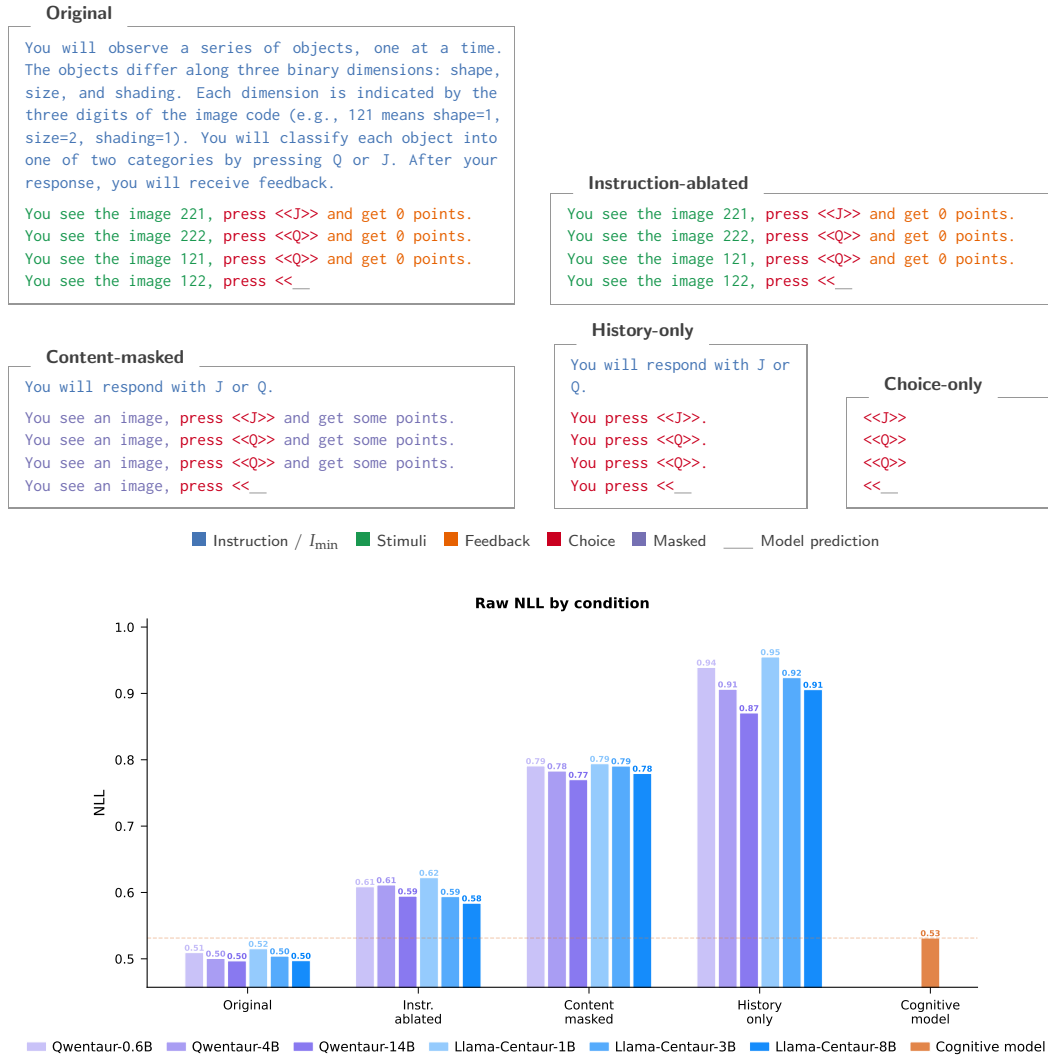


Figure 45: **Supervised learning: Medin categorization** (Levering et al., 2020). No retention ratio or chance baseline is shown because trials mix binary classification with 9-point rating responses, so no single k defines a uniform-guessing floor.

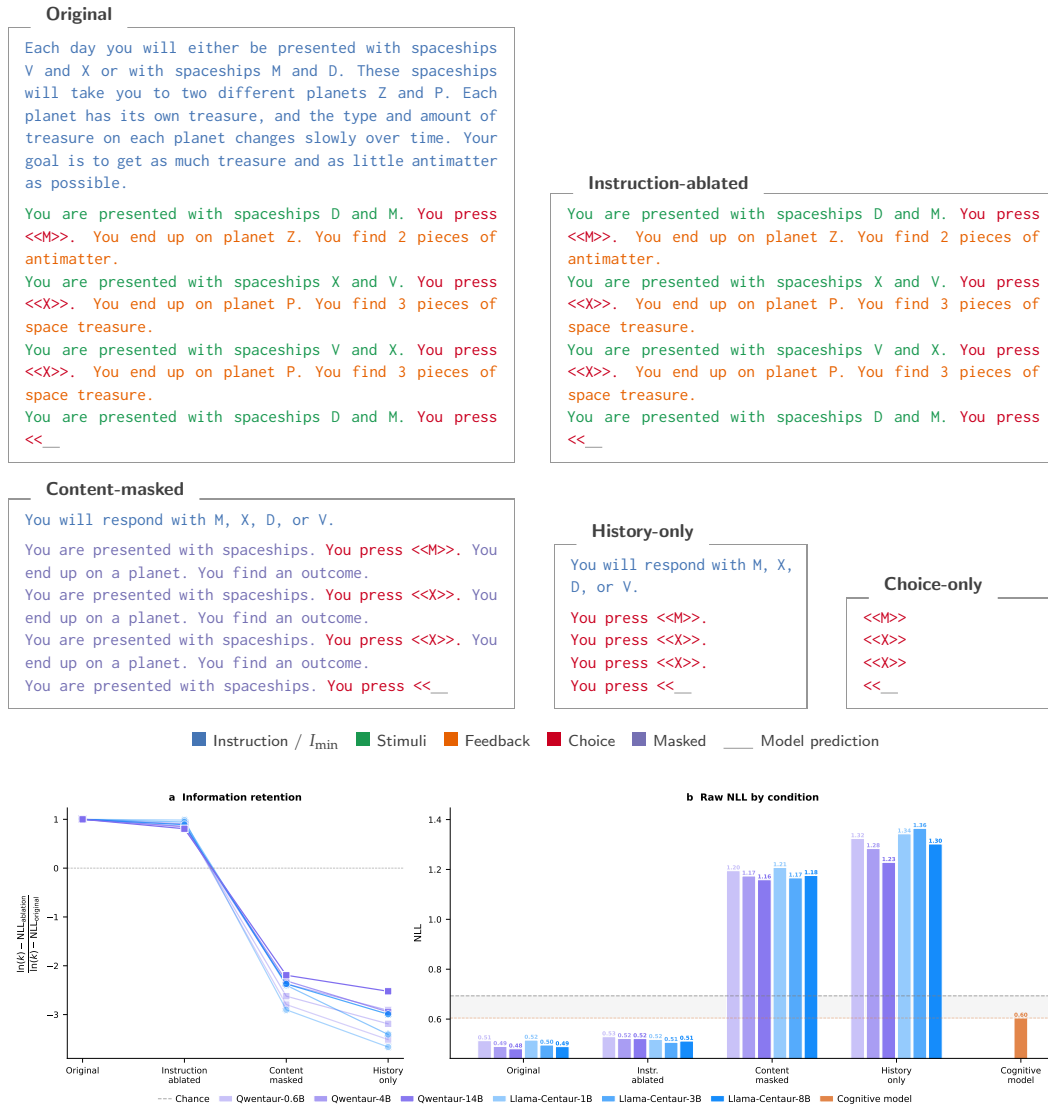


Figure 46: MDP: Model-based two-step task (Kool et al., 2016).

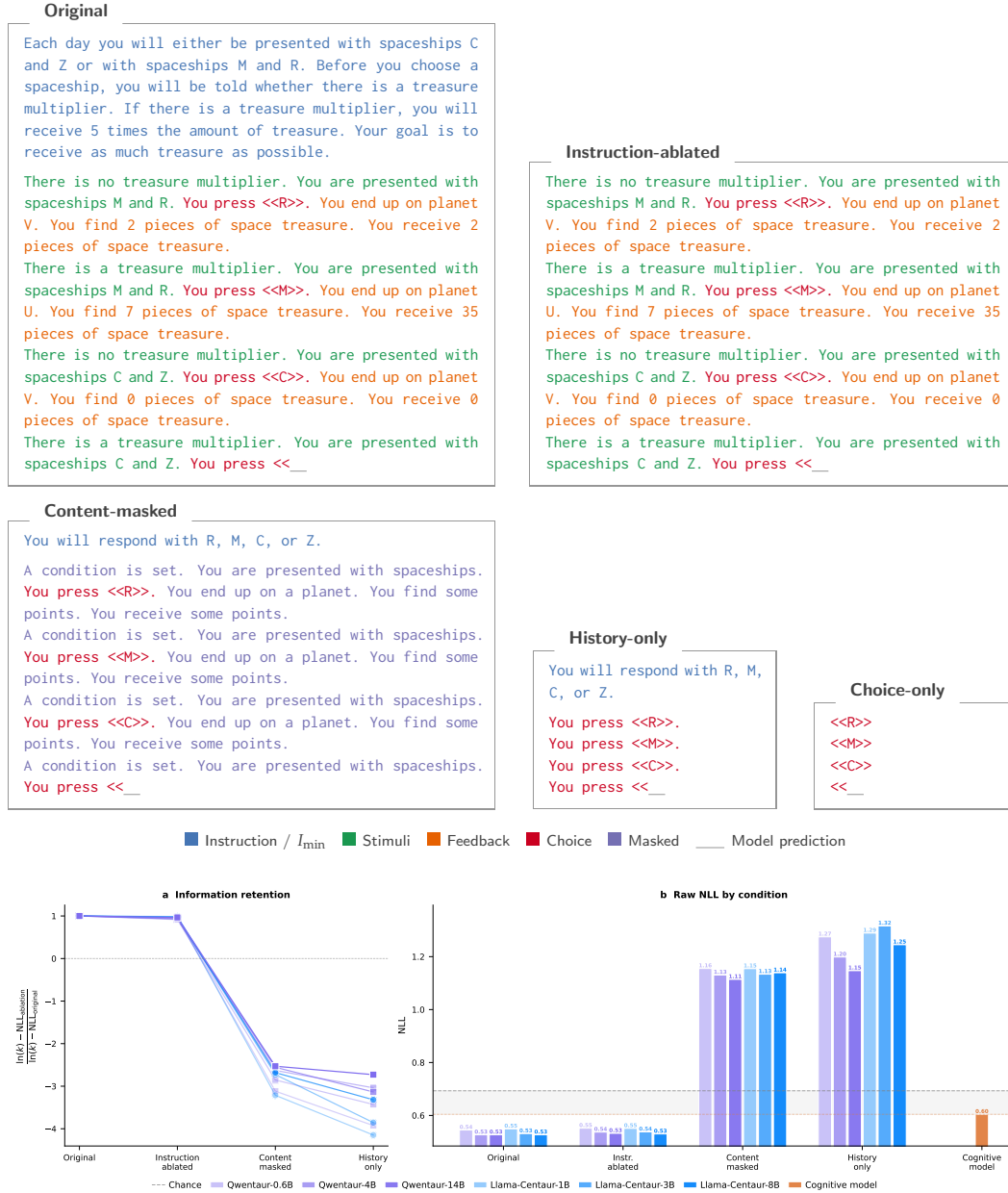


Figure 47: MDP: Two-step effort cost (Kool et al., 2017).

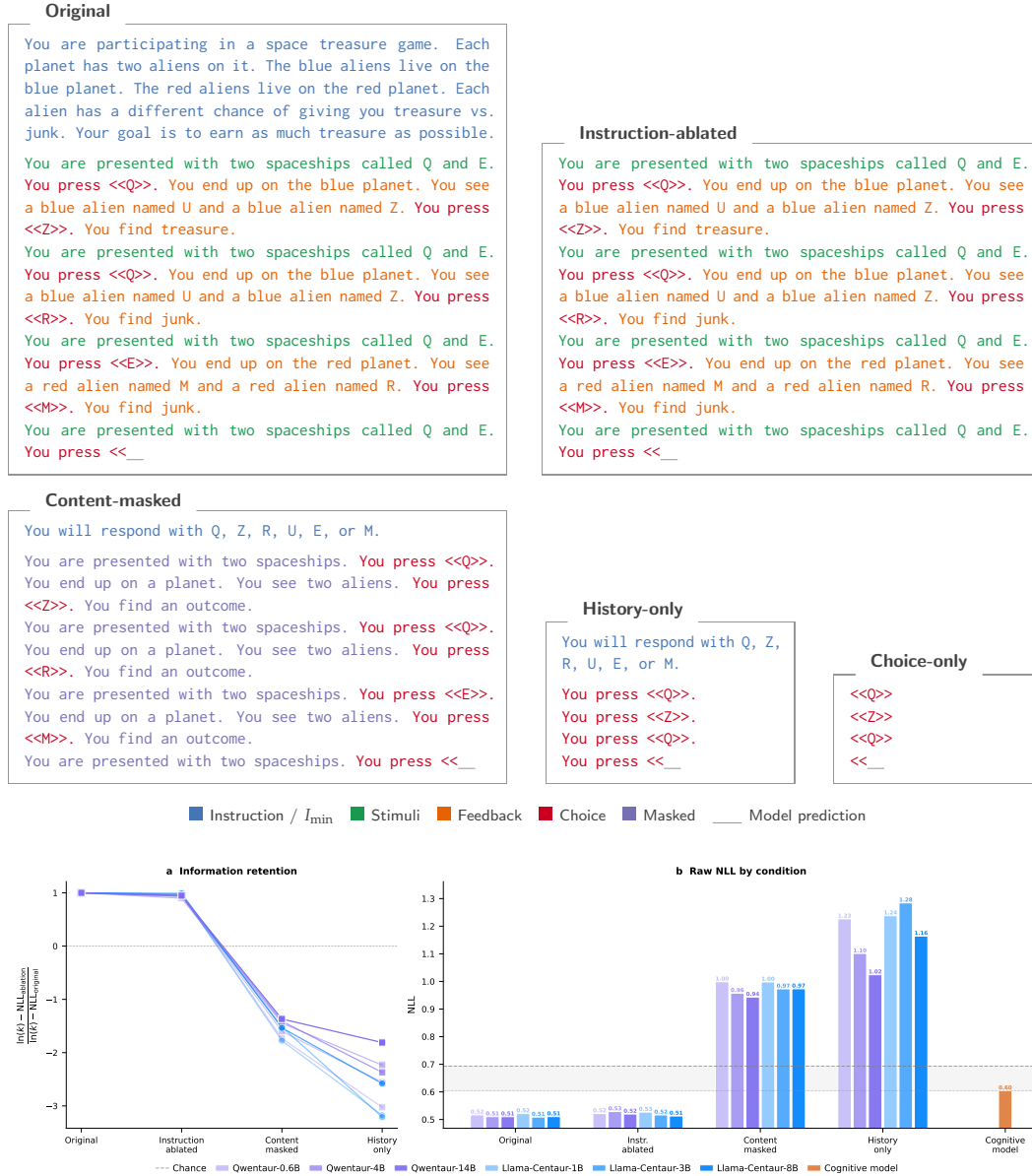


Figure 48: MDP: Two-step with aliens (Zorowitz & Niv, 2023).



Figure 49: MDP: Subway navigation (Tomov et al., 2020).



Figure 50: MDP: Multi-task reinforcement learning (Tomov et al., 2021).

E.3 Order Permutation (non-sequential tasks)

Figures 51 and 52 show original against permuted trial orderings for the two experiments that anchor opposite ends of the exchangeability spectrum.

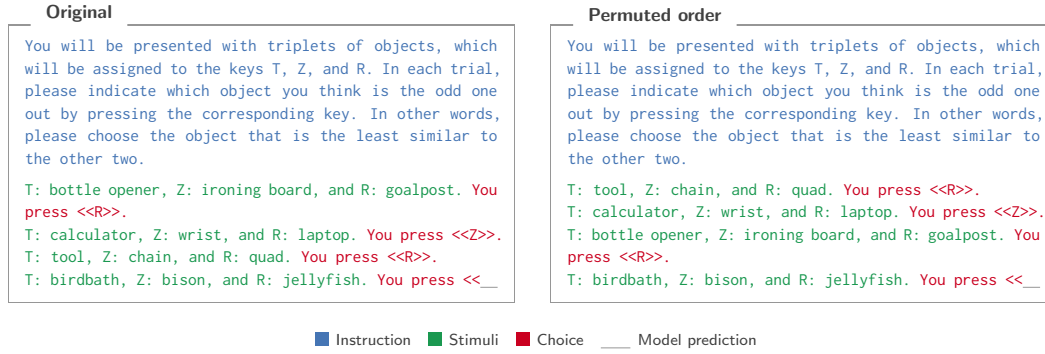


Figure 51: **Miscellaneous: THINGS odd-one-out** (Hebart et al., 2023).

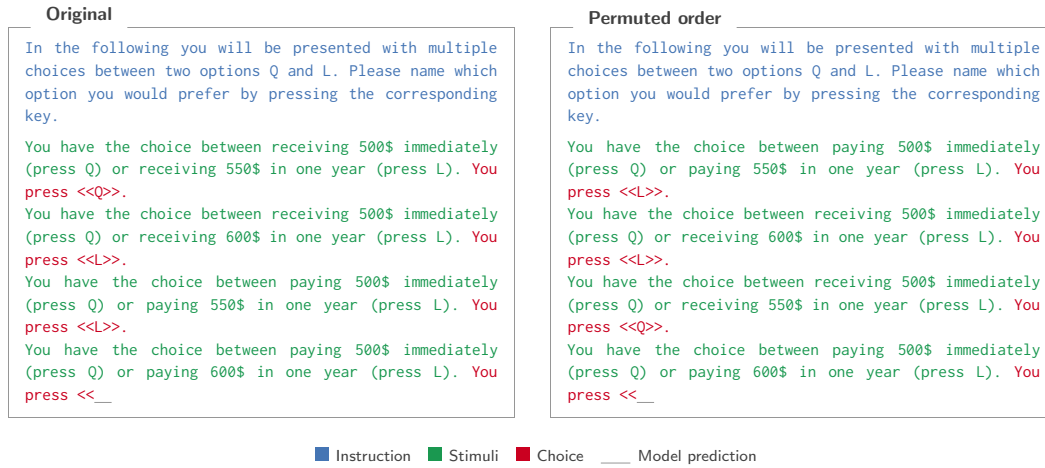


Figure 52: **Decision-making: Intertemporal choice** (Ruggeri et al., 2022).